

6 Control Charts for Variables

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The supplemental material is on the textbook Website www.wiley.com/college/montgomery.

CHAPTER OVERVIEW AND LEARNING OBJECTIVES

A quality characteristic that is measured on a numerical scale is called a **variable**. Examples include dimensions such as length or width, temperature, and volume. This chapter presents Shewhart control charts for these types of quality characteristics. The $\bar{x}$ and R control charts are widely used to monitor the mean and variability of variables. Several variations of the $\bar{x}$ and R charts are also given, including a procedure to adapt them to individual measurements. The chapter concludes with typical applications of variables control charts.

After careful study of this chapter you should be able to do the following:

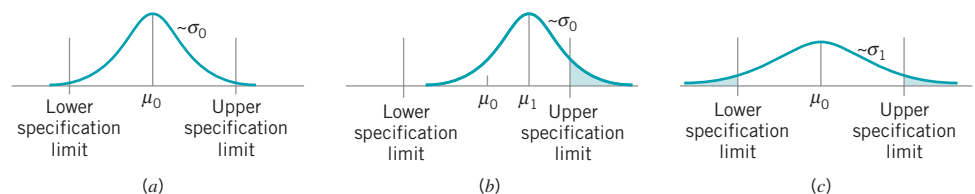
1. Understand the statistical basis of Shewhart control charts for variables
2. Know how to design variables control charts
3. Know how to set up and use $\bar{x}$ and R control charts
4. Know how to estimate process capability from the control chart information
5. Know how to interpret patterns on $\bar{x}$ and R control charts
6. Know how to set up and use $\bar{x}$ and s or s^2 control charts
7. Know how to set up and use control charts for individual measurements
8. Understand the importance of the normality assumption for individuals control charts and know how to check this assumption
9. Understand the rational subgroup concept for variables control charts
10. Determine the average run length for variables control charts

6.1 Introduction

Many quality characteristics can be expressed in terms of a **numerical measurement**. For example, the diameter of a bearing could be measured with a micrometer and expressed in millimeters. A single measurable quality characteristic, such as a dimension, weight, or volume, is called a **variable**. Control charts for variables are used extensively. Control charts are one of the primary tools used in the analyze and control steps of DMAIC.

When dealing with a quality characteristic that is a variable, it is usually necessary to monitor both the mean value of the quality characteristic and its variability. Control of the process average or mean quality level is usually done with the control chart for means, or the **$\bar{x}$ control chart**. Process variability can be monitored with either a control chart for the standard deviation, called the **s control chart**, or a control chart for the range, called an **R control chart**. The R chart is more widely used. Usually, separate $\bar{x}$ and R charts are maintained for each quality characteristic of interest. (However, if the quality characteristics are closely related, this can sometimes cause misleading results; refer to Chapter 12 of Part IV.) The $\bar{x}$ and R (or s) charts are among the most important and useful on-line statistical process monitoring and control techniques.

It is important to maintain control over both the process mean and process variability. Figure 6.1 illustrates the output of a production process. In Fig. 6.1a both the mean μ and standard deviation σ are in control at their nominal values (say, μ_0 and σ_0); consequently, most of the process output falls within the specification limits. However, in Fig. 6.1b the mean has shifted to a value $\mu_1 > \mu_0$, resulting in a higher fraction of nonconforming product. In Fig. 6.1c the process standard deviation has shifted to a value $\sigma_1 > \sigma_0$. This also results in higher process fallout, even though the process mean is still at the nominal value.



■ **FIGURE 6.1** The need for controlling both process mean and process variability. (a) Mean and standard deviation at nominal levels. (b) Process mean $\mu_1 > \mu_0$. (c) Process standard deviation $\sigma_1 > \sigma_0$.

6.2 Control Charts for $\bar{x}$ and R

6.2.1 Statistical Basis of the Charts

Suppose that a quality characteristic is normally distributed with mean μ and standard deviation σ , where both μ and σ are known. If $x_1, x_2, \dots, x_n$ is a sample of size n , then the average of this sample is

$$\bar{x} = \frac{x_1 + x_2 + \dots + x_n}{n}$$

and we know that $\bar{x}$ is normally distributed with mean μ and standard deviation $\sigma_{\bar{x}} = \sigma/\sqrt{n}$. Furthermore, the probability is $1 - \alpha$ that any sample mean will fall between

$$\mu + Z_{\alpha/2}\sigma_{\bar{x}} = \mu + Z_{\alpha/2}\frac{\sigma}{\sqrt{n}} \quad \text{and} \quad \mu - Z_{\alpha/2}\sigma_{\bar{x}} = \mu - Z_{\alpha/2}\frac{\sigma}{\sqrt{n}} \quad (6.1)$$

Therefore, if μ and σ are known, equation (6.1) could be used as upper and lower control limits on a control chart for sample means. As noted in Chapter 5, it is customary to replace $Z_{\alpha/2}$ by 3, so that three-sigma limits are employed. If a sample mean falls outside of these limits, it is an indication that the process mean is no longer equal to μ .

We have assumed that the distribution of the quality characteristic is normal. However, the above results are still approximately correct even if the underlying distribution is non-normal, because of the central limit theorem. We discuss the effect of the normality assumption on variables control charts in Section 6.2.5.

In practice, we usually will not know μ and σ . Therefore, they must be estimated from preliminary samples or subgroups taken when the process is thought to be in control. These estimates should usually be based on at least 20 to 25 samples. Suppose that m samples are available, each containing n observations on the quality characteristic. Typically, n will be small, often either 4, 5, or 6. These small sample sizes usually result from the construction of rational subgroups and from the fact that the sampling and inspection costs associated with variables measurements are usually relatively large. Let $\bar{x}_1, \bar{x}_2, \dots, \bar{x}_m$ be the average of each sample. Then the best estimator of μ , the process average, is the grand average—say,

$$\bar{\bar{x}} = \frac{\bar{x}_1 + \bar{x}_2 + \dots + \bar{x}_m}{m} \quad (6.2)$$

Thus, $\bar{\bar{x}}$ would be used as the center line on the $\bar{x}$ chart.

To construct the control limits, we need an estimate of the standard deviation σ . Recall from Chapter 4 (Section 4.2) that we may estimate σ from either the standard deviations or the ranges of the m samples. For the present, we will use the range method. If $x_1, x_2, \dots, x_n$ is a sample of size n , then the range of the sample is the difference between the largest and smallest observations; that is,

$$R = x_{\max} - x_{\min}$$

Let $R_1, R_2, \dots, R_m$ be the ranges of the m samples. The average range is

$$\bar{R} = \frac{R_1 + R_2 + \dots + R_m}{m} \quad (6.3)$$

We may now give the formulas for constructing the control limits on the $\bar{x}$ chart. They are as follows:

Control Limits for the $\bar{x}$ Chart

$$\begin{aligned} \text{UCL} &= \bar{\bar{x}} + A_2 \bar{R} \\ \text{Center line} &= \bar{\bar{x}} \\ \text{LCL} &= \bar{\bar{x}} - A_2 \bar{R} \end{aligned} \quad (6.4)$$

The constant A_2 is tabulated for various sample sizes in Appendix Table VI.

Process variability may be monitored by plotting values of the sample range R on a control chart. The center line and control limits of the R chart are as follows:

Control Limits for the R Chart

$$\begin{aligned} \text{UCL} &= D_4 \bar{R} \\ \text{Center line} &= \bar{R} \\ \text{LCL} &= D_3 \bar{R} \end{aligned} \quad (6.5)$$

The constants D_3 and D_4 are tabulated for various values of n in Appendix Table VI.

Development of Equations (6.4) and (6.5). The development of the equations for computing the control limits on the $\bar{x}$ and R control charts is relatively easy. In Chapter 4 (Section 4.2) we observed that there is a well-known relationship between the range of a sample from a normal distribution and the standard deviation of that distribution. The random variable $W = R/\sigma$ is called the **relative range**. The parameters of the distribution of W are a function of the sample size n . The mean of W is d_2 . Consequently, an estimator of σ is $\hat{\sigma} = R/d_2$. Values of d_2 for various sample sizes are given in Appendix Table VI. Therefore, if $\bar{R}$ is the average range of the m preliminary samples, we may use

$$\hat{\sigma} = \frac{\bar{R}}{d_2} \quad (6.6)$$

to estimate σ . This is an unbiased estimator of σ .

If we use $\bar{\bar{x}}$ as an estimator of μ and $\bar{R}/d_2$ as an estimator of σ , then the parameters of the $\bar{x}$ chart are

$$\begin{aligned} \text{UCL} &= \bar{\bar{x}} + \frac{3}{d_2\sqrt{n}} \bar{R} \\ \text{Center line} &= \bar{\bar{x}} \\ \text{LCL} &= \bar{\bar{x}} - \frac{3}{d_2\sqrt{n}} \bar{R} \end{aligned} \quad (6.7)$$

If we define

$$A_2 = \frac{3}{d_2\sqrt{n}} \quad (6.8)$$

then equation (6.7) reduces to equation (6.4).

Now consider the R chart. The center line will be $\bar{R}$. To determine the control limits, we need an estimate of σ_R . Assuming that the quality characteristic is normally distributed, $\hat{\sigma}_R$ can be found from the distribution of the relative range $W = R/\sigma$. The standard deviation of W , say d_3 , is a known function of n . Thus, since

$$R = W\sigma$$

the standard deviation of R is

$$\sigma_R = d_3\sigma$$

Since σ is unknown, we may estimate σ_R by

$$\hat{\sigma}_R = d_3 \frac{\bar{R}}{d_2} \quad (6.9)$$

Consequently, the parameters of the R chart with the usual three-sigma control limits are

$$\begin{aligned} \text{UCL} &= \bar{R} + 3\hat{\sigma}_R = \bar{R} + 3d_3 \frac{\bar{R}}{d_2} \\ \text{Center line} &= \bar{R} \\ \text{LCL} &= \bar{R} - 3\hat{\sigma}_R = \bar{R} - 3d_3 \frac{\bar{R}}{d_2} \end{aligned} \quad (6.10)$$

If we let

$$D_3 = 1 - 3 \frac{d_3}{d_2} \quad \text{and} \quad D_4 = 1 + 3 \frac{d_3}{d_2}$$

equation (6.10) reduces to equation (6.5).

Phase I Application of $\bar{x}$ and R Charts. In **phase I control chart usage**, when preliminary samples are used to construct $\bar{x}$ and R control charts, it is customary to treat the control limits obtained from equations (6.4) and (6.5) as **trial control limits**. They allow us to determine whether the process was in control when the m initial samples were selected. To determine whether the process was in control when the preliminary samples were collected, plot the values of $\bar{x}$ and R from each sample on the charts and analyze the resulting display. If all points plot inside the control limits and no systematic behavior is evident, we conclude

that the process was in control in the past, and the trial control limits are suitable for controlling current or future production. It is highly desirable to have 20–25 samples or subgroups of size n (typically n is between 3 and 5) to compute the trial control limits. We can, of course, work with fewer data but the control limits are not as reliable.

Suppose that one or more of the values of either $\bar{x}$ or R plot out of control when compared to the trial control limits. Clearly, if control limits for current or future production are to be meaningful, they must be based on data from a process that is in control. Therefore, when the hypothesis of past control is rejected, it is necessary to **revise** the trial control limits. This is done by examining each of the out-of-control points, looking for an assignable cause. If an assignable cause is found, the point is discarded and the trial control limits are recalculated, using only the remaining points. Then these remaining points are reexamined for control. (Note that points that were in control initially may now be out of control, because the new trial control limits will generally be tighter than the old ones.) This process is continued until all points plot in control, at which point the trial control limits are adopted for current use.

In some cases, it may not be possible to find an assignable cause for a point that plots out of control. There are two courses of action open to us. The first of these is to eliminate the point, just as if an assignable cause had been found. There is no analytical justification for choosing this action, other than that points that are outside of the control limits are likely to have been drawn from a probability distribution characteristic of an out-of-control state. The alternative is to retain the point (or points) considering the trial control limits as appropriate for current control. Of course, if the point really does represent an out-of-control condition, the resulting control limits will be too wide. However, if there are only one or two such points, this will not distort the control chart significantly. If future samples still indicate control, then the unexplained points can probably be safely dropped.

Occasionally, when the initial sample values of $\bar{x}$ and R are plotted against the trial control limits, many points will plot out of control. Clearly, if we arbitrarily drop the out-of-control points, we will have an unsatisfactory situation, as few data will remain with which we can recompute reliable control limits. We also suspect that this approach would ignore much useful information in the data. On the other hand, searching for an assignable cause for *each* out-of-control point is unlikely to be successful. We have found that when many of the initial samples plot out of control against the trial limits, it is better to concentrate on **the patterns on control charts** formed by these points. Such a pattern will almost always exist. Usually, the assignable cause associated with the pattern of out-of-control points is fairly easy to identify. Removal of this process problem usually results in a major process improvement.

6.2.2 Development and Use of $\bar{x}$ and R Charts

In the previous section we presented the statistical background for $\bar{x}$ and R control charts. We now illustrate the construction and application of these charts. We also discuss some guidelines for using these charts in practice.

EXAMPLE 6.1 $\bar{x}$ and R Charts for a Manufacturing Process

A hard-bake process (see Section 5.3.1) is used in conjunction with photolithography in semiconductor manufacturing. We wish to establish statistical control of the flow width of the resist in this process using $\bar{x}$ and R charts. Twenty-five samples,

each of size five wafers, have been taken when we think the process is in control. The interval of time between samples or subgroups is one hour. The flow width measurement data (in microns) from these samples are shown in Table 6.1.

(continued)

■ TABLE 6.1

Flow Width Measurements (microns) for the Hard-Bake Process

Sample Number	Wafers					$\bar{x}_i$	R_i
	1	2	3	4	5		
1	1.3235	1.4128	1.6744	1.4573	1.6914	1.5119	0.3679
2	1.4314	1.3592	1.6075	1.4666	1.6109	1.4951	0.2517
3	1.4284	1.4871	1.4932	1.4324	1.5674	1.4817	0.1390
4	1.5028	1.6352	1.3841	1.2831	1.5507	1.4712	0.3521
5	1.5604	1.2735	1.5265	1.4363	1.6441	1.4882	0.3706
6	1.5955	1.5451	1.3574	1.3281	1.4198	1.4492	0.2674
7	1.6274	1.5064	1.8366	1.4177	1.5144	1.5805	0.4189
8	1.4190	1.4303	1.6637	1.6067	1.5519	1.5343	0.2447
9	1.3884	1.7277	1.5355	1.5176	1.3688	1.5076	0.3589
10	1.4039	1.6697	1.5089	1.4627	1.5220	1.5134	0.2658
11	1.4158	1.7667	1.4278	1.5928	1.4181	1.5242	0.3509
12	1.5821	1.3355	1.5777	1.3908	1.7559	1.5284	0.4204
13	1.2856	1.4106	1.4447	1.6398	1.1928	1.3947	0.4470
14	1.4951	1.4036	1.5893	1.6458	1.4969	1.5261	0.2422
15	1.3589	1.2863	1.5996	1.2497	1.5471	1.4083	0.3499
16	1.5747	1.5301	1.5171	1.1839	1.8662	1.5344	0.6823
17	1.3680	1.7269	1.3957	1.5014	1.4449	1.4874	0.3589
18	1.4163	1.3864	1.3057	1.6210	1.5573	1.4573	0.3153
19	1.5796	1.4185	1.6541	1.5116	1.7247	1.5777	0.3062
20	1.7106	1.4412	1.2361	1.3820	1.7601	1.5060	0.5240
21	1.4371	1.5051	1.3485	1.5670	1.4880	1.4691	0.2185
22	1.4738	1.5936	1.6583	1.4973	1.4720	1.5390	0.1863
23	1.5917	1.4333	1.5551	1.5295	1.6866	1.5592	0.2533
24	1.6399	1.5243	1.5705	1.5563	1.5530	1.5688	0.1156
25	1.5797	1.3663	1.6240	1.3732	1.6887	1.5264	0.3224
						$\Sigma \bar{x}_i = 37.6400$	$\Sigma R_i = 8.1302$
						$\bar{\bar{x}} = 1.5056$	$\bar{R} = 0.32521$

SOLUTION

When setting up $\bar{x}$ and R control charts, it is best to begin with the R chart. Because the control limits on the $\bar{x}$ chart depend on the process variability, unless process variability is in control, these limits will not have much meaning. Using the data in Table 6.1, we find that the center line for the R chart is

$$\bar{R} = \frac{\sum_{i=1}^{25} R_i}{25} = \frac{8.1302}{25} = 0.32521$$

For samples of $n = 5$, we find from Appendix Table VI that $D_3 = 0$ and $D_4 = 2.114$. Therefore, the control limits for the R chart are, using equation (6.5),

$$\text{LCL} = \bar{R}D_3 = 0.32521(0) = 0$$

$$\text{UCL} = \bar{R}D_4 = 0.32521(2.114) = 0.68749$$

The R chart is shown in Fig. 6.2b. Both control charts in Fig. 6.2 were constructed by Minitab. Note that the UCL for the R chart is reported as 0.6876, because Minitab uses more decimal

places in the calculation than we did. When the 25 sample ranges are plotted on the R chart in Fig. 6.2b there is no indication of an out-of-control condition.

Since the R chart indicates that process variability is in control, we may now construct the $\bar{x}$ chart. The center line is

$$\bar{\bar{x}} = \frac{\sum_{i=1}^{25} \bar{x}_i}{25} = \frac{37.6400}{25} = 1.5056$$

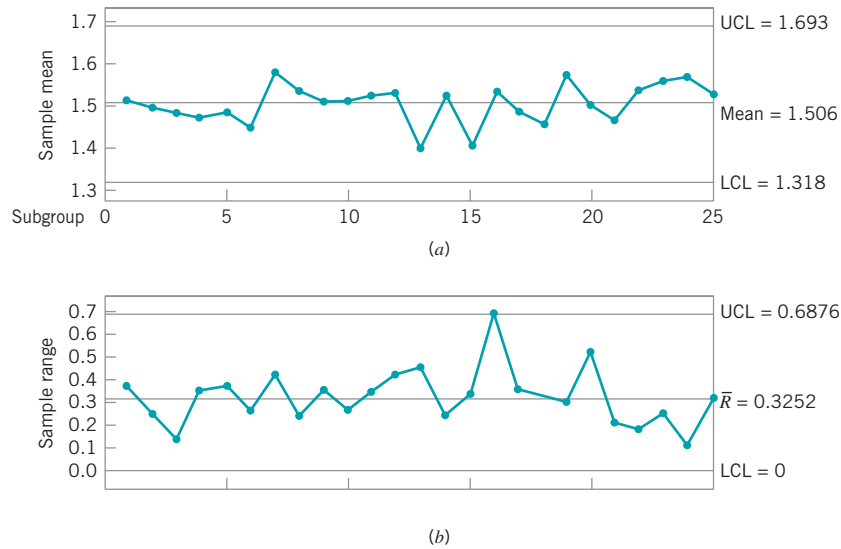
To find the control limits on the $\bar{x}$ chart, we use $A_2 = 0.577$ from Appendix Table VI for samples of size $n = 5$ and equation (6.4) to find

$$UCL = \bar{\bar{x}} + A_2 \bar{R} = 1.5056 + (0.577)(0.32521) = 1.69325$$

and

$$LCL = \bar{\bar{x}} - A_2 \bar{R} = 1.5056 - (0.577)(0.32521) = 1.31795$$

The $\bar{x}$ chart is shown in Fig. 6.2a. When the preliminary sample averages are plotted on this chart, no indication of an out-of-control condition is observed. Therefore, since both the $\bar{x}$ and R charts exhibit control, we would conclude that the process is in control at the stated levels and adopt the trial control limits for use in phase II, where monitoring of future production is of interest.



■ **FIGURE 6.2** $\bar{x}$ and R charts (from Minitab) for flow width in the hard-bake process.

Estimating Process Capability. The $\bar{x}$ and R charts provide information about the performance or **process capability** of the process. From the $\bar{x}$ chart, we may estimate the mean flow width of the resist in the hard-bake process as $\bar{\bar{x}} = 1.5056$ microns. The process standard deviation may be estimated using equation (6.6); that is,

$$\hat{\sigma} = \frac{\bar{R}}{d_2} = \frac{0.32521}{2.326} = 0.1398 \text{ microns}$$

where the value of d_2 for samples of size five is found in Appendix Table VI. The specification limits on flow width are 1.50 ± 0.50 microns. The control chart data may be used to describe the capability of the process to produce wafers relative to these specifications. Assuming that flow width is a normally distributed random variable, with mean 1.5056 and

standard deviation 0.1398, we may estimate the fraction of nonconforming wafers produced as

$$\begin{aligned}
 p &= P\{x < 1.00\} + P\{x > 2.00\} \\
 &= \Phi\left(\frac{1.00 - 1.5056}{0.1398}\right) + 1 - \Phi\left(\frac{2.00 - 1.5056}{0.1398}\right) \\
 &= \Phi(-3.61660) + 1 - \Phi(3.53648) \\
 &\approx 0.00015 + 1 - 0.99980 \\
 &\approx 0.00035
 \end{aligned}$$

That is, about 0.035 percent [350 parts per million (ppm)] of the wafers produced will be outside of the specifications.

Another way to express process capability is in terms of the **process capability ratio (PCR)** C_p , which for a quality characteristic with both upper and lower specification limits (USL and LSL, respectively) is

$$C_p = \frac{USL - LSL}{6\sigma} \quad (6.11)$$

Note that the 6σ spread of the process is the basic definition of process capability. Since σ is usually unknown, we must replace it with an estimate. We frequently use $\hat{\sigma} = \bar{R}/d_2$ as an estimate of σ , resulting in an estimate $\hat{C}_p$ of C_p . For the hard-bake process, since $\bar{R}/d_2 = \hat{\sigma} = 0.1398$, we find that

$$\hat{C}_p = \frac{2.00 - 1.00}{6(0.1398)} = \frac{1.00}{0.8388} = 1.192$$

This implies that the “natural” tolerance limits in the process (three-sigma above and below the mean) are inside the lower and upper specification limits. Consequently, a moderately small number of nonconforming wafers will be produced. The PCR C_p may be interpreted another way. The quantity

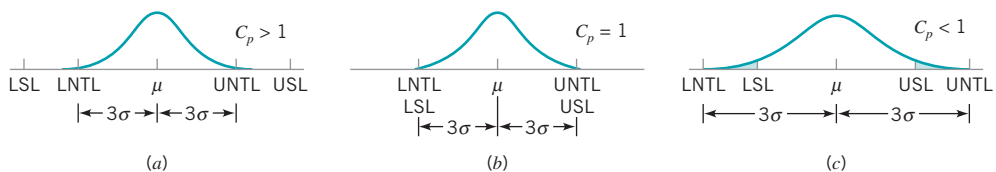
$$P = \left(\frac{1}{C_p}\right) 100\%$$

is simply the percentage of the specification band that the process uses up. For the hard-bake process an estimate of P is

$$\hat{P} = \left(\frac{1}{\hat{C}_p}\right) 100\% = \left(\frac{1}{1.192}\right) 100\% = 83.89$$

That is, the process uses up about 84% of the specification band.

Figure 6.3 illustrates three cases of interest relative to the PCR C_p and process specifications. In Fig. 6.3a the PCR C_p is greater than unity. This means that the process uses up much less than 100% of the tolerance band. Consequently, relatively few non-conforming units



■ FIGURE 6.3 Process fallout and the process capability ratio C_p .

will be produced by this process. Figure 6.3*b* shows a process for which the PCR $C_p = 1$; that is, the process uses up all the tolerance band. For a normal distribution this would imply about 0.27% (or 2700 ppm) nonconforming units. Finally, Fig. 6.3*c* presents a process for which the PCR $C_p < 1$; that is, the process uses up more than 100% of the tolerance band. In this case the process is very yield-sensitive, and a large number of nonconforming units will be produced.

Note that all the cases in Fig. 6.3 assume that the process is centered at the midpoint of the specification band. In many situations this will not be the case, and as we will see in Chapter 8 (which is devoted to a more extensive treatment of process capability analysis), some modification of the PCR C_p is necessary to describe this situation adequately.

Revision of Control Limits and Center Lines. The effective use of any control chart will require **periodic revision** of the control limits and center lines. Some practitioners establish regular periods for review and revision of control chart limits, such as every week, every month, or every 25, 50, or 100 samples. When revising control limits, remember that it is highly desirable to use at least 25 samples or subgroups (some authorities recommend 200–300 individual observations) in computing control limits.

Sometimes the user will replace the center line of the $\bar{x}$ chart with a **target value**, say $\bar{x}_0$. If the R chart exhibits control, this can be helpful in shifting the process average to the desired value, particularly in processes where the mean may be changed by a fairly simple adjustment of a manipulatable variable in the process. If the mean is not easily influenced by a simple process adjustment, then it is likely to be a complex and unknown function of several process variables and a target value $\bar{x}_0$ may not be helpful, as use of that value could result in many points outside the control limits. In such cases, we would not necessarily know whether the point was really associated with an assignable cause or whether it plotted outside the limits because of a poor choice for the center line.

When the R chart is out of control, we often eliminate the out-of-control points and recompute a revised value of $\bar{R}$. This value is then used to determine new limits and center line on the R chart and new limits on the $\bar{x}$ chart. This will tighten the limits on both charts, making them consistent with a process standard deviation σ consistent with use of the *revised* $\bar{R}$ in the relationship R/d_2 . This estimate of σ could be used as the basis of a preliminary analysis of process capability.

Phase II Operation of the $\bar{x}$ and R Charts. Once a set of reliable control limits is established, we use the control chart for monitoring future production. This is called **phase II control chart usage**.

Twenty additional samples of wafers from the hard-bake process were collected after the control charts were established and the sample values of $\bar{x}$ and R plotted on the control charts immediately after each sample was taken. The data from these new samples are shown in Table 6.2, and the continuations of the $\bar{x}$ and R charts are shown in Fig. 6.4. The control charts indicate that the process is in control, until the $\bar{x}$ -value from the 43rd sample is plotted. Since this point (as well as the $\bar{x}$ -value from sample 45) plots above the upper control limit, we would suspect that an assignable cause has occurred at or before that time. The general pattern of points on the $\bar{x}$ chart from about subgroup 38 onward is indicative of a shift in the process mean.

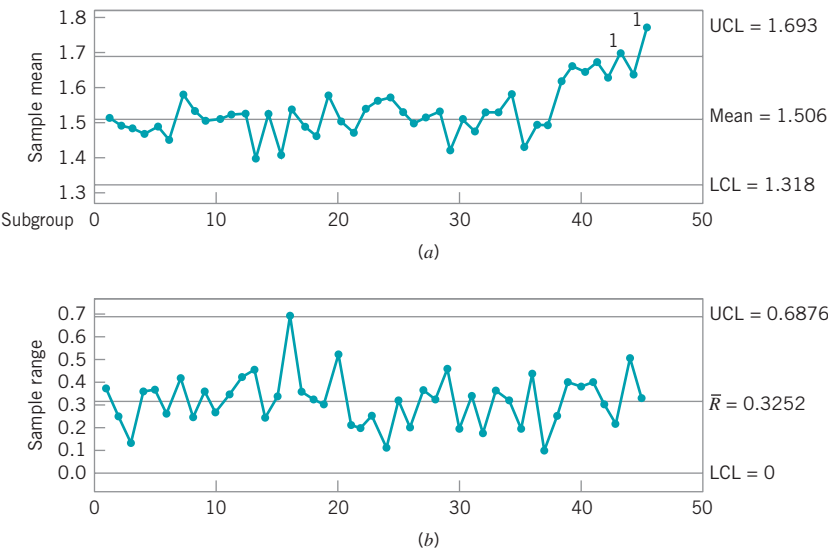
Once the control chart is established and is being used in on-line process monitoring, one is often tempted to use the sensitizing rules (or Western Electric rules) discussed in Chapter 5 (Section 5.3.6) to speed up shift detection. Here, for example, the use of such rules would likely result in the shift being detected around sample 40. However, recall the discussion from Section 5.3.3 in which we discouraged the routine use of these sensitizing rules

■ TABLE 6.2
Additional Samples for Example 6.1

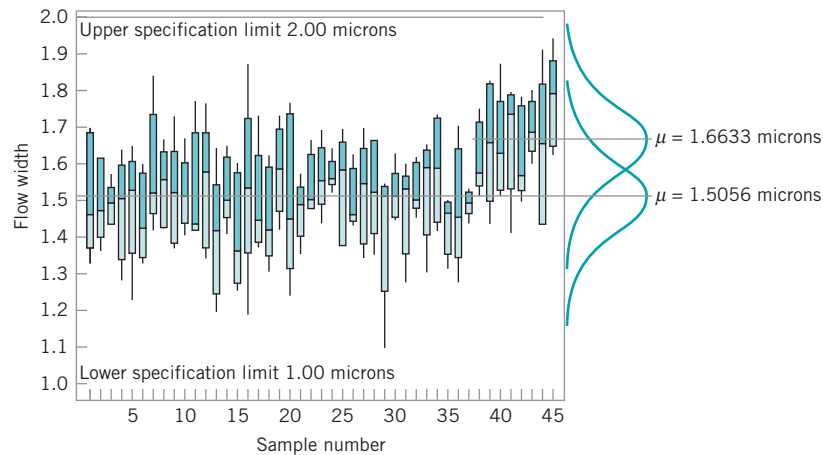
Sample Number	Wafers					$\bar{x}_i$	R_i
	1	2	3	4	5		
26	1.4483	1.5458	1.4538	1.4303	1.6206	1.4998	0.1903
27	1.5435	1.6899	1.5830	1.3358	1.4187	1.5142	0.3541
28	1.5175	1.3446	1.4723	1.6657	1.6661	1.5332	0.3215
29	1.5454	1.0931	1.4072	1.5039	1.5264	1.4152	0.4523
30	1.4418	1.5059	1.5124	1.4620	1.6263	1.5097	0.1845
31	1.4301	1.2725	1.5945	1.5397	1.5252	1.4724	0.3220
32	1.4981	1.4506	1.6174	1.5837	1.4962	1.5292	0.1668
33	1.3009	1.5060	1.6231	1.5831	1.6454	1.5317	0.3445
34	1.4132	1.4603	1.5808	1.7111	1.7313	1.5793	0.3181
35	1.3817	1.3135	1.4953	1.4894	1.4596	1.4279	0.1818
36	1.5765	1.7014	1.4026	1.2773	1.4541	1.4824	0.4241
37	1.4936	1.4373	1.5139	1.4808	1.5293	1.4910	0.0920
38	1.5729	1.6738	1.5048	1.5651	1.7473	1.6128	0.2425
39	1.8089	1.5513	1.8250	1.4389	1.6558	1.6560	0.3861
40	1.6236	1.5393	1.6738	1.8698	1.5036	1.6420	0.3662
41	1.4120	1.7931	1.7345	1.6391	1.7791	1.6716	0.3811
42	1.7372	1.5663	1.4910	1.7809	1.5504	1.6252	0.2899
43	1.5971	1.7394	1.6832	1.6677	1.7974	1.6970	0.2003
44	1.4295	1.6536	1.9134	1.7272	1.4370	1.6321	0.4839
45	1.6217	1.8220	1.7915	1.6744	1.9404	1.7700	0.3187

for on-line monitoring of a stable process because they greatly increase the occurrence of false alarms.

In examining control chart data, it is sometimes helpful to construct a run chart of the individual observations in each sample. This chart is sometimes called a **tier chart** or **tolerance diagram**. This may reveal some pattern in the data, or it may show that a particular



■ FIGURE 6.4 Continuation of the $\bar{x}$ and R charts in Example 6.1.

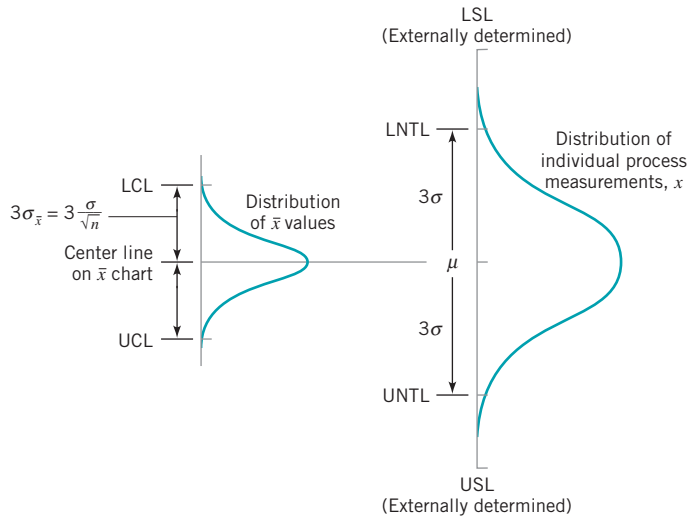


■ **FIGURE 6.5** Tier chart constructed using the Minitab box plot procedure for the flow width data.

value of $\bar{x}$ or R was produced by one or two unusual observations in the sample. A box plot is usually a very simple way to construct the tier diagram.

A tier chart of the flow width data observations is shown in Fig. 6.5. This chart does not indicate that the out-of-control signals were generated by unusual individual observations, but instead, they probably resulted from a shift in the mean around the time that sample 38 was taken. The average of the averages of samples 38 through 45 is 1.6633 microns. The specification limits of 1.50 ± 0.50 microns are plotted in Fig. 6.5, along with a sketch of the normal distribution that represents process output when the process mean equals the in-control value 1.5056 microns. A sketch of the normal distribution representing process output at the new apparent mean diameter of 1.6633 microns is also shown in Fig. 6.5. It is obvious that a much higher percentage of nonconforming wafers will be produced at this new mean flow rate. Since the process is out of control, a search for the cause of this shift in the mean must be conducted. The out-of-control-action plan (**OCAP**) for this control chart, shown in Fig. 5.6, would play a key role in these activities by directing operating personnel through a series of sequential activities to find the assignable cause. Often additional input and support from engineers, management, and the quality engineering staff is necessary to find and eliminate assignable causes.

Control Limits, Specification Limits, and Natural Tolerance Limits. A point that should be emphasized is that there is no connection or relationship between the **control limits** on the $\bar{x}$ and R charts and the **specification limits** on the process. The control limits are driven by the natural variability of the process (measured by the process standard deviation σ), that is, by the **natural tolerance limits of the process**. It is customary to define the upper and lower natural tolerance limits, say UNTL and LNTL, as 3σ above and below the process mean. The specification limits, on the other hand, are determined externally. They may be set by management, the manufacturing engineers, the customer, or by product developers/designers. One should have knowledge of inherent process variability when setting specifications, but **remember that there is no mathematical or statistical relationship between the control limits and specification limits**. The situation is summarized in Fig. 6.6. We have encountered practitioners who have plotted specification limits on the $\bar{x}$ control chart. This practice is completely incorrect and should not be done. When dealing with plots of *individual* observations (not averages), as in Fig. 6.5, it is helpful to plot the specification limits on that chart.



■ **FIGURE 6.6** Relationship of natural tolerance limits, control limits, and specification limits.

Rational Subgroups. Rational subgroups play an important role in the use of $\bar{x}$ and R control charts. Defining a rational subgroup in practice may be easier with a clear understanding of the functions of the two types of control charts. The $\bar{x}$ chart monitors the average quality level in the process. Therefore, samples should be selected in such a way that maximizes the chances for shifts in the process average to occur between samples, and thus to show up as out-of-control points on the $\bar{x}$ chart. The R chart, on the other hand, measures the variability *within* a sample. Therefore, samples should be selected so that variability within samples measures only chance or random causes. Another way of saying this is that the $\bar{x}$ chart monitors **between-sample variability** (variability in the process over time), and the R chart measures **within-sample variability** (the instantaneous process variability at a given time).

An important aspect of this is evident from carefully examining how the control limits for the $\bar{x}$ and R charts are determined from past data. The estimate of the process standard deviation σ used in constructing the control limits is calculated from the variability **within** each sample (i.e., from the individual sample ranges). Consequently, the estimate of σ reflects only **within-sample variability**. It is not correct to estimate σ based on the usual quadratic estimator—say,

$$s = \sqrt{\frac{\sum_{i=1}^m \sum_{j=1}^n (x_{ij} - \bar{\bar{x}})^2}{mn - 1}}$$

where x_{ij} is the j th observation in the i th sample, because if the sample *means* differ, then this will cause s to be too large. Consequently, σ will be overestimated. Pooling all of the preliminary data in this manner to estimate σ is not a good practice because it potentially combines *both between-sample and within-sample variability*. The control limits must be based on only **within-sample variability**. Refer to the supplemental text material for more details.

Guidelines for the Design of the Control Chart. To design the $\bar{x}$ and R charts, we must specify the sample size, control limit width, and frequency of sampling to be used. It is not possible to give an exact solution to the problem of control chart design, unless the analyst has detailed information about both the statistical characteristics of the control chart

tests and the economic factors that affect the problem. A complete solution of the problem involves knowledge of the cost of sampling, the costs of investigating and possibly correcting the process in response to out-of-control signals, and the costs associated with producing a product that does not meet specifications. Given this kind of information, an economic decision model could be constructed to allow economically optimum control chart design. In Chapter 10 (Section 10.6) we briefly discuss this approach to the problem. However, it is possible to give some general guidelines now that will aid in control chart design.

If the $\bar{x}$ chart is being used primarily to detect moderate to large process shifts—say, on the order of 2σ or larger—then relatively small samples of size $n = 4, 5$, or 6 are reasonably effective. On the other hand, if we are trying to detect small shifts, then larger sample sizes of possibly $n = 15$ to $n = 25$ are needed. When smaller samples are used, there is less risk of a process shift occurring while a sample is taken. If a shift does occur while a sample is taken, the sample average can obscure this effect. Consequently, this is an argument for using as small a sample size as is consistent with the magnitude of the process shift that one is trying to detect. An alternative to increasing the sample size is to use warning limits and other sensitizing procedures to enhance the ability of the control chart to detect small process shifts. However, as we discussed in Chapter 5, we do not favor the routine use of these sensitizing rules. If you are interested in small shifts, use the CUSUM or EWMA charts in Chapter 9.

The R chart is relatively insensitive to shifts in the process standard deviation for small samples. For example, samples of size $n = 5$ have only about a 40% chance of detecting on the first sample a shift in the process standard deviation from σ to 2σ . Larger samples would seem to be more effective, but we also know that the range method for estimating the standard deviation drops dramatically in efficiency as n increases. Consequently, for large n —say, $n > 10$ or 12 —it is probably best to use a control chart for s or s^2 instead of the R chart. Details of the construction of these charts are shown in Sections 6.3.1 and 6.3.2.

From a statistical point of view, the operating-characteristic curves of the $\bar{x}$ and R charts can be helpful in choosing the sample size. They provide a feel for the magnitude of process shift that will be detected with a stated probability for any sample size n . These operating-characteristic curves are discussed in Section 6.2.6.

The problem of choosing the sample size and the frequency of sampling is one of **allocating sampling effort**. Generally, the decision maker will have only a limited number of resources to allocate to the inspection process. The available strategies will usually be either to take small, frequent samples or to take larger samples less frequently. For example, the choice may be between samples of size 5 every half hour or samples of size 20 every two hours. It is impossible to say which strategy is best in all cases, but current industry practice favors small, frequent samples. The general belief is that if the interval between samples is too great, too much defective product will be produced before another opportunity to detect the process shift occurs. From economic considerations, if the cost associated with producing defective items is high, smaller, more frequent samples are better than larger, less frequent ones. Variable sample interval and variable sample size schemes could, of course, be used. Refer to Chapter 10.

The rate of production also influences the choice of sample size and sampling frequency. If the rate of production is high—say, 50,000 units per hour—then more frequent sampling is called for than if the production rate is extremely slow. At high rates of production, many nonconforming units of product will be produced in a very short time when process shifts occur. Furthermore, at high production rates, it is sometimes possible to obtain fairly large samples economically. For example, if we produce 50,000 units per hour, it does not take an appreciable difference in time to collect a sample of size 20 compared to a sample of size 5. If per unit inspection and testing costs are not excessive, high-speed production processes are often monitored with moderately large sample sizes.

The use of three-sigma control limits on the $\bar{x}$ and R control charts is a widespread practice. There are situations, however, when departures from this customary choice of control limits

are helpful. For example, if false alarms or type I errors (when an out-of-control signal is generated when the process is really in control) are very expensive to investigate, then it may be best to use wider control limits than three-sigma—perhaps as wide as 3.5-sigma. However, if the process is such that out-of-control signals are quickly and easily investigated with a minimum of lost time and cost, then narrower control limits, perhaps at 2.5-sigma, are appropriate.

Changing Sample Size on the $\bar{x}$ and R Charts. We have presented the development of $\bar{x}$ and R charts assuming that the sample size n is constant from sample to sample. However, there are situations in which the sample size n is not constant. One situation is that of **variable sample size on control charts**; that is, each sample may consist of a different number of observations. The $\bar{x}$ and R charts are generally not used in this case because they lead to a changing center line on the R chart, which is difficult to interpret for many users. The $\bar{x}$ and s charts in Section 6.3.2 would be preferable in this case.

Another situation is that of making a **permanent** (or **semipermanent**) **change** in the sample size because of cost or because the process has exhibited good stability and fewer resources are being allocated for process monitoring. In this case, it is easy to recompute, the new control limits directly from the old ones without collecting additional samples based on the new sample size. Let

$$\begin{aligned}\bar{R}_{\text{old}} &= \text{average range for the old sample size} \\ \bar{R}_{\text{new}} &= \text{average range for the new sample size} \\ n_{\text{old}} &= \text{old sample size} \\ n_{\text{new}} &= \text{new sample size} \\ d_2(\text{old}) &= \text{factor } d_2 \text{ for the old sample size} \\ d_2(\text{new}) &= \text{factor } d_2 \text{ for the new sample size}\end{aligned}$$

For the $\bar{x}$ chart the new control limits are

$$\begin{aligned}\text{UCL} &= \bar{\bar{x}} + A_2 \left[\frac{d_2(\text{new})}{d_2(\text{old})} \right] \bar{R}_{\text{old}} \\ \text{LCL} &= \bar{\bar{x}} - A_2 \left[\frac{d_2(\text{new})}{d_2(\text{old})} \right] \bar{R}_{\text{old}}\end{aligned}\tag{6.12}$$

where the center line $\bar{\bar{x}}$ is unchanged and the factor A_2 is selected for the *new* sample size. For the R chart, the new parameters are

$$\begin{aligned}\text{UCL} &= D_4 \left[\frac{d_2(\text{new})}{d_2(\text{old})} \right] \bar{R}_{\text{old}} \\ \text{CL} &= \bar{R}_{\text{new}} = \left[\frac{d_2(\text{new})}{d_2(\text{old})} \right] \bar{R}_{\text{old}} \\ \text{LCL} &= \max \left\{ 0, D_3 \left[\frac{d_2(\text{new})}{d_2(\text{old})} \right] \bar{R}_{\text{old}} \right\}\end{aligned}\tag{6.13}$$

where D_3 and D_4 are selected for the *new* sample size.

EXAMPLE 6.2 Changing Sample Size

To illustrate the above procedure, consider the $\bar{x}$ and R charts developed for the hard-bake process in Example 6.1. These charts were based on a sample size of five wafers. Suppose

that since the process exhibits good control, the process engineering personnel want to reduce the sample size to three wafers. Set up the new control charts.

SOLUTION

From Example 6.1, we know that

$$n_{\text{old}} = 5 \quad \bar{R}_{\text{old}} = 0.32521$$

and from Appendix Table VI we have

$$d_2(\text{old}) = 2.326 \quad d_2(\text{new}) = 1.693$$

Therefore, the new control limits on the $\bar{x}$ chart are found from equation (6.12) as

$$\begin{aligned} \text{UCL} &= \bar{\bar{x}} + A_2 \left[\frac{d_2(\text{new})}{d_2(\text{old})} \right] \bar{R}_{\text{old}} \\ &= 1.5056 + (1.023) \left[\frac{1.693}{2.326} \right] (0.32521) \\ &= 1.5056 + 0.2422 = 1.7478 \end{aligned}$$

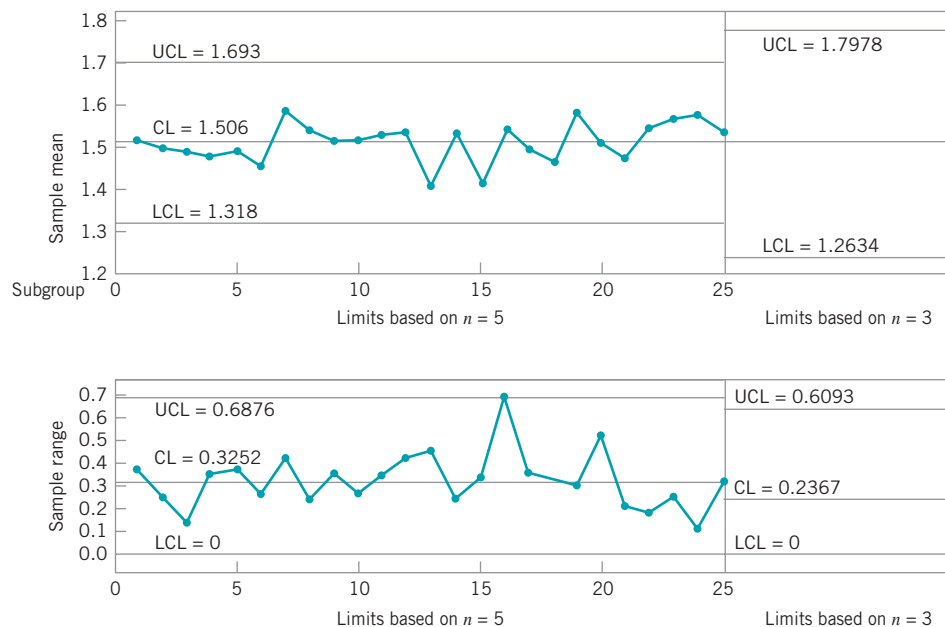
and

$$\begin{aligned} \text{LCL} &= \bar{\bar{x}} - A_2 \left[\frac{d_2(\text{new})}{d_2(\text{old})} \right] \bar{R}_{\text{old}} \\ &= 1.5056 - (1.023) \left[\frac{1.693}{2.326} \right] (0.32521) \\ &= 1.5056 - 0.2422 = 1.2634 \end{aligned}$$

For the R chart, the new parameters are given by equation (6.13):

$$\begin{aligned} \text{UCL} &= D_4 \left[\frac{d_2(\text{new})}{d_2(\text{old})} \right] \bar{R}_{\text{old}} \\ &= (2.574) \left[\frac{1.693}{2.326} \right] (0.32521) \\ &= 0.6093 \\ \text{CL} &= \bar{R}_{\text{new}} = \left[\frac{d_2(\text{new})}{d_2(\text{old})} \right] \bar{R}_{\text{old}} \\ &= \left[\frac{1.693}{2.326} \right] (0.32521) \\ &= 0.2367 \\ \text{LCL} &= \max \left\{ 0, D_3 \left[\frac{d_2(\text{new})}{d_2(\text{old})} \right] \bar{R}_{\text{old}} \right\} \\ &= 0 \end{aligned}$$

Figure 6.7 shows the new control limits. Note that the effect of reducing the sample size is to *increase* the width of the limits on the $\bar{x}$ chart (because $\sigma/\sqrt{n}$ is smaller when $n = 5$ than when $n = 3$) and to *lower* the center line and the upper control limit on the R chart (because the expected range from a sample of $n = 3$ is smaller than the expected range from a sample of $n = 5$).



■ **Figure 6.7**
Recalculated control limits for the hard-bake process in Example 6.1 to reflect changing the sample size from $n = 5$ to $n = 3$.

Probability Limits on the $\bar{x}$ and R Charts. It is customary to express the control limits on the $\bar{x}$ and R charts as a multiple of the standard deviation of the statistic plotted on the charts. If the multiple chosen is k , then the limits are referred to as k -sigma limits, the usual choice being $k = 3$. As mentioned in Chapter 5, however, it is also possible to define the control limits by specifying the type I error level for the test. These are called **probability limits for control charts** and are used extensively in the United Kingdom and some parts of Western Europe.

It is easy to choose probability limits for the $\bar{x}$ chart. Since $\bar{x}$ is approximately normally distributed, we may obtain a desired type I error of α by choosing the multiple of sigma for the control limit as $k = Z_{\alpha/2}$, where $Z_{\alpha/2}$ is the upper $\alpha/2$ percentage point of the standard normal distribution. Note that the usual **three-sigma control limits** imply that the type I error probability is $\alpha = 0.0027$. If we choose $\alpha = 0.002$, for example, as most writers who recommend probability limits do, then $Z_{\alpha/2} = Z_{0.001} = 3.09$. Consequently, there is very little difference between such control limits and three-sigma control limits.

We may also construct R charts using probability limits. If $\alpha = 0.002$, the 0.001 and 0.999 percentage points of the distribution of the relative range $W = R/\sigma$ are required. These points obviously depend on the subgroup size n . Denoting these points by $W_{0.001}(n)$ and $W_{0.999}(n)$, and estimating σ by $\bar{R}/d_2$, we would have the 0.001 and 0.999 limits for R as $W_{0.001}(n)(\bar{R}/d_2)$ and $W_{0.999}(n)(\bar{R}/d_2)$. If we let $D_{0.001} = W_{0.001}(n)/d_2$ and $D_{0.999} = W_{0.999}(n)/d_2$, then the probability limits for the R chart are

$$\text{UCL} = D_{0.999}\bar{R}$$

$$\text{LCL} = D_{0.001}\bar{R}$$

Tables of pairs of values $(D_{0.001}, D_{0.999})$, $(D_{0.005}, D_{0.995})$, and $(D_{0.025}, D_{0.975})$ for $2 \leq n \leq 10$ are in Grant and Leavenworth (1980). These control limits will not differ substantially from the customary three-sigma limits. However, for sample sizes $3 \leq n \leq 6$, they will produce a positive lower control limit for the R chart whereas the conventional three-sigma limits do not.

6.2.3 Charts Based on Standard Values

When it is possible to specify standard values for the process mean and standard deviation, we may use these standards to establish the control charts for $\bar{x}$ and R without analysis of past data. Suppose that the standards given are μ and σ . Then the parameters of the $\bar{x}$ chart are

$$\begin{aligned}\text{UCL} &= \mu + 3\frac{\sigma}{\sqrt{n}} \\ \text{Center line} &= \mu \\ \text{LCL} &= \mu - 3\frac{\sigma}{\sqrt{n}}\end{aligned}\tag{6.14}$$

The quantity $3/\sqrt{n} = A$, say, is a constant that depends on n , which has been tabulated in Appendix Table VI. Consequently, we could write the parameters of the $\bar{x}$ chart as

$$\begin{aligned}\text{UCL} &= \mu + A\sigma \\ \text{Center line} &= \mu \\ \text{LCL} &= \mu - A\sigma\end{aligned}\tag{6.15}$$

To construct the R chart with a standard value of σ , recall that $\sigma = R/d_2$, where d_2 is the mean of the distribution of the relative range. Furthermore, the standard deviation of R is $\sigma_R = d_3\sigma$, where d_3 is the standard deviation of the distribution of the relative range. Therefore, the parameters of the control chart are

$$\begin{aligned} \text{UCL} &= d_2\sigma + 3d_3\sigma \\ \text{Center line} &= d_2\sigma \\ \text{LCL} &= d_2\sigma - 3d_3\sigma \end{aligned} \tag{6.16}$$

It is customary to define the constants

$$\begin{aligned} D_1 &= d_2 - 3d_3 \\ D_2 &= d_2 + 3d_3 \end{aligned}$$

These constants are tabulated in Appendix Table VI. Thus, the parameters of the R chart with standard σ given are

$$\begin{aligned} \text{UCL} &= D_2\sigma \\ \text{Center line} &= d_2\sigma \\ \text{LCL} &= D_1\sigma \end{aligned} \tag{6.17}$$

One must exercise considerable care when standard values of μ and σ are given. It may be that these standards are not really applicable to the process, and as a result, the $\bar{x}$ and R charts will produce many out-of-control signals relative to the specified standards. If the process is really in control at some *other* mean and standard deviation, then the analyst may spend considerable effort looking for assignable causes that do not exist. Standard values of σ seem to give more trouble than standard values of μ . In processes where the mean of the quality characteristic is controlled by adjustments to the machine, standard or target values of μ are sometimes helpful in achieving management goals with respect to process performance.

6.2.4 Interpretation of $\bar{x}$ and R Charts

We have noted previously that a control chart can indicate an out-of-control condition even though no single point plots outside the control limits, if the pattern of the plotted points exhibits nonrandom or systematic behavior. In many cases, the pattern of the plotted points will provide useful diagnostic information on the process, and this information can be used to make process modifications that reduce variability (the goal of statistical process control). Furthermore, these patterns occur fairly often in phase I (retrospective study of past data), and their elimination is frequently crucial in bringing a process into control.

In this section, we briefly discuss **interpretation of control charts**, focusing on some of the more common patterns that appear on $\bar{x}$ and R charts and some of the process characteristics

that may produce the patterns. To effectively interpret $\bar{x}$ and R charts, the analyst must be familiar with both the statistical principles underlying the control chart and the process itself. Additional information on the interpretation of patterns on control charts is in the Western Electric *Statistical Quality Control Handbook* (1956, pp. 149–183).

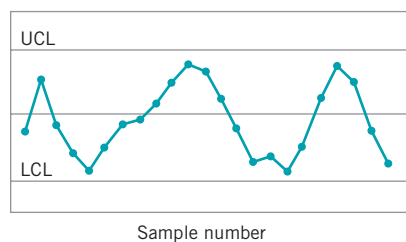
In interpreting patterns on the $\bar{x}$ chart, we must first determine whether or not the R chart is in control. Some assignable causes show up on *both* the $\bar{x}$ and R charts. If both the $\bar{x}$ and R charts exhibit a nonrandom pattern, the best strategy is to eliminate the R chart assignable causes first. In many cases, this will automatically eliminate the nonrandom pattern on the $\bar{x}$ chart. Never attempt to interpret the $\bar{x}$ chart when the R chart indicates an out-of-control condition.

Cyclic patterns occasionally appear on the control chart. A typical example is shown in Fig. 6.8. Such a pattern on the $\bar{x}$ chart may result from systematic environmental changes such as temperature, operator fatigue, regular rotation of operators and/or machines, or fluctuation in voltage or pressure or some other variable in the production equipment. R charts will sometimes reveal cycles because of maintenance schedules, operator fatigue, or tool wear resulting in excessive variability. In one study in which this author was involved, systematic variability in the fill volume of a metal container was caused by the on–off cycle of a compressor in the filling machine.

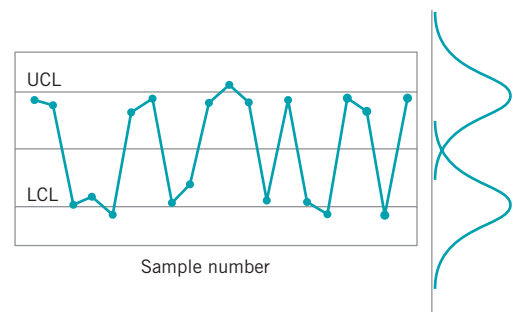
A **mixture** is indicated when the plotted points tend to fall near or slightly outside the control limits, with relatively few points near the center line, as shown in Fig. 6.9. A mixture pattern is generated by two (or more) overlapping distributions generating the process output. The probability distributions that could be associated with the mixture pattern in Fig. 6.9 are shown on the right-hand side of that figure. The severity of the mixture pattern depends on the extent to which the distributions overlap. Sometimes mixtures result from “overcontrol,” where the operators make process adjustments too often, responding to random variation in the output rather than systematic causes. A mixture pattern can also occur when output product from several sources (such as parallel machines) is fed into a common stream which is then sampled for process monitoring purposes.

A **shift in process level** is illustrated in Fig. 6.10. These shifts may result from the introduction of new workers; changes in methods, raw materials, or machines; a change in the inspection method or standards; or a change in either the skill, attentiveness, or motivation of the operators. Sometimes an improvement in process performance is noted following introduction of a control chart program, simply because of motivational factors influencing the workers.

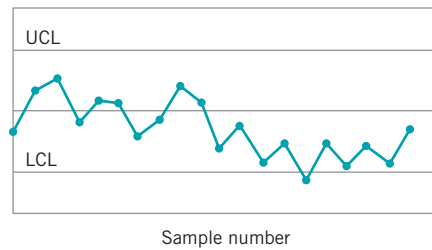
A **trend**, or continuous movement in one direction, is shown on the control chart in Fig. 6.11. Trends are usually due to a gradual wearing out or deterioration of a tool or some other critical process component. In chemical processes they often occur because of settling



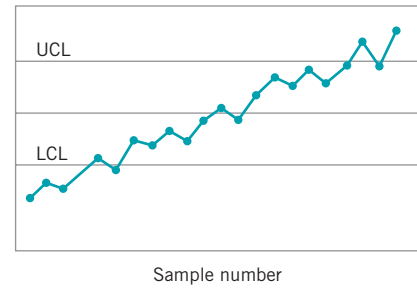
■ **FIGURE 6.8** Cycles on a control chart.



■ **FIGURE 6.9** A mixture pattern.



■ **FIGURE 6.10** A shift in process level.

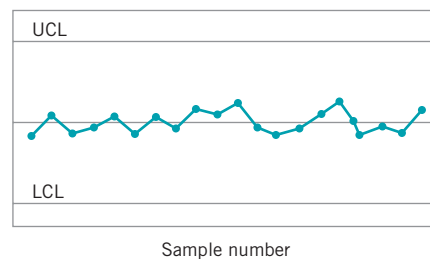


■ **FIGURE 6.11** A trend in process level.

or separation of the components of a mixture. They can also result from human causes, such as operator fatigue or the presence of supervision. Finally, trends can result from seasonal influences, such as temperature. When trends are due to tool wear or other systematic causes of deterioration, this may be directly incorporated into the control chart model. A device useful for monitoring and analyzing processes with trends is the **regression control chart** [see Mandel (1969)]. The **modified control chart**, discussed in Chapter 9, is also used when the process exhibits tool wear.

Stratification, or a tendency for the points to cluster artificially around the center line, is illustrated in Fig. 6.12. We note that there is a marked lack of natural variability in the observed pattern. One potential cause of stratification is incorrect calculation of control limits. This pattern may also result when the sampling process collects one or more units from several different underlying distributions within each subgroup. For example, suppose that a sample of size 5 is obtained by taking one observation from each of five parallel processes. If the largest and smallest units in each sample are relatively far apart because they come from two different distributions, then R will be incorrectly inflated, causing the limits on the $\bar{x}$ chart to be too wide. In this case R incorrectly measures the variability *between* the different underlying distributions, in addition to the chance cause variation that it is intended to measure.

In interpreting patterns on the $\bar{x}$ and R charts, one should consider the two charts jointly. If the underlying distribution is normal, then the random variables $\bar{x}$ and R computed from the same sample are statistically independent. Therefore, $\bar{x}$ and R should behave independently on the control chart. If there is correlation between the $\bar{x}$ and R values—that is, if the points on the two charts “follow” each other—then this indicates that the underlying distribution is skewed. If specifications have been determined assuming normality, then those analyses may be in error.



■ **FIGURE 6.12** A stratification pattern.

6.2.5 The Effect of Non-normality on $\bar{x}$ and R Charts

There often is an assumption that links **normality** and **control charts** in the development of the performance properties of $\bar{x}$ and R control charts; that is, that the underlying distribution of the quality characteristic is **normal**. In many situations, we may have reason to doubt the validity of this assumption. For example, we may know that the underlying distribution is not normal, because we have collected extensive data that indicate the normality assumption is inappropriate. Now if we know the form of the underlying distribution, it is possible to derive the sampling distributions of $\bar{x}$ and R (or some other measure of process variability) and to obtain exact probability limits for the control charts. This approach could be difficult in some cases, and most analysts would probably prefer to use the standard approach based on the normality assumption if they believed that the effect of departure from this assumption was not serious. However, we may know nothing about the form of the underlying distribution, and then our only choice may be to use the normal theory results. Obviously, in either case, we would be interested in knowing the effect of departures from normality on the usual control charts for $\bar{x}$ and R .

Several authors have investigated the effect of departures from normality on control charts. Burr (1967) notes that the usual normal theory control limit constants are very robust to the normality assumption and can be employed unless the population is extremely non-normal. Schilling and Nelson (1976), Chan, Hapuarachchi, and Macpherson (1988), and Yourstone and Zimmer (1992) have also studied the effect of non-normality on the control limits of the $\bar{x}$ chart. Schilling and Nelson investigated the uniform, right triangular, gamma (with $\lambda = 1$ and $r = \frac{1}{2}, 1, 2, 3$, and 4), and two bimodal distributions formed as mixtures of two normal distributions. Their study indicates that, in most cases, samples of size 4 or 5 are sufficient to ensure reasonable robustness to the normality assumption.

The worst cases observed were for small values of r in the gamma distribution [$r = \frac{1}{2}$ and $r = 1$ (the exponential distribution)]. For example, they report the actual α -risk to be 0.014 or less if $n \geq 4$ for the gamma distribution with $r = \frac{1}{2}$, as opposed to a theoretical value of 0.0027 for the normal distribution.

While the use of three-sigma control limits on the $\bar{x}$ chart will produce an α -risk of 0.0027 if the underlying distribution is normal, the same is not true for the R chart. The sampling distribution of R is not symmetric, even when sampling from the normal distribution, and the long tail of the distribution is on the high or positive side. Thus, symmetric three-sigma limits are only an approximation, and the α -risk on such an R chart is *not* 0.0027. (In fact, for $n = 4$, it is $\alpha = 0.00461$.) Furthermore, the R chart is more sensitive to departures from normality than the $\bar{x}$ chart.

Once again, it is important to remember the role of theory and assumptions such as normality and independence. These are fundamental to the study of the performance of the $\bar{x}$ control chart which is very useful for assessing its suitability for phase II, but plays a much less important role in phase I. In fact, these considerations are not a primary concern in phase I.

6.2.6 The Operating-Characteristic Function

The ability of the $\bar{x}$ and R charts to detect shifts in process quality is described by their operating-characteristic (OC) curves. In this section, we present these OC curves for charts used for phase II monitoring of a process.

Consider the **operating-characteristic (OC) curve for an $\bar{x}$ control chart**. The standard deviation σ is assumed known and constant. If the mean shifts from the in-control value—say, μ_0 —to another value $\mu_1 = \mu_0 + k\sigma$, the probability of *not* detecting this shift on the first subsequent sample or the β -risk is

$$\beta = P\{LCL \leq \bar{x} \leq UCL | \mu = \mu_1 = \mu_0 + k\sigma\} \quad (6.18)$$

Since $\bar{x} \sim N(\mu, \sigma^2/n)$, and the upper and lower control limits are $\text{UCL} = \mu_0 + L\sigma/\sqrt{n}$ and $\text{LCL} = \mu_0 - L\sigma/\sqrt{n}$, we may write equation (6.18) as

$$\begin{aligned}\beta &= \Phi\left[\frac{\text{UCL} - (\mu_0 + k\sigma)}{\sigma/\sqrt{n}}\right] - \Phi\left[\frac{\text{LCL} - (\mu_0 + k\sigma)}{\sigma/\sqrt{n}}\right] \\ &= \Phi\left[\frac{\mu_0 + L\sigma/\sqrt{n} - (\mu_0 + k\sigma)}{\sigma/\sqrt{n}}\right] - \Phi\left[\frac{\mu_0 - L\sigma/\sqrt{n} - (\mu_0 + k\sigma)}{\sigma/\sqrt{n}}\right]\end{aligned}$$

where Φ denotes the standard normal cumulative distribution function. This reduces to

$$\beta = \Phi(L - k\sqrt{n}) - \Phi(-L - k\sqrt{n}) \quad (6.19)$$

To illustrate the use of equation (6.19), suppose that we are using an $\bar{x}$ chart with $L = 3$ (the usual three-sigma limits), the sample size $n = 5$, and we wish to determine the probability of detecting a shift to $\mu_1 = \mu_0 + 2\sigma$ on the first sample following the shift. Then, since $L = 3$, $k = 2$, and $n = 5$, we have

$$\begin{aligned}\beta &= \Phi[3 - 2\sqrt{5}] - \Phi[-3 - 2\sqrt{5}] \\ &= \Phi(-1.47) - \Phi(-7.37) \\ &\cong 0.0708\end{aligned}$$

This is the β -risk, or the probability of not detecting such a shift. The probability that such a shift *will* be detected on the first subsequent sample is $1 - \beta = 1 - 0.0708 = 0.9292$.

To construct the OC curve for the $\bar{x}$ chart, plot the β -risk against the magnitude of the shift we wish to detect expressed in standard deviation units for various sample sizes n . These probabilities may be evaluated directly from equation (6.19). This OC curve is shown in Fig. 6.13 for the case of three-sigma limits ($L = 3$).

Figure 6.13 indicates that for the typical sample sizes 4, 5, and 6, the $\bar{x}$ chart is not particularly effective in detecting a small shift—say, those on the order of 1.5σ or less—on the first sample following the shift. For example, if the shift is 1.0σ and $n = 5$, then from Fig. 6.13, we have $\beta = 0.75$, approximately. Thus, the probability that the shift will be detected on the first sample is only $1 - \beta = 0.25$. However, the probability that the shift is detected on

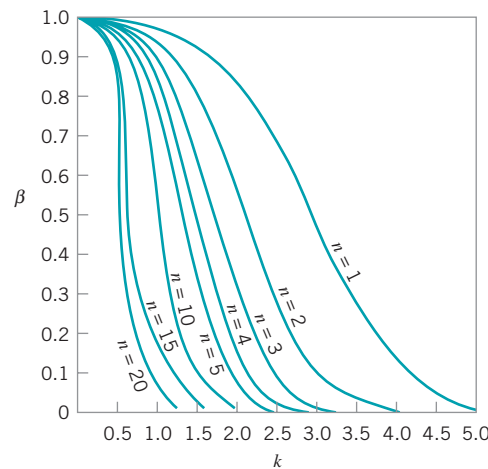


FIGURE 6.13 Operating-characteristic curves for the $\bar{x}$ chart with three-sigma limits. $\beta = P$ (not detecting a shift of $k\sigma$ in the mean on the first sample following the shift).

the second sample is $\beta(1 - \beta) = 0.75(0.25) = 0.19$, whereas the probability that it is detected on the third sample is $\beta^2(1 - \beta) = (0.75^2)0.25 = 0.14$. Thus, the probability that the shift will be detected on the r th subsequent sample is simply $1 - \beta$ times the probability of not detecting the shift on each of the initial $r - 1$ samples, or

$$\beta^{r-1}(1 - \beta)$$

In general, the expected number of samples taken before the shift is detected is simply the **average run length**, or

$$\text{ARL} = \sum_{r=1}^{\infty} r\beta^{r-1}(1 - \beta) = \frac{1}{1 - \beta}$$

Therefore, in our example, we have

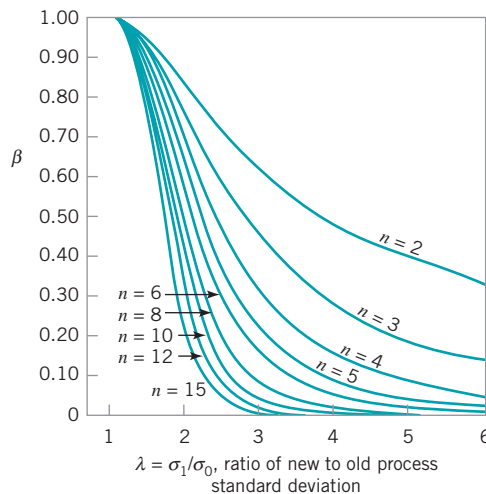
$$\text{ARL} = \frac{1}{1 - \beta} = \frac{1}{0.25} = 4$$

In other words, the expected number of samples taken to detect a shift of 1.0σ with $n = 5$ is four.

The above discussion provides a supportive argument for the use of small sample sizes on the $\bar{x}$ chart. Even though small sample sizes often result in a relatively large β -risk, because samples are collected and tested periodically, there is a very good chance that the shift will be detected reasonably quickly, although perhaps not on the first sample following the shift.

To construct the OC curve for the R chart, the distribution of the relative range $W = R/\sigma$ is employed. Suppose that the in-control value of the standard deviation is σ_0 . Then the OC curve plots the probability of not detecting a shift to a new value of σ —say, $\sigma_1 > \sigma_0$ —on the first sample following the shift. Figure 6.14 presents the OC curve, in which β is plotted against $\lambda = \sigma_1/\sigma_0$ (the ratio of new to old process standard deviation) for various values of n .

From examining Fig. 6.14, we observe that the R chart is not very effective in detecting process shifts for small sample sizes. For example, if the process standard deviation doubles (i.e., $\lambda = \sigma_1/\sigma_0 = 2$), which is a fairly large shift, then samples of size 5 have only about a 40% chance of detecting this shift on each subsequent sample. Most quality engineers believe that the R chart is insensitive to small or moderate shifts for the usual subgroup sizes of $n = 4, 5$, or 6 . If $n > 10$ or 12 , the s chart discussed in Section 6.3.1 should generally be used instead of the R chart.



■ **FIGURE 6.14** Operating-characteristic curves for the R chart with three-sigma limits. (Adapted from A. J. Duncan, "Operating Characteristics of R Charts," *Industrial Quality Control*, vol. 7, no. 5, pp. 40–41, 1951, with permission of the American Society for Quality Control.)

The OC curves in Figs. 6.13 and 6.14 assume that the $\bar{x}$ and R charts are used for on-line process monitoring; that is, phase II process monitoring. It is occasionally useful to study the statistical performance of a chart used to analyze past data (phase I). This can give some indication of how the number of preliminary subgroups used to establish the control chart affects the chart's ability to detect out-of-control conditions that existed when the data were collected. It is from such analytical studies, as well as practical experience, that the recommendation to use at least 20 to 25 preliminary subgroups in establishing $\bar{x}$ and R charts has evolved.

6.2.7 The Average Run Length for the $\bar{x}$ Chart

For any **Shewhart control chart**, we have noted previously that the ARL can be expressed as

$$ARL = \frac{1}{P(\text{one point plots out of control})}$$

or

$$ARL_0 = \frac{1}{\alpha} \quad (6.20)$$

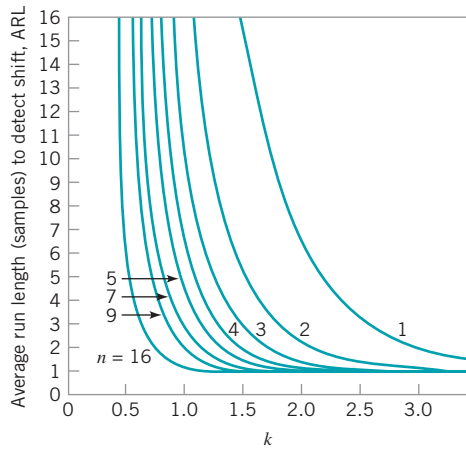
for the in-control ARL and

$$ARL_1 = \frac{1}{1 - \beta} \quad (6.21)$$

for the out-of-control ARL. These results are actually rather intuitive. If the observations plotted on the control chart are independent, then the number of points that must be plotted until the first point exceeds a control limit is a geometric random variable with parameter p (see Chapter 3). The mean of this geometric distribution is simply $1/p$, the average run length.

Since it is relatively easy to develop a general expression for β for the $\bar{x}$ control chart to detect a shift in the mean of $k\sigma$ [see equation (6.19)], then it is not difficult to construct a set of ARL curves for the $\bar{x}$ chart. Figure 6.15 presents these ARL curves for sample sizes of $n = 1, 2, 3, 4, 5, 7, 9$, and 16 for the $\bar{x}$ control chart, where the ARL is in terms of the expected number of *samples* taken in order to detect the shift. To illustrate the use of Fig. 6.15, note that if we wish to detect a shift of 1.5σ using a sample size of $n = 3$, then the average number of samples required will be $ARL_1 = 3$. Note also that we could reduce the ARL_1 to approximately 1 if we increased the sample size to $n = 16$.

Recall our discussion in Chapter 5 (Section 5.3.3) indicating that ARLs are subject to some criticism as performance measures for control charts. We noted there (and above as well) that the distribution of run length for a Shewhart control chart is geometric and that this can be a very skewed distribution, so the average (that is, the ARL) may not be the best measure of a typical run length. Another issue concerning ARL relates to the fact that the computations for a specific control chart are usually based on **estimates** of the process



■ **FIGURE 6.15** Average run length (samples) for the $\bar{x}$ chart with three-sigma limits, where the process mean shifts by $k\sigma$. (Adapted from *Modern Methods for Quality Control and Improvement*, by H. M. Wadsworth, K. S. Stephens, and A. B. Godfrey, 2nd edition, John Wiley & Sons, 2002.)

parameters. This results in inflation of both ARL_0 and ARL_1 . For example, suppose that the center line of the chart is estimated perfectly, but the process standard deviation is overestimated by 10%. This would result in $ARL_0 = 517$, considerably longer than the nominal or “theoretical” value of 370. With a normally distributed process, we are just as likely to underestimate the process standard deviation by 10%, and this would result in $ARL_0 = 268$, a value considerably shorter than 370. The average is $(268 + 517)/2 = 392.5$, suggesting that errors in estimating the process standard deviation result in ARLs that are overestimated. There is a very comprehensive literature review and discussion of this topic in the paper by Jensen et al. (2006).

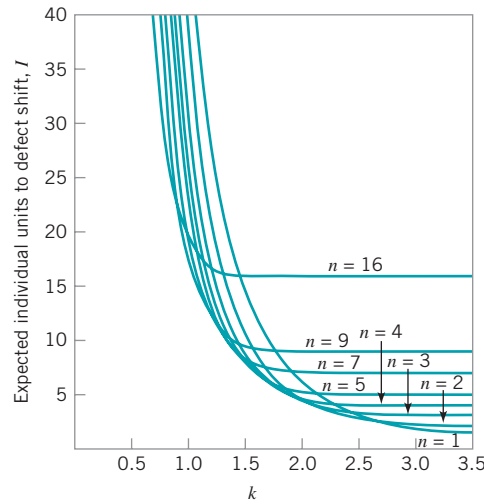
Two other performance measures based on ARL are sometimes of interest. The average time to signal is the number of time periods that occur until a signal is generated on the control chart. If samples are taken at equally spaced intervals of time h , then the **average time to signal** or the **ATS** is

$$ATS = ARL \, h \tag{6.22}$$

It may also be useful to express the ARL in terms of the expected number of individual *units* sampled—say, I —rather than the number of samples taken to detect a shift. If the sample size is n , the relationship between I and ARL is

$$I = n \, ARL \tag{6.23}$$

Figure 6.16 presents a set of curves that plot the expected number of individual units I that must be sampled for the $\bar{x}$ chart to detect a shift of $k\sigma$. To illustrate the use of the curve, note that to detect a shift of 1.5σ , an $\bar{x}$ chart with $n = 16$ will require that approximately 16 units be sampled, whereas if the sample size is $n = 3$, only about 9 units will be required, on the average.



■ **FIGURE 6.16** Average run length (individual units) for the $\bar{x}$ chart with three-sigma limits, where the process mean shifts by $k\sigma$. (Adapted from *Modern Methods for Quality Control and Improvement*, by H. M. Wadsworth, K. S. Stephens, and A. B. Godfrey, 2nd edition, John Wiley & Sons, 2002.)

6.3 Control Charts for $\bar{x}$ and s

Although $\bar{x}$ and R charts are widely used, it is occasionally desirable to estimate the process standard deviation directly instead of indirectly through the use of the range R . This leads to control charts for $\bar{x}$ and s , where s is the sample standard deviation.¹ Generally, $\bar{x}$ and s charts are preferable to their more familiar counterparts, $\bar{x}$ and R charts, when either

1. The sample size n is moderately large—say, $n > 10$ or 12. (Recall that the range method for estimating σ loses statistical efficiency for moderate to large samples.)
2. The sample size n is variable.

In this section, we illustrate the construction and operation of $\bar{x}$ and s control charts. We also show how to deal with variable sample size and discuss an alternative to the s chart.

6.3.1 Construction and Operation of $\bar{x}$ and s Charts

Setting up and operating control charts for $\bar{x}$ and s requires about the same sequence of steps as those for $\bar{x}$ and R charts, except that for each sample we must calculate the sample average $\bar{x}$ and the sample standard deviation s . Table 6.3 presents the inside diameter measurements of forged automobile engine piston rings. Each sample or subgroup consists of five piston rings. We have calculated the sample average and sample standard deviation for each of the 25 samples. We will use these data to illustrate the construction and operation of $\bar{x}$ and s charts.

If σ^2 is the unknown variance of a probability distribution, then an unbiased estimator of σ^2 is the sample variance

$$s^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n - 1}$$

¹Some authors refer to the s chart as the σ chart.

■ TABLE 6.3

Inside Diameter Measurements (mm) for Automobile Engine Piston Rings

Sample Number	Observations					$\bar{x}_i$	s_i
1	74.030	74.002	74.019	73.992	74.008	74.010	0.0148
2	73.995	73.992	74.001	74.011	74.004	74.001	0.0075
3	73.988	74.024	74.021	74.005	74.002	74.008	0.0147
4	74.002	73.996	73.993	74.015	74.009	74.003	0.0091
5	73.992	74.007	74.015	73.989	74.014	74.003	0.0122
6	74.009	73.994	73.997	73.985	73.993	73.996	0.0087
7	73.995	74.006	73.994	74.000	74.005	74.000	0.0055
8	73.985	74.003	73.993	74.015	73.988	73.997	0.0123
9	74.008	73.995	74.009	74.005	74.004	74.004	0.0055
10	73.998	74.000	73.990	74.007	73.995	73.998	0.0063
11	73.994	73.998	73.994	73.995	73.990	73.994	0.0029
12	74.004	74.000	74.007	74.000	73.996	74.001	0.0042
13	73.983	74.002	73.998	73.997	74.012	73.998	0.0105
14	74.006	73.967	73.994	74.000	73.984	73.990	0.0153
15	74.012	74.014	73.998	73.999	74.007	74.006	0.0073
16	74.000	73.984	74.005	73.998	73.996	73.997	0.0078
17	73.994	74.012	73.986	74.005	74.007	74.001	0.0106
18	74.006	74.010	74.018	74.003	74.000	74.007	0.0070
19	73.984	74.002	74.003	74.005	73.997	73.998	0.0085
20	74.000	74.010	74.013	74.020	74.003	74.009	0.0080
21	73.982	74.001	74.015	74.005	73.996	74.000	0.0122
22	74.004	73.999	73.990	74.006	74.009	74.002	0.0074
23	74.010	73.989	73.990	74.009	74.014	74.002	0.0119
24	74.015	74.008	73.993	74.000	74.010	74.005	0.0087
25	73.982	73.984	73.995	74.017	74.013	73.998	0.0162
						$\Sigma = 1850.028$	0.2351
						$\bar{\bar{x}} = 74.001$	$\bar{s} = 0.0094$

However, the sample standard deviation s is *not* an unbiased estimator of σ . In Chapter 4 (Section 4.2) we observed that if the underlying distribution is normal, then s actually estimates $c_4\sigma$, where c_4 is a constant that depends on the sample size n . Furthermore, the standard deviation of s is $\sigma\sqrt{1 - c_4^2}$. This information can be used to establish control charts on $\bar{x}$ and s .

Consider the case where a standard value is given for σ . Since $E(s) = c_4\sigma$, the center line for the chart is $c_4\sigma$. The three-sigma control limits for s are then

$$\text{UCL} = c_4\sigma + 3\sigma\sqrt{1 - c_4^2}$$

$$\text{LCL} = c_4\sigma - 3\sigma\sqrt{1 - c_4^2}$$

It is customary to define the two constants

$$B_5 = c_4 - 3\sqrt{1 - c_4^2} \quad \text{and} \quad B_6 = c_4 + 3\sqrt{1 - c_4^2} \quad (6.24)$$

Consequently, the parameters of the s chart with a standard value for σ given become

$$\begin{aligned} \text{UCL} &= B_6\sigma \\ \text{Center line} &= c_4\sigma \\ \text{LCL} &= B_5\sigma \end{aligned} \quad (6.25)$$

Values of B_5 and B_6 are tabulated for various sample sizes in Appendix Table VI. The parameters of the corresponding $\bar{x}$ chart are given in equation (6.15), Section 6.2.3.

If no standard is given for σ , then it must be estimated by analyzing past data. Suppose that m preliminary samples are available, each of size n , and let s_i be the standard deviation of the i th sample. The average of the m standard deviations is

$$\bar{s} = \frac{1}{m} \sum_{i=1}^m s_i$$

The statistic $\bar{s}/c_4$ is an unbiased estimator of σ . Therefore, the parameters of the s chart would be

$$\begin{aligned} \text{UCL} &= \bar{s} + 3 \frac{\bar{s}}{c_4} \sqrt{1 - c_4^2} \\ \text{Center line} &= \bar{s} \\ \text{LCL} &= \bar{s} - 3 \frac{\bar{s}}{c_4} \sqrt{1 - c_4^2} \end{aligned}$$

We usually define the constants

$$B_3 = 1 - \frac{3}{c_4} \sqrt{1 - c_4^2} \quad \text{and} \quad B_4 = 1 + \frac{3}{c_4} \sqrt{1 - c_4^2} \quad (6.26)$$

Consequently, we may write the parameters of the s chart as

$$\begin{aligned} \text{UCL} &= B_4\bar{s} \\ \text{Center line} &= \bar{s} \\ \text{LCL} &= B_3\bar{s} \end{aligned} \quad (6.27)$$

Note that $B_4 = B_6/c_4$ and $B_3 = B_5/c_4$.

When $\bar{s}/c_4$ is used to estimate σ , we may define the control limits on the corresponding $\bar{x}$ chart as

$$\begin{aligned} \text{UCL} &= \bar{\bar{x}} + \frac{3\bar{s}}{c_4\sqrt{n}} \\ \text{Center line} &= \bar{\bar{x}} \\ \text{LCL} &= \bar{\bar{x}} - \frac{3\bar{s}}{c_4\sqrt{n}} \end{aligned}$$

Let the constant $A_3 = 3/(c_4\sqrt{n})$. Then the $\bar{x}$ chart parameters become

$$\begin{aligned} \text{UCL} &= \bar{\bar{x}} + A_3\bar{s} \\ \text{Center line} &= \bar{\bar{x}} \\ \text{LCL} &= \bar{\bar{x}} - A_3\bar{s} \end{aligned} \quad (6.28)$$

The constants B_3 , B_4 , and A_3 for construction of $\bar{x}$ and s charts from past data are listed in Appendix Table VI for various sample sizes.

Note that we have assumed that the sample standard deviation is defined as

$$s = \sqrt{\frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n-1}} \quad (6.29)$$

Some authors define s with n in the denominator of equation (6.29) instead of $n-1$. When this is the case, the definitions of the constants c_4 , B_3 , B_4 , and A_3 are altered. The corresponding constants based on the use of n in calculating s are called c_2 , B_1 , B_2 , and A_1 , respectively. See Bowker and Lieberman (1972) for their definitions.

Traditionally, quality engineers have preferred the R chart to the s chart because of the simplicity of calculating R from each sample. The availability of hand-held calculators with automatic calculation of s and computers at workstations to implement control charts on site have eliminated any computational difficulty.

EXAMPLE 6.3 $\bar{x}$ and s Charts for the Piston Ring Data

Construct and interpret $\bar{x}$ and s charts using the piston ring inside diameter measurements in Table 6.3.

SOLUTION

The grand average and the average standard deviation are

$$\bar{\bar{x}} = \frac{1}{25} \sum_{i=1}^{25} \bar{x}_i = \frac{1}{25} (1850.028) = 74.001$$

and

$$\bar{s} = \frac{1}{25} \sum_{i=1}^{25} s_i = \frac{1}{25} (0.2351) = 0.0094$$

respectively. Consequently, the parameters for the $\bar{x}$ chart are

$$\text{UCL} = \bar{\bar{x}} + A_3\bar{s} = 74.001 + (1.427)(0.0094) = 74.014$$

$$\text{CL} = \bar{\bar{x}} = 74.001$$

$$\text{LCL} = \bar{\bar{x}} - A_3\bar{s} = 74.001 - (1.427)(0.0094) = 73.988$$

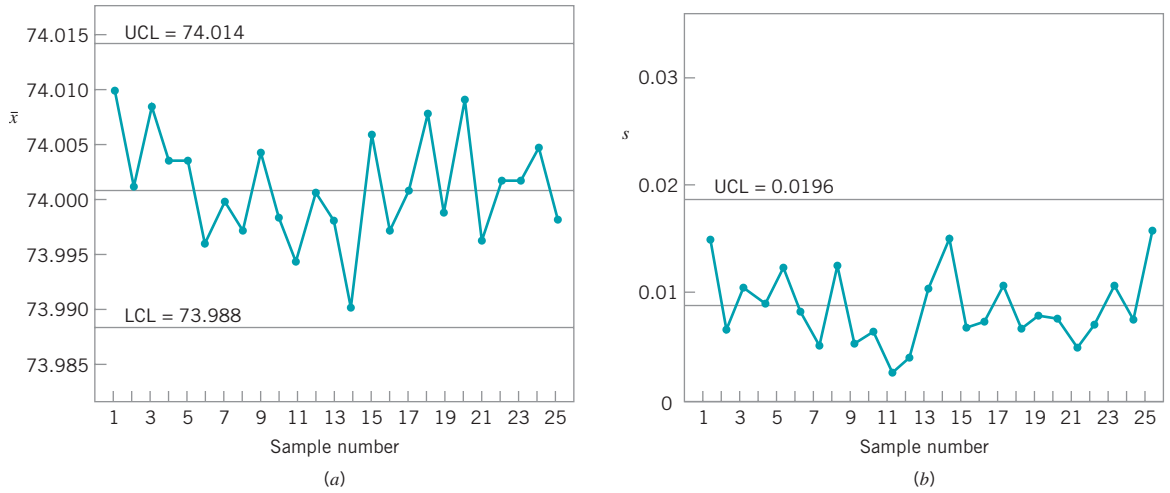
and for the s chart

$$\text{UCL} = B_4\bar{s} = (2.089)(0.0094) = 0.0196$$

$$\text{CL} = \bar{s} = 0.0094$$

$$\text{LCL} = B_3\bar{s} = (0)(0.0094) = 0$$

The control charts are shown in Fig. 6.17. There is no indication that the process is out of control, so those limits could be adopted for phase II monitoring of the process.



■ **FIGURE 6.17** The $\bar{x}$ and s control charts for Example 6.3. (a) The $\bar{x}$ chart with control limits based on $\bar{s}$. (b) The s control chart.

Estimation of σ . We can estimate the process standard deviation using the fact that s/c_4 is an unbiased estimate of σ . Therefore, since $c_4 = 0.9400$ for samples of size 5, our estimate of the process standard deviation is

$$\hat{\sigma} = \frac{\bar{s}}{c_4} = \frac{0.0094}{0.9400} = 0.01$$

6.3.2 The $\bar{x}$ and s Control Charts with Variable Sample Size

The $\bar{x}$ and s control charts are relatively easy to apply in cases where the sample sizes are variable. In this case, we should use a weighted average approach in calculating $\bar{\bar{x}}$ and $\bar{\bar{s}}$. If n_i is the number of observations in the i th sample, then use

$$\bar{\bar{x}} = \frac{\sum_{i=1}^m n_i \bar{x}_i}{\sum_{i=1}^m n_i} \quad (6.30)$$

and

$$\bar{\bar{s}} = \left[\frac{\sum_{i=1}^m (n_i - 1) s_i^2}{\sum_{i=1}^m n_i - m} \right]^{1/2} \quad (6.31)$$

as the center lines on the $\bar{x}$ and s control charts, respectively. The control limits would be calculated from equations (6.27) and (6.28), respectively, but the constants A_3 , B_3 , and B_4 will depend on the sample size used in each individual subgroup.

EXAMPLE 6.4

 $\bar{x}$ and s Chart for the Piston Rings, Variable Sample Size

Consider the data in Table 6.4, which is a modification of the piston-ring data used in Example 6.3. Note that the sample sizes vary from $n = 3$ to $n = 5$. Use the procedure described on page 255 to set up the $\bar{x}$ and s control charts.

SOLUTION

The weighted grand mean and weighted average standard deviation are computed from equations (6.30) and (6.31) as follows:

$$\bar{\bar{x}} = \frac{\sum_{i=1}^{25} n_i \bar{x}_i}{\sum_{i=1}^{25} n_i} = \frac{5(74.010) + 3(73.996) + \cdots + 5(73.998)}{5 + 3 + \cdots + 5}$$

$$= \frac{8362.075}{113} = 74.001$$

TABLE 6.4
 Inside Diameter Measurements (mm) on Automobile Engine Piston Rings

Sample Number	Observations					$\bar{x}_i$	s_i
1	74.030	74.002	74.019	73.992	74.008	74.010	0.0148
2	73.995	73.992	74.001			73.996	0.0046
3	73.988	74.024	74.021	74.005	74.002	74.008	0.0147
4	74.002	73.996	73.993	74.015	74.009	74.003	0.0091
5	73.992	74.007	74.015	73.989	74.014	74.003	0.0122
6	74.009	73.994	73.997	73.985		73.996	0.0099
7	73.995	74.006	73.994	74.000		73.999	0.0055
8	73.985	74.003	73.993	74.015	73.988	73.997	0.0123
9	74.008	73.995	74.009	74.005		74.004	0.0064
10	73.998	74.000	73.990	74.007	73.995	73.998	0.0063
11	73.994	73.998	73.994	73.995	73.990	73.994	0.0029
12	74.004	74.000	74.007	74.000	73.996	74.001	0.0042
13	73.983	74.002	73.998			73.994	0.0100
14	74.006	73.967	73.994	74.000	73.984	73.990	0.0153
15	74.012	74.014	73.998			74.008	0.0087
16	74.000	73.984	74.005	73.998	73.996	73.997	0.0078
17	73.994	74.012	73.986	74.005		73.999	0.0115
18	74.006	74.010	74.018	74.003	74.000	74.007	0.0070
19	73.984	74.002	74.003	74.005	73.997	73.998	0.0085
20	74.000	74.010	74.013			74.008	0.0068
21	73.982	74.001	74.015	74.005	73.996	74.000	0.0122
22	74.004	73.999	73.990	74.006	74.009	74.002	0.0074
23	74.010	73.989	73.990	74.009	74.014	74.002	0.0119
24	74.015	74.008	73.993	74.000	74.010	74.005	0.0087
25	73.982	73.984	73.995	74.017	74.013	73.998	0.0162

and

$$\begin{aligned}\bar{s} &= \left[\frac{\sum_{i=1}^{25} (n_i - 1) s_i^2}{\sum_{i=1}^{25} n_i - 25} \right]^{1/2} = \left[\frac{4(0.0148)^2 + 2(0.0046)^2 + \cdots + 4(0.0162)^2}{5 + 3 + \cdots + 5 - 25} \right]^{1/2} \\ &= \left[\frac{0.009324}{88} \right]^{1/2} = 0.0103\end{aligned}$$

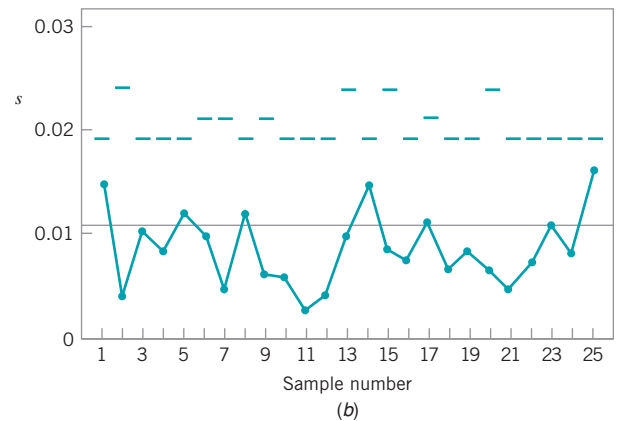
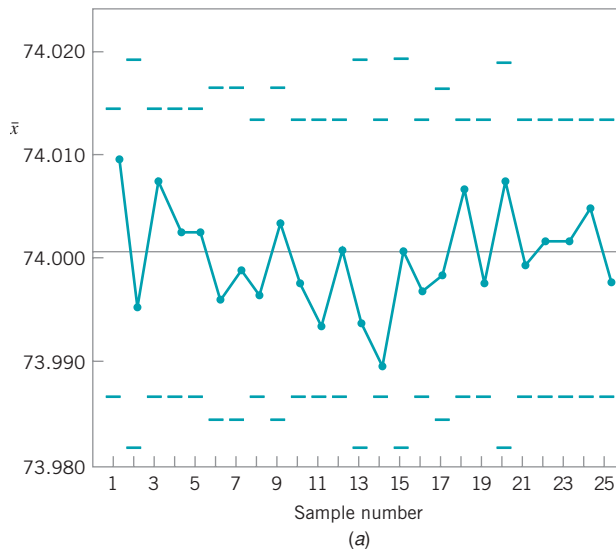
Therefore, the center line of the $\bar{x}$ chart is $\bar{\bar{x}} = 74.001$, and the center line of the s chart is $\bar{s} = 0.0103$. The control limits may now be easily calculated. To illustrate, consider the first sample. The limits for the $\bar{x}$ chart are

$$\begin{aligned}\text{UCL} &= 74.001 + 1.427(0.0103) = 74.016 \\ \text{CL} &= 74.001 \\ \text{LCL} &= 74.001 - 1.427(0.0103) = 73.986\end{aligned}$$

The control limits for the s chart are

$$\begin{aligned}\text{UCL} &= 2.089(0.0103) = 0.022 \\ \text{CL} &= 0.0103 \\ \text{LCL} &= 0(0.0103) = 0\end{aligned}$$

Note that we have used the values of A_3 , B_3 , and B_4 for $n_1 = 5$. The limits for the second sample would use the values of these constants for $n_2 = 3$. The control limit calculations for all 25 samples are summarized in Table 6.5. The control charts are plotted in Fig. 6.18.



■ **FIGURE 6.18** The (a) $\bar{x}$ and (b) s control charts for piston-ring data with variable sample size, Example 6.4.

(continued)

■ TABLE 6.5

Computation of Control Limits for $\bar{x}$ and s Charts with Variable Sample Size

Sample	n	$\bar{x}$	s	A_3	$\bar{x}$ Chart		B_3	B_4	s Chart	
					LCL	UCL			LCL	UCL
1	5	74.010	0.0148	1.427	73.986	74.016	0	2.089	0	0.022
2	3	73.996	0.0046	1.954	73.981	74.021	0	2.568	0	0.026
3	5	74.008	0.0147	1.427	73.986	74.016	0	2.089	0	0.022
4	5	74.003	0.0091	1.427	73.986	74.016	0	2.089	0	0.022
5	5	74.003	0.0122	1.427	73.986	74.016	0	2.089	0	0.022
6	4	73.996	0.0099	1.628	73.984	74.018	0	2.266	0	0.023
7	4	73.999	0.0055	1.628	73.984	74.018	0	2.266	0	0.023
8	5	73.997	0.0123	1.427	73.986	74.016	0	2.089	0	0.022
9	4	74.004	0.0064	1.628	73.984	74.018	0	2.266	0	0.023
10	5	73.998	0.0063	1.427	73.986	74.016	0	2.089	0	0.022
11	5	73.994	0.0029	1.427	73.986	74.016	0	2.089	0	0.022
12	5	74.001	0.0042	1.427	73.986	74.016	0	2.089	0	0.022
13	3	73.994	0.0100	1.954	73.981	74.021	0	2.568	0	0.026
14	5	73.990	0.0153	1.427	73.986	74.016	0	2.089	0	0.022
15	3	74.008	0.0087	1.954	73.981	74.021	0	2.568	0	0.026
16	5	73.997	0.0078	1.427	73.986	74.016	0	2.089	0	0.022
17	4	73.999	0.0115	1.628	73.984	74.018	0	2.226	0	0.023
18	5	74.007	0.0070	1.427	73.986	74.016	0	2.089	0	0.022
19	5	73.998	0.0085	1.427	73.986	74.016	0	2.089	0	0.022
20	3	74.008	0.0068	1.954	73.981	74.021	0	2.568	0	0.026
21	5	74.000	0.0122	1.427	73.986	74.016	0	2.089	0	0.022
22	5	74.002	0.0074	1.427	73.986	74.016	0	2.089	0	0.022
23	5	74.002	0.0119	1.427	73.986	74.016	0	2.089	0	0.022
24	5	74.005	0.0087	1.427	73.986	74.016	0	2.089	0	0.022
25	5	73.998	0.0162	1.427	73.986	74.016	0	2.089	0	0.022

An alternative to using variable-width control limits on the $\bar{x}$ and s control charts is to base the control limit calculations on an **average sample size $\bar{n}$** . If the n_i are not very different, this approach may be satisfactory in some situations; it is particularly helpful if the charts are to be used in a presentation to management. Since the average sample size n_i may not be an integer, a useful alternative is to base these approximate control limits on a modal (most common) sample size.

Estimation of σ . We may estimate the process standard deviation, σ , from the individual sample values s_i . First, average all the values of s_i for which $n_i = 5$ (the most frequently occurring value of n_i). This gives

$$\bar{s} = \frac{0.1715}{17} = 0.0101$$

The estimate of the process σ is then

$$\hat{\sigma} = \frac{\bar{s}}{c_4} = \frac{0.0101}{0.9400} = 0.01$$

where the value of c_4 used is for samples of size $n = 5$.

6.3.3 The s^2 Control Chart

Most quality engineers use either the R chart or the s chart to monitor process variability, with s preferable to R for moderate to large sample sizes. Some practitioners recommend a control chart based directly on the sample variance s^2 , the **s^2 control chart**. The parameters for the s^2 control chart are

$$\begin{aligned} \text{UCL} &= \frac{\bar{s}^2}{n-1} \chi_{\alpha/2, n-1}^2 \\ \text{Center line} &= \bar{s}^2 \\ \text{LCL} &= \frac{\bar{s}^2}{n-1} \chi_{1-(\alpha/2), n-1}^2 \end{aligned} \quad (6.32)$$

where $\chi_{\alpha/2}^2$, and $\chi_{(\alpha/2), n-1}^2$ denote the upper and lower $\alpha/2$ percentage points of the chi-square distribution with $n - 1$ degrees of freedom, and $\bar{s}^2$ is an average sample variance obtained from the analysis of preliminary data. A standard value σ^2 could be used in equation (6.32) instead of $\bar{s}^2$ if one were available. Note that this control chart is defined with probability limits.

6.4 The Shewhart Control Chart for Individual Measurements

There are many situations in which the sample size used for process monitoring is $n = 1$; that is, the sample consists of an individual unit. Some examples of these situations are as follows:

1. Automated inspection and measurement technology is used, and every unit manufactured is analyzed so there is no basis for rational subgrouping.
2. Data comes available relatively slowly, and it is inconvenient to allow sample sizes of $n > 1$ to accumulate before analysis. The long interval between observations will cause problems with rational subgrouping. This occurs frequently in both manufacturing and nonmanufacturing situations.
3. Repeat measurements on the process differ only because of laboratory or analysis error, as in many chemical processes.
4. Multiple measurements are taken on the same unit of product, such as measuring oxide thickness at several different locations on a wafer in semiconductor manufacturing.
5. In process plants, such as papermaking, measurements on some parameter such as coating thickness *across* the roll will differ very little and produce a standard deviation that is much too small if the objective is to control coating thickness *along* the roll.

In such situations, the **control chart for individual units** is useful. (The cumulative sum and exponentially weighted moving-average control charts discussed in Chapter 9 will be a better alternative in phase II or when the magnitude of the shift in process mean that is of interest is small.) In many applications of the *individuals control chart*, we use the *moving range* two successive observations as the basis of estimating the process variability. The moving range is defined as

$$MR_i = |x_i - x_{i-1}|$$

It is also possible to establish a **moving range control chart**. The procedure is illustrated in the following example.

EXAMPLE 6.5

Loan Processing Costs

The mortgage loan processing unit of a bank monitors the costs of processing loan applications. The quantity tracked is the average weekly processing costs, obtained by dividing total weekly costs by the number of loans processed during

the week. The processing costs for the most recent 20 weeks are shown in Table 6.6. Set up individual and moving range control charts for these data.

SOLUTION

To set up the control chart for individual observations, note that the sample average cost of the 20 observations is $\bar{x} = 300.5$ and that the average of the moving ranges of two observations is $\overline{MR} = 7.79$. To set up the moving range chart, we use $D_3 = 0$ and $D_4 = 3.267$ for $n = 2$. Therefore, the moving range chart has center line $\overline{MR} = 7.79$, LCL = 0, and $UCL = D_4\overline{MR} = (3.267)7.79 = 25.45$. The control chart (from Minitab) is shown in Fig. 6.19*b*. Notice that no points are out of control.

For the control chart for individual measurements, the parameters are

$$UCL = \bar{x} + 3 \frac{\overline{MR}}{d_2}$$
$$\text{Center line} = \bar{x}$$
$$LCL = \bar{x} - 3 \frac{\overline{MR}}{d_2}$$

(6.33)

■ TABLE 6.6
Costs of Processing Mortgage Loan Applications

Weeks	Cost <i>x</i>	Moving Range MR
1	310	
2	288	22
3	297	9
4	298	1
5	307	9
6	303	4
7	294	9
8	297	3
9	308	11
10	306	2
11	294	12
12	299	5
13	297	2
14	299	2
15	314	15
16	295	19
17	293	2
18	306	13
19	301	5
20	304	3
	$\bar{x} = 300.5$	$\overline{MR} = 7.79$

If a moving range of $n = 2$ observations is used, then $d_2 = 1.128$. For the data in Table 6.6, we have

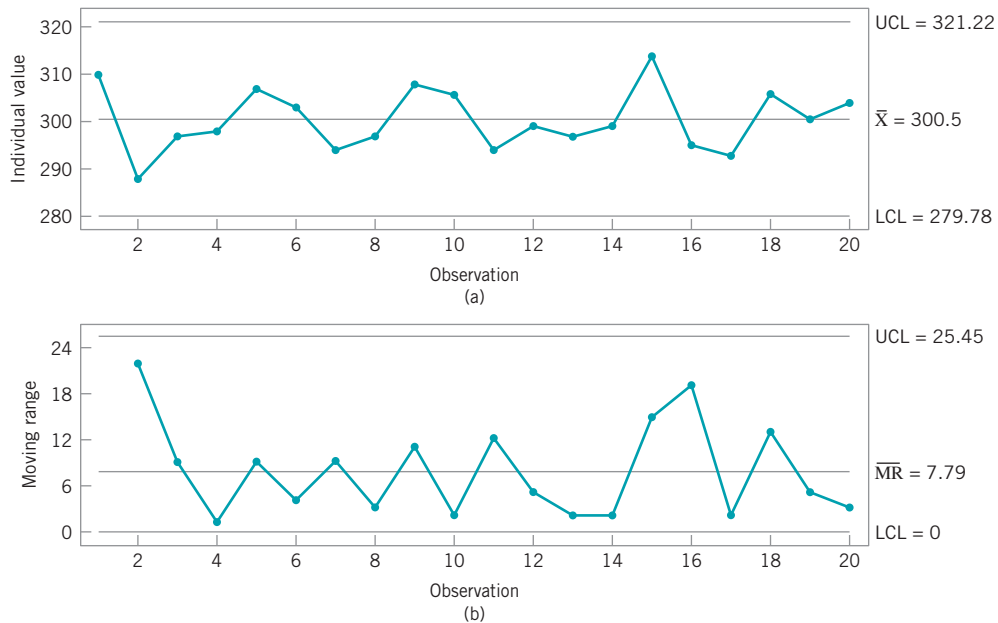
$$UCL = \bar{x} + 3 \frac{\overline{MR}}{d_2} = 300.5 + 3 \frac{7.79}{1.128} = 321.22$$

$$\text{Center line} = \bar{x} = 300.088$$

$$LCL = \bar{x} - 3 \frac{\overline{MR}}{d_2} = 300.5 - 3 \frac{7.79}{1.128} = 279.78$$

The control chart for individual cost values is shown in Fig. 6.19a. There are no out of control observations on the individuals control chart.

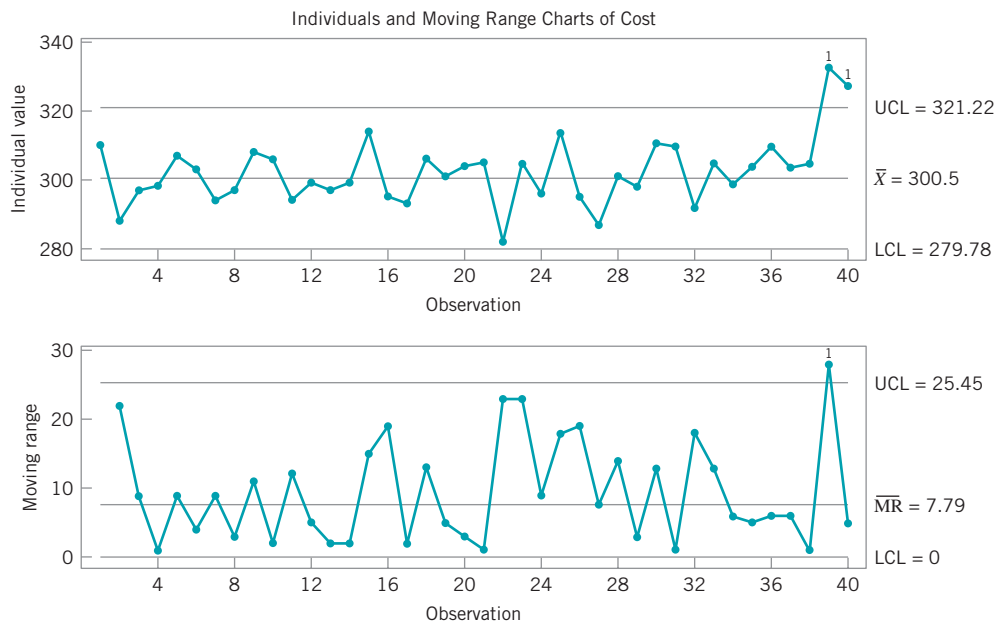
The interpretation of the individuals control chart is very similar to the interpretation of the ordinary $\bar{x}$ control chart. A shift in the process mean will result in a single point or a series of points that plot outside the control limits on the control chart for individuals. Sometimes a point will plot outside the control limits on both the individuals chart and the moving range chart. This will often occur because a large value of x will also lead to a large value of the moving range for that sample. This is very typical behavior for the individuals and moving range control charts. It is most likely an indication that the mean is out of control and not an indication that both the mean and the variance of the process are out of control.



■ **FIGURE 6.19** Control charts for (a) individual observations on cost and for (b) the moving range.

Phase II Operation and Interpretation of the Charts. Table 6.7 contains data on mortgage application processing costs for weeks 21–40. These data are plotted in Fig. 6.20 on the continuation of the control chart for individuals and the moving range control chart developed in Example 6.5. As this figure makes clear, an upward shift in cost has occurred around week 39, since there is an obvious “shift in process level” pattern on the chart for individuals followed by another out-of-control signal at week 40. Note that the moving range chart also reacts to this level shift with a single large spike at week 39. This spike on the moving range chart is sometimes helpful in identifying exactly where a process shift in the mean has occurred. Clearly one should look for possible assignable causes around weeks 39. Possible causes could include an unusual number of applications requiring additional manual underwriting work, or possibly new underwriters working in the process, or possibly temporary underwriters replacing regular employees taking vacations.

Some care should be exercised in interpreting patterns on the moving range chart. The moving ranges are correlated, and this correlation may often induce a pattern of runs or cycles



■ **FIGURE 6.20** Continuation of the control chart for individuals and the moving range using the additional data in Table 6.7.

■ **TABLE 6.7**
Costs of Processing Mortgage Loan Applications, Weeks 21–40

Week	Cost x	Week	Cost x
21	305	31	310
22	282	32	292
23	305	33	305
24	296	34	299
25	314	35	304
26	295	36	310
27	287	37	304
28	301	38	305
29	298	39	333
30	311	40	328

on the chart. Such a pattern is evident on the moving range chart in Fig. 6.21. The individual measurements on the x chart are assumed to be uncorrelated, however, and any apparent pattern on this chart should be carefully investigated.

Some authorities have recommended not constructing and plotting the moving range chart. They point out (correctly, we add) that the moving range chart cannot really provide useful information about a shift in process variability. For example, see Rigdon, Cruthis, and Champ (1994). In effect, shifts in the process mean also show up on the moving range chart. Our feeling is that as long as the analyst is careful in interpretation and relies primarily on the individuals chart, little trouble will ensue from plotting both charts.

Average Run Lengths. Crowder (1987b) has studied the average run length of the combined control chart for individuals and moving range chart. He produces ARLs for various settings of the control limits and shifts in the process mean and standard deviation. In general, his work shows that the ARL_0 of the combined procedure will generally be much less than the ARL_0 of a standard Shewhart control chart when the process is in control (recall that ARL_0 for a

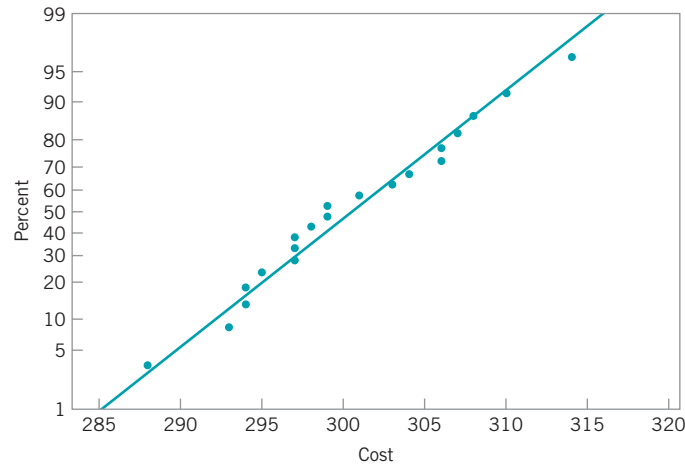


FIGURE 6.21
Normal probability plot of the mortgage application processing cost data from Table 6.6, Example 6.5.

Shewhart chart is 370 samples), if we use the conventional three-sigma limits on the charts. In general, results closer to the Shewhart in-control ARL are obtained if we use three-sigma limits on the chart for individuals and compute the upper control limit on the moving range chart from

$$UCL = \overline{DMR}$$

where the constant D should be chosen such that $4 \leq D \leq 5$.

One can get a very good idea about the ability of the individuals control chart to detect process shifts by looking at the OC curves in Fig. 6.13 or the ARL curves in Fig. 6.15. For an individuals control chart with three-sigma limits, we can compute the following:

Size of Shift	β	ARL ₁
1σ	0.9772	43.96
2σ	0.8413	6.30
3σ	0.5000	2.00

Note that the ability of the individuals control chart to detect small shifts is very poor. For instance, consider a continuous chemical process in which samples are taken every hour. If a shift in the process mean of about one standard deviation occurs, the information above tells us that it will take about 44 samples, on the average, to detect the shift. This is nearly two full days of continuous production in the out-of-control state, a situation that has potentially devastating economic consequences. This limits the usefulness of the individuals control chart in phase II process monitoring.

Some individuals have suggested that control limits narrower than three-sigma be used on the chart for individuals to enhance its ability to detect small process shifts. This is a dangerous suggestion, as narrower limits will dramatically reduce the value of ARL_0 and increase the occurrence of false alarms to the point where the charts are ignored and hence become useless. If we are interested in detecting small shifts in phase II, then the correct approach is to use either the cumulative sum control chart or the exponentially weighted moving average control chart (see Chapter 9).

Normality. Our discussion in this section has made an assumption that the observations follow a normal distribution. Borror, Montgomery, and Runger (1999) have studied the phase II performance of the Shewhart control chart for individuals when the process data are not normal. They investigated various gamma distributions to represent skewed process data and t distributions to represent symmetric normal-like data. They found that the in-control ARL is dramatically

affected by non-normal data. For example, if the individuals chart has three-sigma limits so that for normal data $ARL_0 = 370$, the actual ARL_0 for gamma-distributed data is between 45 and 97, depending on the shape of the gamma distribution (more highly skewed distributions yield poorer performance). For the t distribution, the ARL_0 values range from 76 to 283 as the degrees of freedom increase from 4 to 50 (that is, as the t becomes more like the normal distribution).

In the face of these results, we conclude that if the process shows evidence of even moderate departure from normality, the control limits given here may be entirely inappropriate for phase II process monitoring. One approach to dealing with the problem of non-normality would be to determine the control limits for the individuals control chart based on the percentiles of the correct underlying distribution. These percentiles could be obtained from a histogram if a large sample (at least 100 but preferably 200 observations) were available, or from a probability distribution fit to the data. See Willemain and Runger (1996) for details on designing control charts from empirical reference distributions. Another approach would be to transform the original variable to a new variable that is approximately normally distributed, and then apply control charts to the new variable. Borror, Montgomery, and Runger (1999) show how a properly designed EWMA control chart is very insensitive to the normality assumption. This approach will be discussed in Chapter 9.

It is important to check the normality assumption when using the control chart for individuals. A simple way to do this is with the normal probability plot. Figure 6.21 is the normal probability plot for the mortgage application processing cost data in Table 6.6. There is no obvious problem with the normality assumption in these data. However, remember that the normal probability plot is but a **crude check** of the normality assumption and the individuals control chart is very sensitive to non-normality. Furthermore, mean shifts could show up as a problem with normality on the normal probability plot. Process stability is needed to properly interpret the plot. We suggest that the Shewhart individuals chart be used with extreme caution, particularly in phase II process monitoring.

EXAMPLE 6.6 The Use of Transformations

Table 6.8 presents consecutive measurements on the resistivity of 25 silicon wafers after an epitaxial layer is deposited in a single-wafer deposition process. Construct an individuals control chart for this process.

TABLE 6.8
Resistivity Data for Example 6.6

Sample, i	Resistivity (x_i)	$\ln(x_i)$	MR	Sample, i	Resistivity (x_i)	$\ln(x_i)$	MR
1	216	5.37528		14	242	5.48894	0.23791
2	290	5.66988	0.29460	15	168	5.12396	0.36498
3	236	5.46383	0.20605	16	360	5.88610	0.76214
4	228	5.42935	0.03448	17	226	5.42053	0.46557
5	244	5.49717	0.06782	18	253	5.53339	0.11286
6	210	5.34711	0.15006	19	380	5.94017	0.40678
7	139	4.93447	0.41264	20	131	4.87520	1.06497
8	310	5.73657	0.80210	21	173	5.15329	0.27809
9	240	5.48064	0.25593	22	224	5.41165	0.25836
10	211	5.35186	0.12878	23	195	5.27300	0.13865
11	175	5.16479	0.18707	24	199	5.29330	0.02030
12	447	6.10256	0.93777	25	226	5.42053	0.12723
13	307	5.72685	0.37571				

$\overline{\ln(x_i)} = 5.44402 \quad \overline{MR} = 0.33712$

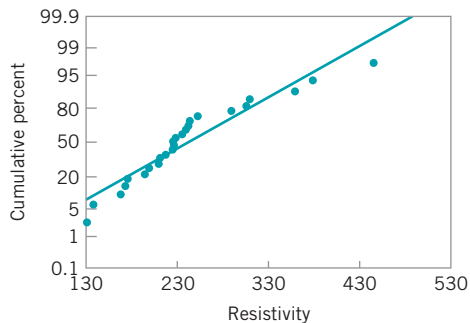
SOLUTION

A normal probability plot of the resistivity measurements is shown in Fig. 6.22. This plot was constructed by Minitab, which fits the line to the points by least squares (not the best method). It is clear from inspection of the normal probability plot that the normality assumption for resistivity is at best questionable, so it would be dangerous to apply an individuals control chart to the original process data.

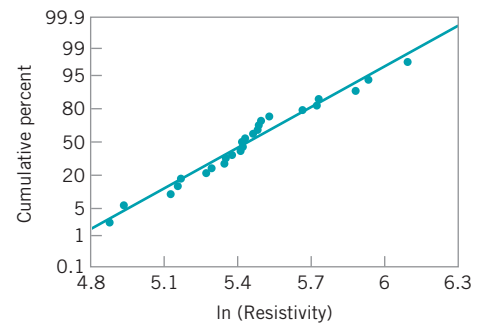
Figure 6.22 indicates that the distribution of resistivity has a long tail to the right, and consequently we would expect the log transformation (or a similar transformation) to produce a distribution that is closer to normal. The nat-

ural log of resistivity is shown in column three of Table 6.8, and the normal probability plot of the natural log of resistivity is shown in Fig. 6.23. Clearly the log transformation has resulted in a new variable that is more nearly approximated by a normal distribution than were the original resistivity measurements.

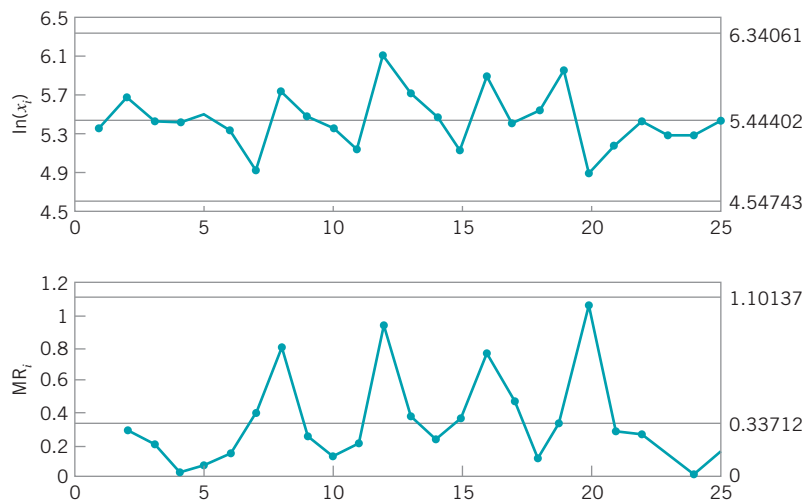
The last column of Table 6.8 shows the moving ranges of the natural log of resistivity. Figure 6.24 presents the individuals and moving range control charts for the natural log of resistivity. Note that there is no indication of an out-of-control process.



■ **FIGURE 6.22** Normal probability plot of resistivity.



■ **FIGURE 6.23** Normal probability plot of ln (resistivity).



■ **FIGURE 6.24** Individuals and moving range control charts on ln (resistivity), Example 6.6

More about Estimating σ . Very often in practice we use moving ranges in estimating σ for the individuals control chart. Recall that moving ranges are defined as $MR_i = |(x_i - x_{i-1})|$, $i = 2, 3, \dots, m$. More properly, this statistic should be called a moving range of **span two** since the number of observations used to calculate the range in the moving window could be increased. The most common estimator is the one we used in Example 6.5, based on the **average moving range** $\overline{MR} = \sum_{i=2}^m MR_i / (m - 1)$ and can be written as

$$\hat{\sigma}_1 = 0.8865 \overline{MR}$$

where the constant 0.8865 is the reciprocal of d_2 for samples of size 2. For in-control processes, Cryer and Ryan (1990), among others, have pointed out that a more efficient estimator is one based on the sample standard deviation

$$\hat{\sigma}_2 = \frac{s}{c_4}$$

Both of these estimators are unbiased, assuming that no assignable causes are present in the sequence of m individual observations.

If assignable causes are present, then both $\hat{\sigma}_1$ and $\hat{\sigma}_2$ result in biased estimates of the process standard deviation. To illustrate, suppose that in the sequence of individual observations

$$x_1, x_2, \dots, x_t, x_{t+1}, \dots, x_m$$

the process is in control with mean μ_0 and standard deviation σ for the first t observations, but between x_t and x_{t+1} an assignable cause occurs that results in a sustained shift in the process mean to a new level $\mu = \mu_0 + \delta\sigma$ and the mean remains at this new level for the remaining sample observations $x_{t+1}, \dots, x_m$. Under these conditions, Woodall and Montgomery (2000–01) show that

$$E(s^2) = \sigma^2 + \frac{t(m-1)}{m(m-1)} (\delta\sigma)^2$$

In fact, this result holds for any case in which the mean of t of the observations is μ_0 and the mean of the remaining observations is $\mu_0 + \delta\sigma$, since the order of the observations is not relevant in computing s^2 . Therefore, s^2 is biased upwards, and consequently $\hat{\sigma}_2 = S/c_4$ will tend to overestimate σ . Note that the extent of the bias in $\hat{\sigma}_2$ depends on the magnitude of the shift in the mean ($\delta\sigma$), the time period following which the shift occurs (t), and the number of available observations (m). Now the moving range is only impacted by the shift in the mean during one period ($t + 1$), so the bias in $\hat{\sigma}_1$ depends only on the shift magnitude and m . If $1 < t < m - 1$, the bias in $\hat{\sigma}_1$ will always be smaller than the bias in $\hat{\sigma}_2$. Cruthis and Rigdon (1992–93) show how the ratio

$$F^* = \left(\frac{\hat{\sigma}_1}{\hat{\sigma}_2} \right)^2$$

can be used to determine whether the process was in control when both estimates were calculated. They use simulation to obtain the approximate 90th, 95th, 99th, and 99.9th percentiles of the distribution of F^* for sample sizes $m = 10(5)100$, assuming that the process is in control. Since this is an empirical reference distribution, observed values of F^* that exceed one of these percentiles is an indication that the process was not in control over the time period during which the m observations were collected.

One way to reduce or possibly eliminate the bias in estimating σ when a sustained shift in the mean is present is to base the estimator on the **median** of the moving ranges of span two, as suggested by Clifford (1959) and Bryce, Gaudard, and Joiner (1997–98). This estimator is

$$\hat{\sigma}_3 = 1.047 \overline{\text{MR}}$$

where $\overline{\text{MR}}$ is the median of the span-two moving ranges, and 1.047 is the reciprocal of the control chart constant d_4 for subgroups of size two defined such that $E(\tilde{R}) = d_4\sigma$ and $\tilde{R}$ is the median range. A table of d_4 values is in Wadsworth, Stephens, and Godfrey (2002). Essentially, only one of the span-two moving ranges should be affected by the sustained shift, and this single large moving range will have little impact on the value of the median moving range, certainly much less impact than it will have on the average moving range. Constructing an individuals control chart using the median moving range to estimate σ is an option in Minitab.

Now suppose that the assignable cause affects a single observation rather than causing a sustained shift in the mean. If there is a single observation that has mean $\mu_0 + \delta\sigma$, then

$$E(s^2) = \sigma^2 + \frac{1}{m}(\delta\sigma)^2$$

and this observation will affect two of the span-two moving ranges. If there are two adjacent such observations, then

$$E(s^2) = \sigma^2 + \frac{2(m-2)}{m(m-1)}(\delta\sigma)^2$$

and two of the span-two moving ranges will be affected by the out-of-control observations. Thus, when the assignable cause affects one or only a few adjacent observations, we expect the bias in s^2 to be smaller than when a sustained shift occurs. However, if an assignable cause producing a sustained shift in the mean occurs either very early in the sequence of observations or very late, it will produce much the same effect as an assignable cause affecting only one or a few adjacent points.

Some authors have suggested basing the estimate of σ on moving ranges of span greater than two, and some computer programs for control charts offer this as an option. It is easy to show that this will always lead to potentially increased bias in the estimate of σ when assignable causes are present. Note that if one uses a span-three moving range and there is a single observation whose mean is affected by the assignable cause, then this single observation will affect up to three of the moving ranges. Thus, a span-three moving range will result in more bias in the estimate of σ than will the moving range of span two. Furthermore, two span-three moving ranges will be affected by a sustained shift. In general, if one uses a span- w moving range and there is a single observation whose mean is affected by the assignable cause, up to w of these moving ranges will be impacted by this observation. Furthermore, if there is a sustained shift in the mean, up to $w - 1$ of the moving ranges will be affected by the shift in the mean. Consequently, increasing the span of a moving average beyond two results in increasing the bias in the estimate of σ if assignable causes that either produce sustained shifts in the process mean or that affect the mean of a single observation (or a few adjacent ones) are present. In fact, Wetherill and Brown (1991) advise plotting the estimate of σ versus the span of the moving average used to obtain the estimate. A sharply rising curve indicates the presence of assignable causes. For more discussion of using ranges to estimate process variability, see Woodall and Montgomery (2000–01).

■ TABLE 6.9
Formulas for Control Charts, Standards Given

Chart	Center Line	Control Limits
$\bar{x}$ (μ and σ given)	μ	$\mu \pm A\sigma$
R (σ given)	$d_2\sigma$	$UCL = D_2\sigma, LCL = D_1\sigma$
s (σ given)	$c_4\sigma$	$UCL = B_6\sigma, LCL = B_5\sigma$

■ TABLE 6.10
Formulas for Control Charts, Control Limits Based on Past Data
(No Standards Given)

Chart	Center Line	Control Limits
$\bar{x}$ (using R)	$\bar{\bar{x}}$	$\bar{\bar{x}} \pm A_2\bar{R}$
$\bar{x}$ (using s)	$\bar{\bar{x}}$	$\bar{\bar{x}} \pm A_3\bar{s}$
R	$\bar{\bar{R}}$	$UCL = D_4\bar{R}, LCL = D_3\bar{R}$
s	$\bar{\bar{s}}$	$UCL = B_4\bar{s}, LCL = B_3\bar{s}$

6.5 Summary of Procedures for $\bar{x}$, R , And s Charts

It is convenient to summarize in one place the various computational formulas for the major types of variables control charts discussed in this chapter. Table 6.9 summarizes the formulas for $\bar{x}$, R , and s charts when standard values for μ and σ are given. Table 6.10 provides the corresponding summary when no standard values are given and trial control limits must be established from analysis of past data. The constants given for the s chart assume that $n - 1$ is used in the denominator of s . All constants are tabulated for various sample sizes in Appendix Table VI.

6.6 Applications of Variables Control Charts

There are many interesting applications of variables control charts. In this section, a few of them will be described to give additional insights into how the control chart works, as well as ideas for further applications.

EXAMPLE 6.7 Using Control Charts to Improve Suppliers' Processes

A large aerospace manufacturer purchased an aircraft component from two suppliers. These components frequently exhibited excessive variability on a key dimension that made it impossible to assemble them into the final product. This problem always resulted in expensive rework costs and occasionally caused delays in finishing the assembly of an airplane.

The materials receiving group performed 100% inspection of these parts in an effort to improve the situation. They maintained $\bar{x}$ and R charts on the dimension of interest for both suppliers. They found that the fraction of nonconforming units was about the same for both suppliers, but for very different reasons. Supplier A could produce parts with mean dimension equal to the required value, but the process was out of statistical control. Supplier B could maintain good statistical control

and, in general, produced a part that exhibited considerably less variability than parts from supplier A, but its process was centered so far off the nominal required dimension that many parts were out of specification.

This situation convinced the procurement organization to work with both suppliers, persuading supplier A to implement SPC and to begin working at continuous improvement, and assisting supplier B to find out why his process was consistently centered incorrectly. Supplier B's problem was ultimately tracked to some incorrect code in an NC (numerical-controlled) machine, and the use of SPC at supplier A resulted in considerable reduction in variability over a six-month period. As a result of these actions, the problem with these parts was essentially eliminated.

7 Control Charts for Attributes

CHAPTER OUTLINE

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|-------|---|-------|--|
| 7.1 | INTRODUCTION | 7.3.3 | Demerit Systems |
| 7.2 | THE CONTROL CHART FOR FRACTION NONCONFORMING | 7.3.4 | The Operating-Characteristic Function |
| 7.2.1 | Development and Operation of the Control Chart | 7.3.5 | Dealing with Low Defect Levels |
| 7.2.2 | Variable Sample Size | 7.3.6 | Nonmanufacturing Applications |
| 7.2.3 | Applications in Transactional and Service Businesses | 7.4 | CHOICE BETWEEN ATTRIBUTES AND VARIABLES CONTROL CHARTS |
| 7.2.4 | The Operating-Characteristic Function and Average Run Length Calculations | 7.5 | GUIDELINES FOR IMPLEMENTING CONTROL CHARTS |
| 7.3 | CONTROL CHARTS FOR NONCONFORMITIES (DEFECTS) | | |
| 7.3.1 | Procedures with Constant Sample Size | | |
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| Supplemental Material for Chapter 7 | |
| S7.1 | Probability Limits on Control Charts |

The supplemental material is on the textbook Website www.wiley.com/college/montgomery.

CHAPTER OVERVIEW AND LEARNING OBJECTIVES

Many quality characteristics cannot be conveniently represented numerically. In such cases, we usually classify each item inspected as either conforming or nonconforming to the specifications on that quality characteristic. The terminology **defective** or **nondefective** is often used to identify these two classifications of product. More recently, the terminology **conforming** and **nonconforming** has become popular. Quality characteristics of this type are called **attributes**. Some examples of quality characteristics that are attributes are the proportion of warped automobile engine connecting rods in a day's production, the number of nonfunctional semiconductor chips on a wafer, the number of errors or mistakes made in completing a loan application, and the number of medical errors made in a hospital.

This chapter presents three widely used attributes control charts. The first of these relates to the fraction of nonconforming or defective product produced by a manufacturing process, and is called the **control chart for fraction nonconforming**, or p chart. In some situations it is more convenient to deal with the number of **defects** or **nonconformities** observed

rather than the fraction nonconforming. The second type of control chart that we study, called the **control chart for nonconformities**, or the c chart, is designed to deal with this case. Finally, we present a **control chart for nonconformities per unit**, or the u chart, which is useful in situations where the average number of nonconformities per unit is a more convenient basis for process control. An excellent supplement to this chapter is the paper by Woodall (1997) that summarizes over 250 papers on attributes control charts and provides a comprehensive bibliography. The chapter concludes with some guidelines for implementing control charts.

After careful study of this chapter you should be able to do the following:

1. Understand the statistical basis of attributes control charts
2. Know how to design attributes control charts
3. Know how to set up and use the p chart for fraction nonconforming
4. Know how to set up and use the np control chart for the number of nonconforming items
5. Know how to set up and use the c control chart for defects
6. Know how to set up and use the u control chart for defects per unit
7. Use attributes control charts with variable sample size
8. Understand the advantages and disadvantages of attributes versus variables control charts
9. Understand the rational subgroup concept for attributes control charts
10. Determine the average run length for attributes control charts

7.1 Introduction

In Chapter 6, we introduced control charts for quality characteristics that are expressed as variables. Although these control charts enjoy widespread application they are not universally applicable because not all quality characteristics can be expressed with variables data. For example, consider a glass container for a liquid product. Suppose we examine a container and classify it into one of the two categories called *conforming* or *nonconforming*, depending on whether the container meets the requirements on one or more quality characteristics. This is an example of **attributes data**, and a control chart for the fraction of nonconforming containers could be established (we show how to do this in Section 7.2). Alternatively, in some processes we may examine a unit of product and count defects or nonconformities on the unit. These types of data are widely encountered in the semiconductor industry, for example. In Section 7.3 we show how to establish control charts for counts, or for the average number of counts per unit.

Attributes charts are generally not as informative as variables charts because there is typically more information in a numerical measurement than in merely classifying a unit as conforming or nonconforming. However, attribute charts do have important applications. They are particularly useful in service industries and in nonmanufacturing quality-improvement efforts because so many of the quality characteristics found in these environments are not easily measured on a numerical scale.

7.2 The Control Chart for Fraction Nonconforming

The **fraction nonconforming** is defined as the ratio of the number of nonconforming items in a population to the total number of items in that population. The items may have *several* quality characteristics that are examined simultaneously by the inspector. If the item does not conform to standard on one or more of these characteristics, it is classified as nonconforming.

We usually express the fraction nonconforming as a decimal, although occasionally the percent nonconforming (which is simply 100% times the fraction nonconforming) is used. When demonstrating or displaying the control chart to production personnel or presenting results to management, the percent nonconforming is often used, as it has more intuitive appeal. Although it is customary to work with fraction nonconforming, we could also analyze the fraction conforming just as easily, resulting in a control chart on **process yield**. For example, many manufacturing organizations operate a yield-management system at each stage of their manufacturing process, with the first-pass yield tracked on a control chart.

The statistical principles underlying the control chart for fraction nonconforming are based on the binomial distribution. Suppose the production process is operating in a stable manner, such that the probability that any unit will not conform to specifications is p , and that successive units produced are independent. Then each unit produced is a realization of a Bernoulli random variable with parameter p . If a random sample of n units of product is selected, and if D is the number of units of product that are nonconforming, then D has a binomial distribution with parameters n and p ; that is,

$$P\{D = x\} = \binom{n}{x} p^x (1-p)^{n-x} \quad x = 0, 1, \dots, n \quad (7.1)$$

From Section 3.2.2 we know that the mean and variance of the random variable D are np and $np(1-p)$, respectively.

The **sample fraction nonconforming** is defined as the ratio of the number of nonconforming units in the sample D to the sample size n ; that is,

$$\hat{p} = \frac{D}{n} \quad (7.2)$$

As noted in Section 3.2.2, the distribution of the random variable $\hat{p}$ can be obtained from the binomial. Furthermore, the mean and variance of $\hat{p}$ are

$$\mu_{\hat{p}} = p \quad (7.3)$$

and

$$\sigma_{\hat{p}}^2 = \frac{p(1-p)}{n} \quad (7.4)$$

respectively. We will now see how this theory can be applied to the development of a control chart for fraction nonconforming. Because the chart monitors the process fraction nonconforming p , it is also called the p chart.

7.2.1 Development and Operation of the Control Chart

In Chapter 5, we discussed the general statistical principles on which the Shewhart control chart is based. If w is a statistic that measures a quality characteristic, and if the mean of w is μ_w and the variance of w is σ_w^2 , then the general model for the Shewhart control chart is as follows:

$$\begin{aligned} \text{UCL} &= \mu_w + L\sigma_w \\ \text{Center line} &= \mu_w \\ \text{LCL} &= \mu_w - L\sigma_w \end{aligned} \quad (7.5)$$

where L is the distance of the control limits from the center line, in multiples of the standard deviation of w . It is customary to choose $L = 3$.

Suppose that the true fraction nonconforming p in the production process is known or is a specified **standard value**. Then from equation (7.5), the center line and control limits of the fraction nonconforming control chart would be as follows:

Fraction Nonconforming Control Chart: Standard Given

$$\begin{aligned} \text{UCL} &= p + 3\sqrt{\frac{p(1-p)}{n}} \\ \text{Center line} &= p \\ \text{LCL} &= p - 3\sqrt{\frac{p(1-p)}{n}} \end{aligned} \quad (7.6)$$

Depending on the values of p and n , sometimes the lower control limit $\text{LCL} < 0$. In these cases, we customarily set $\text{LCL} = 0$ and assume that the control chart only has an upper control limit. The actual operation of this chart would consist of taking subsequent samples of n units, computing the sample fraction nonconforming $\hat{p}$, and plotting the statistic $\hat{p}$ on the chart. As long as $\hat{p}$ remains within the control limits and the sequence of plotted points does not exhibit any systematic nonrandom pattern, we can conclude that the process is in control at the level p . If a point plots outside of the control limits, or if a nonrandom pattern in the plotted points is observed, we can conclude that the process fraction nonconforming has most likely shifted to a new level and the process is out of control.

When the process fraction nonconforming p is not known, then it must be estimated from observed data. The usual procedure is to select m preliminary samples, each of size n . As a general rule, m should be at least 20 or 25. Then if there are D_i nonconforming units in sample i , we compute the fraction nonconforming in the i th sample as

$$\hat{p}_i = \frac{D_i}{n} \quad i = 1, 2, \dots, m$$

and the average of these individual sample fractions nonconforming is

$$\bar{p} = \frac{\sum_{i=1}^m D_i}{mn} = \frac{\sum_{i=1}^m \hat{p}_i}{m} \quad (7.7)$$

The statistic $\bar{p}$ estimates the unknown fraction nonconforming p . The center line and control limits of the control chart for fraction nonconforming are computed as follows:

Fraction Nonconforming Control Chart: No Standard Given

$$\begin{aligned} \text{UCL} &= \bar{p} + 3\sqrt{\frac{\bar{p}(1-\bar{p})}{n}} \\ \text{Center line} &= \bar{p} \\ \text{LCL} &= \bar{p} - 3\sqrt{\frac{\bar{p}(1-\bar{p})}{n}} \end{aligned} \quad (7.8)$$

This control chart is also often called the *p-chart*.

The control limits defined in equation (7.8) should be regarded as **trial control limits**. The sample values of $\hat{p}_i$ from the preliminary subgroups should be plotted against the trial limits to test whether the process was in control when the preliminary data were collected. This is the usual phase I aspect of control chart usage. Any points that exceed the trial control limits should be investigated. If assignable causes for these points are discovered, they should be discarded and new trial control limits determined. Refer to the discussion of trial control limits for the $\bar{x}$ and R charts in Chapter 6.

If the control chart is based on a known or standard value for the fraction nonconforming p , then the calculation of trial control limits is generally unnecessary. However, one should be cautious when working with a standard value for p . Since in practice the true value of p would rarely be known with certainty, we would usually be given a standard value of p that represents a desired or **target value** for the process fraction nonconforming. If this is the case, and future samples indicate an out-of-control condition, we must determine whether the process is out of control at the *target* p but in control at some *other* value of p . For example, suppose we specify a target value of $p = 0.01$, but the process is really in control at a larger value of fraction nonconforming—say, $p = 0.05$. Using the control chart based on $p = 0.01$, we see that many of the points will plot above the upper control limit, indicating an out-of-control condition. However, the process is really out of control only with respect to the target $p = 0.01$. Sometimes it may be possible to “improve” the level of quality by using target values, or to bring a process into control at a particular level of quality performance. In processes where the fraction nonconforming can be controlled by relatively simple process adjustments, target values of p may be useful.

EXAMPLE 7.1 Construction and Operation of a Fraction Nonconforming Control Chart

Frozen orange juice concentrate is packed in 6-oz cardboard cans. These cans are formed on a machine by spinning them from cardboard stock and attaching a metal bottom panel. By inspection of a can, we may determine whether, when filled, it could possibly leak either on the side seam or

around the bottom joint. Such a nonconforming can has an improper seal on either the side seam or the bottom panel. Set up a control chart to improve the fraction of nonconforming cans produced by this machine.

SOLUTION

To establish the control chart, 30 samples of $n = 50$ cans each were selected at half-hour intervals over a three-shift period in which the machine was in continuous operation. The data are shown in Table 7.1.

We construct a phase I control chart using this preliminary data to determine if the process was in control when these data were collected. Since the 30 samples contain $\sum_{i=1}^{30} D_i = 347$ nonconforming cans, we find from equation (7.7),

$$\bar{p} = \frac{\sum_{i=1}^m D_i}{mn} = \frac{347}{(30)(50)} = 0.2313$$

Using $\bar{p}$ as an estimate of the true process fraction nonconforming, we can now calculate the upper and lower control limits as

$$\begin{aligned}\bar{p} \pm 3\sqrt{\frac{\bar{p}(1-\bar{p})}{n}} &= 0.2313 \pm 3\sqrt{\frac{0.2313(0.7687)}{50}} \\ &= 0.2313 \pm 3(0.0596) \\ &= 0.2313 \pm 0.1789\end{aligned}$$

Therefore,

$$UCL = \bar{p} + 3\sqrt{\frac{\bar{p}(1-\bar{p})}{n}} = 0.2313 + 0.1789 = 0.4102$$

and

$$LCL = \bar{p} - 3\sqrt{\frac{\bar{p}(1-\bar{p})}{n}} = 0.2313 - 0.1789 = 0.0524$$

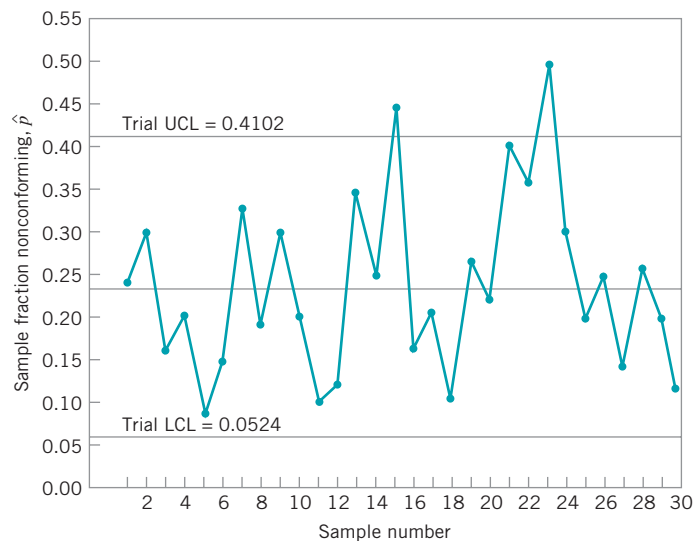
■ TABLE 7.1

Data for Trial Control Limits, Example 7.1, Sample Size $n = 50$

Sample Number	Number of Nonconforming Cans, D_i	Sample Fraction Nonconforming, $\hat{p}_i$	Sample Number	Number of Nonconforming Cans, D_i	Sample Fraction Nonconforming, $\hat{p}_i$
1	12	0.24	17	10	0.20
2	15	0.30	18	5	0.10
3	8	0.16	19	13	0.26
4	10	0.20	20	11	0.22
5	4	0.08	21	20	0.40
6	7	0.14	22	18	0.36
7	16	0.32	23	24	0.48
8	9	0.18	24	15	0.30
9	14	0.28	25	9	0.18
10	10	0.20	26	12	0.24
11	5	0.10	27	7	0.14
12	6	0.12	28	13	0.26
13	17	0.34	29	9	0.18
14	12	0.24	30	6	0.12
15	22	0.44		347	$\bar{p} = 0.2313$
16	8	0.16			

The control chart with center line at $\bar{p} = 0.2313$ and the above upper and lower control limits is shown in Fig. 7.1. The sample fraction nonconforming from each preliminary sample is plotted on this chart. We note that two points,

those from samples 15 and 23, plot above the upper control limit, so the process is not in control. These points must be investigated to see whether an assignable cause can be determined.



■ FIGURE 7.1 Initial phase I fraction nonconforming control chart for the data in Table 7.1.

(continued)

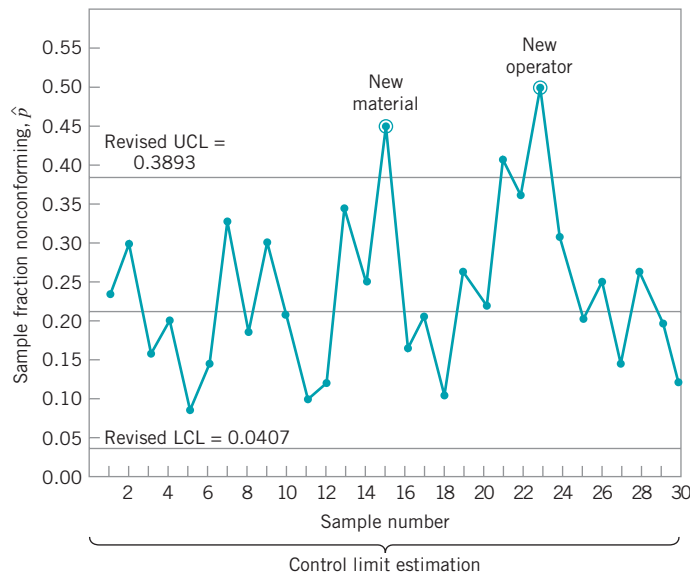
Analysis of the data from sample 15 indicates that a new batch of cardboard stock was put into production during that half-hour period. The introduction of new batches of raw material sometimes causes irregular production performance, and it is reasonable to believe that this has occurred here. Furthermore, during the half-hour period in which

sample 23 was obtained, a relatively inexperienced operator had been temporarily assigned to the machine, and this could account for the high fraction nonconforming obtained from that sample. Consequently, samples 15 and 23 are eliminated, and the new center line and revised control limits are calculated as

$$\begin{aligned}\bar{p} &= \frac{301}{(28)(50)} = 0.2150 \\ \text{UCL} &= \bar{p} + 3\sqrt{\frac{\bar{p}(1-\bar{p})}{n}} = 0.2150 + 3\sqrt{\frac{0.2150(0.7850)}{50}} = 0.3893 \\ \text{LCL} &= \bar{p} - 3\sqrt{\frac{\bar{p}(1-\bar{p})}{n}} = 0.2150 - 3\sqrt{\frac{0.2150(0.7850)}{50}} = 0.0407\end{aligned}$$

The revised center line and control limits are shown on the control chart in Fig. 7.2. Note that we have not dropped samples 15 and 23 from the chart, but they have been excluded from the control limit calculations, and we have noted this directly on the control chart. This annotation of the control chart to indicate unusual points, process adjustments, or the type of investigation made at a particular point in time forms a useful record for future process analysis and should become a standard practice in control chart usage.

Note also that the fraction nonconforming from sample 21 now exceeds the upper control limit. However, analysis of the data does not produce any reasonable or logical assignable cause for this, and we decide to retain the point. Therefore, we conclude that the new control limits in Fig. 7.2 can be used for future samples. Thus, we have concluded the control limit estimation phase (phase I) of control chart usage.



■ FIGURE 7.2 Revised control limits for the data in Table 7.1.

Sometimes examination of control chart data reveals information that affects other points that are not necessarily outside the control limits. For example, if we had found that the temporary operator working when sample 23 was obtained was actually working during the entire two-hour period in which samples 21–24 were obtained, then we should discard all four samples, even if only sample 21 exceeded the control limits, on the grounds that this inexperienced operator probably had some adverse influence on the fraction nonconforming during the entire period.

Before we conclude that the process is in control at this level, we could examine the remaining 28 samples for runs and other nonrandom patterns. The largest run is one of length 5 above the center line, and there are no obvious patterns present in the data. There is no strong evidence of anything other than a random pattern of variation about the center line.

We conclude that the process is in control at the level $p = 0.2150$ and that the revised control limits should be adopted for monitoring current production. However, we note that although the process is in control, the fraction nonconforming is much too high. That is, the process is operating in a stable manner, and no unusual **operator-controllable** problems are present. It is unlikely that the process quality can be improved by action at the workforce level. The nonconforming cans produced are **management controllable** because an intervention by management in the process will be required to improve

performance. Plant management agrees with this observation and directs that, in addition to implementing the control chart program, the engineering staff should analyze the process in an effort to improve the process yield. This study indicates that several adjustments can be made on the machine that should improve its performance.

During the next three shifts following the machine adjustments and the introduction of the control chart, an additional 24 samples of $n = 50$ observations each are collected. These data are shown in Table 7.2, and the sample fractions nonconforming are plotted on the control chart in Fig. 7.3.

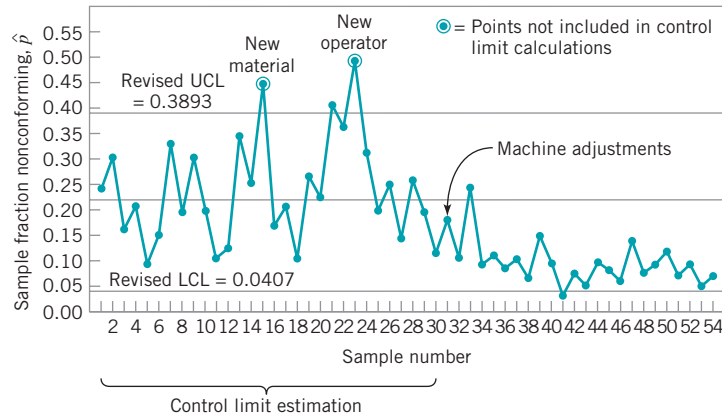
From an examination of Fig. 7.3, our immediate impression is that the process is now operating at a new quality level that is substantially better than the center line level of $\bar{p} = 0.2150$. One point, that from sample 41, is below the lower control limit. No assignable cause for this out-of-control signal can be determined. The only logical reasons for this ostensible change in process performance are the machine adjustments made by the engineering staff and, possibly, the operators themselves. It is not unusual to find that process performance improves following the introduction of formal statistical process-control procedures, often because the operators are more aware of process quality and because the control chart provides a continuing visual display of process performance.

We may formally test the hypothesis that the process fraction nonconforming in this current three-shift period differs

■ **TABLE 7.2**
Orange Juice Concentrate Can Data in Samples of Size $n = 50$

Sample Number	Number of Nonconforming Cans, D_i	Sample Fraction Nonconforming, $\hat{p}_i$	Sample Number	Number of Nonconforming Cans, D_i	Sample Fraction Nonconforming, $\hat{p}_i$
31	9	0.18	44	6	0.12
32	6	0.12	45	5	0.10
33	12	0.24	46	4	0.08
34	5	0.10	47	8	0.16
35	6	0.12	48	5	0.10
36	4	0.08	49	6	0.12
37	6	0.12	50	7	0.14
38	3	0.06	51	5	0.10
39	7	0.14	52	6	0.12
40	6	0.12	53	3	0.06
41	2	0.04	54	5	0.10
42	4	0.08			
43	3	0.06		133	$\bar{p} = 0.1108$

(continued)



■ **FIGURE 7.3** Continuation of the fraction nonconforming control chart, Example 7.1.

from the process fraction nonconforming in the preliminary data, using the procedure given in Section 4.3.4. The hypotheses are

$$H_0: p_1 = p_2$$

$$H_1: p_1 > p_2$$

where p_1 is the process fraction nonconforming from the preliminary data and p_2 is the process fraction nonconforming in the current period. We may estimate p_1 by $\hat{p}_1 = \bar{p} = 0.2150$, and p_2 by

$$\hat{p}_2 = \frac{\sum_{i=31}^{54} D_i}{(50)(24)} = \frac{133}{1200} = 0.1108$$

The (approximate) test statistic for the above hypothesis is, from equation (4.63),

$$Z_0 = \frac{\hat{p}_1 - \hat{p}_2}{\sqrt{\hat{p}(1-\hat{p})\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$$

where

$$\hat{p} = \frac{n_1 \hat{p}_1 + n_2 \hat{p}_2}{n_1 + n_2}$$

In our example, we have

$$\hat{p} = \frac{(1400)(0.2150) + (1200)(0.1108)}{1400 + 1200} = 0.1669$$

and

$$Z_0 = \frac{0.2150 - 0.1108}{\sqrt{(0.1669)(0.8331)\left(\frac{1}{1400} + \frac{1}{1200}\right)}} = 7.10$$

Comparing this to the upper 0.05 point of the standard normal distribution, we find that $Z_0 = 7.10 > Z_{0.05} = 1.645$. Consequently, we reject H_0 and conclude that there has been a significant decrease in the process fallout.

Based on the apparently successful process adjustments, it seems logical to revise the control limits again, using only the most recent samples (numbers 31–54). This results in the new control chart parameters:

$$\text{Center line} = \bar{p} = 0.1108$$

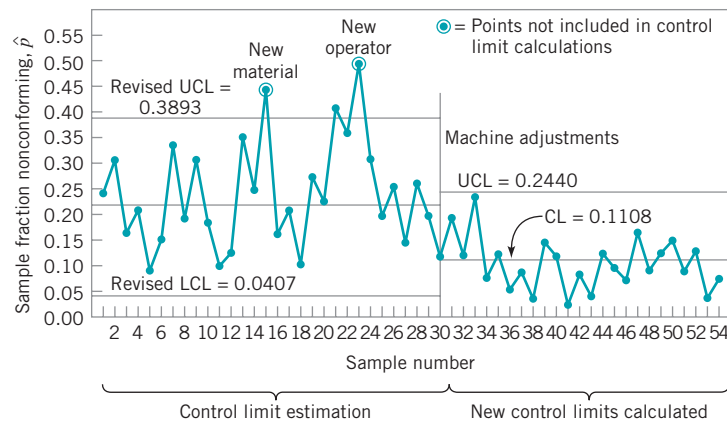
$$\text{UCL} = \bar{p} + 3\sqrt{\frac{\bar{p}(1-\bar{p})}{n}} = 0.1108 + 3\sqrt{\frac{(0.1108)(0.8892)}{50}} = 0.2440$$

$$\text{LCL} = \bar{p} - 3\sqrt{\frac{\bar{p}(1-\bar{p})}{n}} = 0.1108 - 3\sqrt{\frac{(0.1108)(0.8892)}{50}} = -0.0224 = 0$$

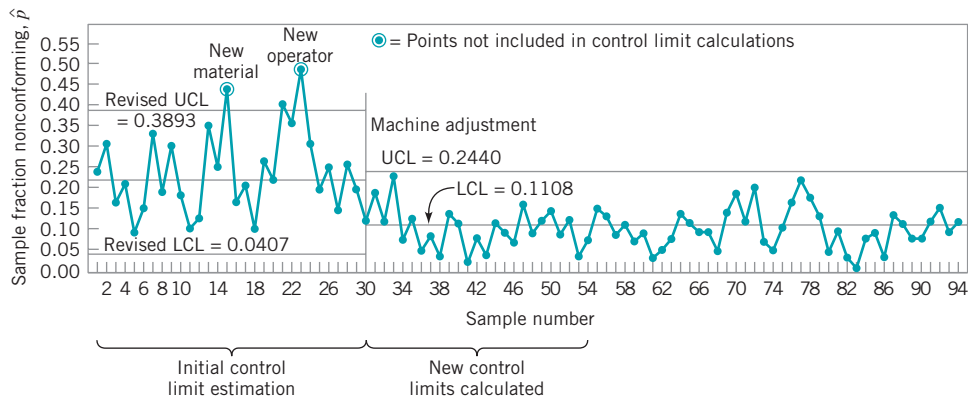
Figure 7.4 shows the control chart with these new parameters. Note that since the calculated lower control limit is less than zero, we have set $LCL = 0$. Therefore, the new control chart will have only an upper control limit. From inspection of Fig. 7.4, we see that all the points would fall inside the revised upper control limit; therefore, we conclude that the process is in control at this new level.

The continued operation of this control chart for the next five shifts is shown in Fig. 7.5. Data for the process during this period are shown in Table 7.3. The control chart does not indicate lack of control. Despite the improvement in yield following the engineering changes in the process and the introduction of the control chart, the process fallout of $\bar{p} = 0.1108$ is still too

high. Further analysis and action will be required to improve the yield. These management interventions may be further adjustments to the machine. **Statistically designed experiments** (see Part IV) are an appropriate way to determine which machine adjustments are critical to further process improvement, and the appropriate magnitude and direction of these adjustments. The control chart should be continued during the period in which the adjustments are made. By marking the time scale of the control chart *when* a process change is made, the control chart becomes a **logbook** in which the timing of process interventions and their subsequent effect on process performance are easily seen. This logbook aspect of control chart usage is extremely important.



■ **FIGURE 7.4** New control limits on the fraction nonconforming control chart, Example 7.1.



■ **FIGURE 7.5** Completed fraction nonconforming control chart, Example 7.1.

(continued)

■ **TABLE 7.3**
New Data for the Fraction Nonconforming Control Chart in Fig. 7.5, $n = 50$

Sample Number	Number of Nonconforming Cans, D_i	Sample Fraction Nonconforming, $\hat{p}_i$	Sample Number	Number of Nonconforming Cans, D_i	Sample Fraction Nonconforming, $\hat{p}_i$
55	8	0.16	75	5	0.10
56	7	0.14	76	8	0.16
57	5	0.10	77	11	0.22
58	6	0.12	78	9	0.18
59	4	0.08	79	7	0.14
60	5	0.10	80	3	0.06
61	2	0.04	81	5	0.10
62	3	0.06	82	2	0.04
63	4	0.08	83	1	0.02
64	7	0.14	84	4	0.08
65	6	0.12	85	5	0.10
66	5	0.10	86	3	0.06
67	5	0.10	87	7	0.14
68	3	0.06	88	6	0.12
69	7	0.14	89	4	0.08
70	9	0.18	90	4	0.08
71	6	0.12	91	6	0.12
72	10	0.20	92	8	0.16
73	4	0.08	93	5	0.10
74	3	0.06	94	6	0.12

Design of the Fraction Nonconforming Control Chart. The fraction nonconforming control chart has three parameters that must be specified: the sample size, the frequency of sampling, and the width of the control limits. Ideally, we should have some general guidelines for selecting those parameters.

It is relatively common to base a control chart for fraction nonconforming on 100% inspection of *all* process output over some convenient period of time, such as a shift or a day. In this case, both sample size and sampling frequency are interrelated. We would generally select a sampling frequency appropriate for the production rate, and this fixes the sample size. Rational subgrouping may also play a role in determining the sampling frequency. For example, if there are three shifts, and we suspect that shifts differ in their general quality level, then we should use the output of each shift as a subgroup rather than pooling the output of all three shifts together to obtain a daily fraction defective.

If we are to select a sample of process output, then we must choose the sample size n . Various rules have been suggested for the choice of n . If p is very small, we should choose n sufficiently large so that we have a high probability of finding at least one nonconforming unit in the sample. Otherwise, we might find that the control limits are such that the presence of only one nonconforming unit in the sample would indicate an out-of-control condition. For example, if $p = 0.01$ and $n = 8$, we find that the upper control limit is

$$\text{UCL} = p + 3\sqrt{\frac{p(1-p)}{n}} = 0.01 + 3\sqrt{\frac{(0.01)(0.99)}{8}} = 0.1155$$

If there is one nonconforming unit in the sample, then $\hat{p} = \frac{1}{8} = 0.1250$, and we can conclude that the process is out of control. Since for any $p > 0$ there is a positive probability

of producing *some* defectives, it is unreasonable in many cases to conclude that the process is out of control on observing a single nonconforming item.

To avoid this pitfall, we can choose the sample size n so that the probability of finding at least one nonconforming unit per sample is at least γ . For example, suppose that $p = 0.01$, and we want the probability of at least one nonconforming unit in the sample to be at least 0.95. If D denotes the number of nonconforming units, then we want to find n such that $P\{D \geq 1\} \geq 0.95$. Using the Poisson approximation to the binomial, we find from the cumulative Poisson table that $\lambda = np$ must exceed 3.00. Consequently, since $p = 0.01$, this implies that the sample size should be 300.

Duncan (1986) has suggested that the sample size should be large enough that we have approximately a 50% chance of detecting a process shift of some specified amount. For example, suppose that $p = 0.01$, and we want the probability of detecting a shift to $p = 0.05$ to be 0.50. Assuming that the normal approximation to the binomial applies, we should choose n so that the upper control limit exactly coincides with the fraction nonconforming in the out-of-control state.¹ If δ is the magnitude of the process shift, then n must satisfy

$$\delta = L\sqrt{\frac{p(1-p)}{n}} \quad (7.9)$$

Therefore,

$$n = \left(\frac{L}{\delta}\right)^2 p(1-p) \quad (7.10)$$

In our example, $p = 0.01$, $\delta = 0.05 - 0.01 = 0.04$, and if three-sigma limits are used, then from equation (7.10),

$$n = \left(\frac{3}{0.04}\right)^2 (0.01)(0.99) = 56$$

If the in-control value of the fraction nonconforming is small, another useful criterion is to choose n large enough so that the control chart will have a positive lower control limit. This ensures that we will have a mechanism to force us to investigate one or more samples that contain an unusually small number of nonconforming items. Since we wish to have

$$\text{LCL} = p - L\sqrt{\frac{p(1-p)}{n}} > 0 \quad (7.11)$$

this implies that

$$n > \frac{(1-p)}{p} L^2 \quad (7.12)$$

For example, if $p = 0.05$ and three-sigma limits are used, the sample size must be

$$n > \frac{0.95}{0.05} (3)^2 = 171$$

Thus, if $n \geq 172$ units, the control chart will have a positive lower control limit.

¹If $\hat{p}$ is approximately normal, then the probability that $\hat{p}$ exceeds the UCL is 0.50 if the UCL equals the out-of-control fraction nonconforming p , due to the symmetry of the normal distribution. See Section 3.4.3 for a discussion of the normal approximation to the binomial.

Three-sigma control limits are usually employed on the control chart for fraction nonconforming on the grounds that they have worked well in practice. As discussed in Section 5.3.2, narrower control limits would make the control chart more sensitive to small shifts in p but at the expense of more frequent “false alarms.” Occasionally, we have seen narrower limits used in an effort to *force* improvement in process quality. Care must be exercised in this, however, as too many false alarms will destroy the operating personnel’s confidence in the control chart program.

We should note that the fraction nonconforming control chart is not a universal model for *all* data on fraction nonconforming. It is based on the binomial probability model; that is, the probability of occurrence of a nonconforming unit is *constant*, and successive units of production are *independent*. In processes where nonconforming units are clustered together, or where the probability of a unit being nonconforming depends on whether or not previous units were nonconforming, the fraction nonconforming control chart is often of little use. In such cases, it is necessary to develop a control chart based on the correct probability model.

Interpretation of Points on the Control Chart for Fraction Nonconforming.

Example 7.1 illustrated how points that plot beyond the control limits are treated, both in establishing the control chart and during its routine operation. Care must be exercised in interpreting points that plot *below* the lower control limit. These points often do not represent a real improvement in process quality. Frequently, they are caused by errors in the inspection process resulting from inadequately trained or inexperienced inspectors or from improperly calibrated test and inspection equipment. We have also seen cases in which inspectors deliberately passed nonconforming units or reported fictitious data. The analyst must keep these warnings in mind when looking for assignable causes if points plot below the lower control limits. Not all downward shifts in p are attributable to improved quality.

The np Control Chart. It is also possible to base a control chart on the number nonconforming rather than the fraction nonconforming. This is often called an **number nonconforming (np) control chart**. The parameters of this chart are as follows.

The np Control Chart

$$\begin{aligned} \text{UCL} &= np + 3\sqrt{np(1-p)} \\ \text{Center line} &= np \\ \text{LCL} &= np - 3\sqrt{np(1-p)} \end{aligned} \quad (7.13)$$

If a standard value for p is unavailable, then $\bar{p}$ can be used to estimate p . Many nonstatistically trained personnel find the np chart easier to interpret than the usual fraction nonconforming control chart.

EXAMPLE 7.2 **An np Control Chart**

Set up an np control chart for the orange juice concentrate can process in Example 7.1.

Using the data in Table 7.1, we found that

$$\bar{p} = 0.2313 \quad n = 50$$

SOLUTION

Therefore, the parameters of the np control chart would be

$$\begin{aligned} \text{UCL} &= n\bar{p} + 3\sqrt{n\bar{p}(1-\bar{p})} \\ &= 50(0.2313) + 3\sqrt{(50)(0.2313)(0.7687)} \\ &= 20.510 \end{aligned}$$

$$\text{Center line} = n\bar{p} = 50(0.2313) = 11.565$$

$$\begin{aligned} \text{LCL} &= n\bar{p} - 3\sqrt{n\bar{p}(1-\bar{p})} \\ &= 50(0.2313) - 3\sqrt{(50)(0.2313)(0.7687)} \\ &= 2.620 \end{aligned}$$

Now in practice, the number of nonconforming units in each sample is plotted on the np control chart, and the number of nonconforming units is an integer. Thus, if 20 units are nonconforming the process is in control, but if 21 occur the process is out of control. Similarly, there are three nonconforming units in the sample and the process is in control, but two nonconforming units would imply an out-of-control process. Some practitioners prefer to use integer values for control limits on the np chart instead of their decimal fraction counterparts. In this example we could choose 2 and 21 as the LCL and UCL, respectively, and the process would be considered out of control if a sample value of np plotted at or beyond the control limits.

7.2.2 Variable Sample Size

In some applications of the control chart for fraction nonconforming, the sample is a 100% inspection of process output over some period of time. Since different numbers of units could be produced in each period, the control chart would then have a variable sample size. There are three approaches to constructing and operating a control chart with a variable sample size.

Variable-Width Control Limits. The first and perhaps the most simple approach is to determine control limits for each individual sample that are based on the specific sample size. That is, if the i th sample is of size n_i , then the upper and lower control limits are $\bar{p} \pm 3\sqrt{\bar{p}(1-\bar{p})/n_i}$. Note that the width of the control limits is inversely proportional to the square root of the sample size.

To illustrate this approach, consider the data in Table 7.4. These data came from the purchasing group of a large aerospace company. This group issues purchase orders to the company's suppliers. The sample sizes in Table 7.4 are the total number of purchase orders issued each week. Obviously, this is not constant. A nonconforming unit is a purchase order with an error. Among the most common errors are specifying incorrect part numbers, wrong delivery dates, and wrong supplier information. Any of these mistakes can result in a purchase order change, which takes time and resources and may result in delayed delivery of material.

For the 25 samples, we calculate

$$\bar{p} = \frac{\sum_{i=1}^{25} D_i}{\sum_{i=1}^{25} n_i} = \frac{234}{2450} = 0.096$$

Consequently, the center line is at 0.096, and the control limits are

$$\text{UCL} = \bar{p} + 3\hat{\sigma}_{\hat{p}} = 0.096 + 3\sqrt{\frac{(0.096)(0.904)}{n_i}}$$

and

$$\text{LCL} = \bar{p} - 3\hat{\sigma}_{\hat{p}} = 0.096 - 3\sqrt{\frac{(0.096)(0.904)}{n_i}}$$

■ TABLE 7.4

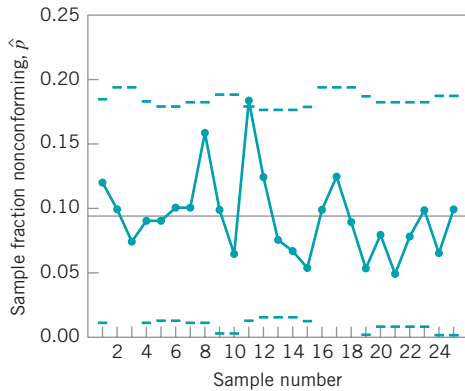
Purchase Order Data for a Control Chart for Fraction Nonconforming with Variable Sample Size

Sample Number, i	Sample Size, n_i	Number of Nonconforming Units, D_i	Sample Fraction Nonconforming, $\hat{p}_i = D_i/n_i$	Standard Deviation	Control Limits	
				$\hat{\sigma}_{\hat{p}} = \sqrt{\frac{(0.096)(0.904)}{n_i}}$	LCL	UCL
1	100	12	0.120	0.029	0.009	0.183
2	80	8	0.100	0.033	0	0.195
3	80	6	0.075	0.033	0	0.195
4	100	9	0.090	0.029	0.009	0.183
5	110	10	0.091	0.028	0.012	0.180
6	110	12	0.109	0.028	0.012	0.180
7	100	11	0.110	0.029	0.009	0.183
8	100	16	0.160	0.029	0.009	0.183
9	90	10	0.110	0.031	0.003	0.189
10	90	6	0.067	0.031	0.003	0.189
11	110	20	0.182	0.028	0.012	0.180
12	120	15	0.125	0.027	0.015	0.177
13	120	9	0.075	0.027	0.015	0.177
14	120	8	0.067	0.027	0.015	0.177
15	110	6	0.055	0.028	0.012	0.180
16	80	8	0.100	0.033	0	0.195
17	80	10	0.125	0.033	0	0.195
18	80	7	0.088	0.033	0	0.195
19	90	5	0.056	0.031	0.003	0.189
20	100	8	0.080	0.029	0.009	0.183
21	100	5	0.050	0.029	0.009	0.183
22	100	8	0.080	0.029	0.009	0.183
23	100	10	0.100	0.029	0.009	0.183
24	90	6	0.067	0.031	0.003	0.189
25	90	9	0.100	0.031	0.003	0.189
	2450	234	2.383			

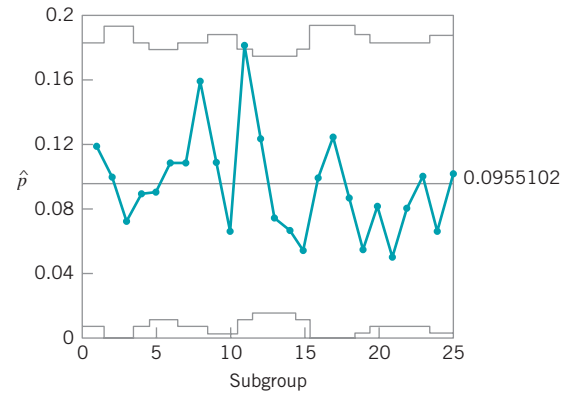
where $\hat{\sigma}_{\hat{p}}$ is the estimate of the standard deviation of the sample fraction nonconforming $\hat{p}$. The calculations to determine the control limits are displayed in the last three columns of Table 7.4. The manually constructed control chart is plotted in Fig. 7.6.

Many popular quality control computer programs will handle the variable sample size case. Figure 7.7 presents the computer-generated control chart corresponding to Fig. 7.6. This control chart was obtained using Minitab.

Control Limits Based on an Average Sample Size. The second approach is to base the control chart on an *average sample size*, resulting in an **approximate set of control limits**. This assumes that future sample sizes will not differ greatly from those previously observed. If this approach is used, the control limits will be *constant*, and the resulting control chart will not look as formidable to operating personnel as the control chart with variable limits. However, if there is an unusually large variation in the size of a particular sample or if a point plots near the approximate control limits, then the *exact* control limits for that point



■ **FIGURE 7.6** Control chart for fraction nonconforming with variable sample size.



■ **FIGURE 7.7** Control chart for fraction nonconforming with variable sample size using Minitab.

should be determined and the point examined relative to that value. For the purchase order data in Table 7.4, we find that the average sample size is

$$\bar{n} = \frac{\sum_{i=1}^{25} n_i}{25} = \frac{2450}{25} = 98$$

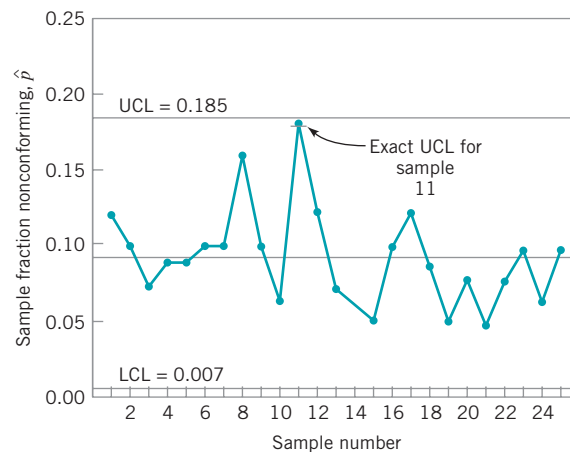
Therefore, the approximate control limits are

$$UCL = \bar{p} + 3\sqrt{\frac{\bar{p}(1-\bar{p})}{\bar{n}}} = 0.096 + 3\sqrt{\frac{(0.096)(0.904)}{98}} = 0.185$$

and

$$LCL = \bar{p} - 3\sqrt{\frac{\bar{p}(1-\bar{p})}{\bar{n}}} = 0.096 - 3\sqrt{\frac{(0.096)(0.904)}{98}} = 0.007$$

The resulting control chart is shown in Fig. 7.8. Note that $\hat{p}$ for sample 11 plots close to the approximate upper control limit, yet it appears to be in control. However, when compared to



■ **FIGURE 7.8** Control chart for fraction nonconforming based on average sample size.

its exact upper control limit (0.180, from Table 7.4), the point indicates an out-of-control condition. Similarly, points that are outside the approximate control limits may be inside their exact control limits. In general, care must be taken in the interpretation of points near the approximate control limits.

We must also be careful in analyzing runs or other apparently abnormal patterns on control charts with variable sample sizes. The problem is that a change in the sample fraction nonconforming $\hat{p}$ must be interpreted relative to the sample size. For example, suppose that $p = 0.20$ and that two successive sample fractions nonconforming are $\hat{p}_i = 0.28$ and $\hat{p}_{i+1} = 0.24$. The first observation seems to indicate poorer quality than the second, since $\hat{p}_i > \hat{p}_{i+1}$. However, suppose that the sample sizes are $n_i = 50$ and $n_{i+1} = 250$. In *standard deviation units*, the first point is 1.41 units above average and the second point is 1.58 units above average. That is, the second point actually represents a *greater* deviation from the standard of $p = 0.20$ than does the first, even though the second point is the smaller of the two. Clearly, looking for runs or other nonrandom patterns is virtually meaningless here.

The Standardized Control Chart. The third approach to dealing with variable sample size is to use a **standardized control chart**, where the points are plotted in standard deviation units. Such a control chart has the center line at zero, and upper and lower control limits of +3 and -3, respectively. The variable plotted on the chart is

$$Z_i = \frac{\hat{p}_i - p}{\sqrt{\frac{p(1-p)}{n_i}}} \quad (7.14)$$

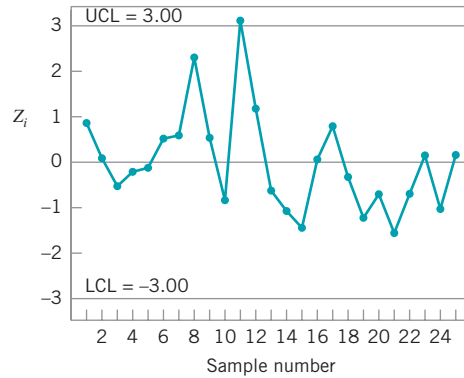
where p (or $\bar{p}$ if no standard is given) is the process fraction nonconforming in the in-control state. The standardized control chart for the purchase order data in Table 7.4 is shown in Fig. 7.9. The calculations associated with this control chart are shown in Table 7.5. Tests for runs and pattern-recognition methods could safely be applied to this chart, because the relative changes from one point to another are all expressed in terms of the same units of measurement.

The standardized control chart is no more difficult to construct or maintain than either of the other two procedures discussed in this section. In fact, many quality control software packages either automatically execute this as a standard feature or can be programmed to plot a standardized control chart. For example, the version of Fig. 7.9 shown in Fig. 7.10 was created using Minitab. Conceptually, however, it may be more difficult for operating personnel to understand and interpret, because reference to the actual process fraction defective has been “lost.” However, if there is large variation in sample size, then runs and pattern-recognition methods can only be safely applied to the standardized control chart. In such a case, it might be advisable to maintain a control chart with individual control limits for each sample (as in Fig. 7.6) for the operating personnel, while simultaneously maintaining a standardized control chart for engineering use.

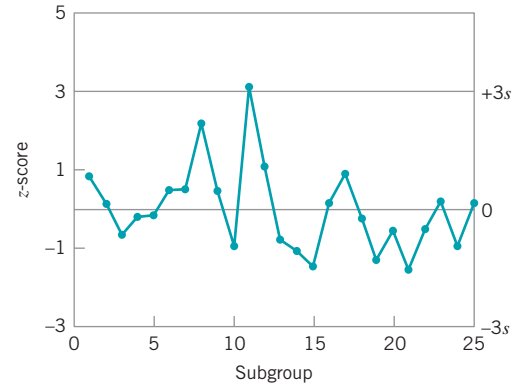
The standardized control chart is also recommended when the length of the production run is short, as in many job-shop settings. Control charts for short production runs are discussed in Chapter 9.

7.2.3 Applications in Transactional and Service Businesses

The control chart for fraction nonconforming is widely used in transactional and service industry applications of statistical process control. In the nonmanufacturing environment,



■ **FIGURE 7.9** Standardized control chart for fraction nonconforming.



■ **FIGURE 7.10** Standardized control chart from Minitab for fraction nonconforming, Table 7.4.

■ **TABLE 7.5**

Calculations for the Standardized Control Chart in Fig. 7.9, $\hat{p} = 0.096$

Sample Number, i	Sample Size, n_i	Number of Nonconforming Units, D_i	Sample Fraction Nonconforming, $\hat{p}_i = D_i/n_i$	Standard Deviation $\hat{\sigma}_p = \sqrt{\frac{(0.096)(0.904)}{n_i}}$	$z_i = \frac{\hat{p}_i - \bar{p}}{\sqrt{\frac{(0.096)(0.904)}{n_i}}}$
1	100	12	0.120	0.029	0.83
2	80	8	0.100	0.033	0.12
3	80	6	0.075	0.033	-0.64
4	100	9	0.090	0.029	-0.21
5	110	10	0.091	0.028	-0.18
6	110	12	0.109	0.028	0.46
7	100	11	0.110	0.029	0.48
8	100	16	0.160	0.029	2.21
9	90	10	0.110	0.031	0.45
10	90	6	0.067	0.031	-0.94
11	110	20	0.182	0.028	3.07
12	120	15	0.125	0.027	1.07
13	120	9	0.075	0.027	-0.78
14	120	8	0.067	0.027	-1.07
15	110	6	0.055	0.028	-1.46
16	80	8	0.100	0.033	0.12
17	80	10	0.125	0.033	0.88
18	80	7	0.088	0.033	-0.24
19	90	5	0.056	0.031	-1.29
20	100	8	0.080	0.029	-0.55
21	100	5	0.050	0.029	-1.59
22	100	8	0.080	0.029	-0.55
23	100	10	0.100	0.029	0.14
24	90	6	0.067	0.031	-0.94
25	90	9	0.100	0.031	0.13

many quality characteristics can be observed on a conforming or nonconforming basis. Examples would include the number of employee paychecks that are in error or distributed late during a pay period, the number of check requests that are not paid within the standard accounting cycle, and the number of deliveries made by a supplier that are not on time.

Many nonmanufacturing applications of the fraction nonconforming control chart will involve the variable sample size case. For example, the total number of check requests during an accounting cycle is most likely not constant, and since information about the timeliness of processing for all check requests is generally available, we would calculate $\hat{p}$ as the ratio of all late checks to the total number of checks processed during the period.

As an illustration consider the purchase order data in Table 7.4. The sample sizes in Table 7.4 are the actual number of purchase orders issued each week. It would be very unusual for this to be exactly the same from week to week. Consequently, a fraction nonconforming control chart with variable sample size was an ideal approach for this situation. The use of this control chart was a key initial step in identifying many of the root causes of the errors on purchase orders and in developing the corrective actions necessary to improve the process.

7.2.4 The Operating-Characteristic Function and Average Run Length Calculations

The operating-characteristic (or OC) function of the fraction nonconforming control chart is a graphical display of the probability of incorrectly accepting the hypothesis of statistical control (i.e., a type II or β -error) against the process fraction nonconforming. The OC curve provides a measure of the **sensitivity** of the control chart—that is, its ability to detect a shift in the process fraction nonconforming from the nominal value $\bar{p}$ to some other value p . The probability of type II error for the fraction nonconforming control chart may be computed from

$$\begin{aligned}\beta &= P\{\hat{p} < \text{UCL}|p\} - P\{\hat{p} \leq \text{LCL}|p\} \\ &= P\{D < n\text{UCL}|p\} - P\{D \leq n\text{LCL}|p\}\end{aligned}\quad (7.15)$$

Since D is a binomial random variable with parameters n and p , the β -error defined in equation (7.15) can be obtained from the cumulative binomial distribution. Note that when the LCL is negative, the second term on the right-hand side of equation (7.15) should be dropped.

Table 7.6 illustrates the calculations required to generate the OC curve for a control chart for fraction nonconforming with parameters $n = 50$, $\text{LCL} = 0.0303$, and $\text{UCL} = 0.3697$. Using these parameters, equation (7.15) becomes

$$\begin{aligned}\beta &= P\{D < (50)(0.3697)|p\} - P\{D \leq (50)(0.0303)|p\} \\ &= P\{D < 18.49|p\} - P\{D \leq 1.52|p\}\end{aligned}$$

However, since D must be an integer, we find that

$$\beta = P\{D \leq 18|p\} - P\{D \leq 1|p\}$$

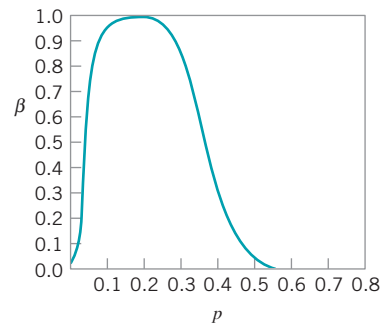
The OC curve is plotted in Fig. 7.11.

■ TABLE 7.6

Calculations^a for Constructing the OC Curve for a Control Chart for Fraction Nonconforming with $n = 50$, $LCL = 0.0303$, and $UCL = 0.3697$

p	$P\{D \leq 18 p\}$	$P\{D \leq 1 p\}$	$\beta = P\{D \leq 18 p\} - P\{D \leq 1 p\}$
0.01	1.0000	0.9106	0.0894
0.03	1.0000	0.5553	0.4447
0.05	1.0000	0.2794	0.7206
0.10	1.0000	0.0338	0.9662
0.15	0.9999	0.0029	0.9970
0.20	0.9975	0.0002	0.9973
0.25	0.9713	0.0000	0.9713
0.30	0.8594	0.0000	0.8594
0.35	0.6216	0.0000	0.6216
0.40	0.3356	0.0000	0.3356
0.45	0.1273	0.0000	0.1273
0.50	0.0325	0.0000	0.0325
0.55	0.0053	0.0000	0.0053

^aThe probabilities in this table were found by evaluating the cumulative binomial distribution. For small p ($p < 0.1$, say) the Poisson approximation could be used, and for larger values of p the normal approximation could be used.



■ FIGURE 7.11 Operating-characteristic curve for the fraction nonconforming control chart with $\bar{p} = 0.20$, $LCL = 0.0303$, and $UCL = 0.3697$.

We may also calculate average run lengths (ARLs) for the fraction nonconforming control chart. Recall from Chapter 5 that for uncorrelated process data the ARL for any Shewhart control chart can be written as

$$ARL = \frac{1}{P(\text{sample point plots out of control})}$$

Thus, if the process is in control, ARL_0 is

$$ARL_0 = \frac{1}{\alpha}$$

and if it is out of control, then

$$ARL_1 = \frac{1}{1 - \beta}$$

These probabilities (α , β) can be calculated directly from the binomial distribution or read from an OC curve.

To illustrate, consider the control chart for fraction nonconforming used in the OC curve calculations in Table 7.6. This chart has parameters $n = 50$, $UCL = 0.3697$, $LCL = 0.0303$, and the center line is $\bar{p} = 0.20$. From Table 7.6 (or the OC curve in Fig. 7.11) we find that if the process is in control with $p = \bar{p}$, the probability of a point plotting in control is 0.9973. Thus, in this case $\alpha = 1 - \beta = 0.0027$, and the value of ARL_0 is

$$ARL_0 = \frac{1}{\alpha} = \frac{1}{0.0027} \approx 370$$

Therefore, if the process is really in control, we will experience a false out-of-control signal about every 370 samples. (This will be approximately true, in general, for any Shewhart control chart with three-sigma limits.) This in-control ARL_0 is generally considered to be satisfactorily large. Now suppose that the process shifts out of control to $p = 0.3$. Table 7.6 indicates that if $p = 0.3$, then $\beta = 0.8594$. Therefore, the value of ARL_1 is

$$ARL_1 = \frac{1}{1 - \beta} = \frac{1}{1 - 0.8594} \approx 7$$

and it will take about seven samples, on the average, to detect this shift with a point outside of the control limits. If this is unsatisfactory, then action must be taken to reduce the out-of-control ARL_1 . Increasing the sample size would result in a smaller value of β and a shorter out-of-control ARL_1 . Another approach would be to reduce the interval *between* samples. That is, if we are currently sampling every hour, it will take about seven hours, on the average, to detect the shift. If we take the sample every half hour, it will require only three and a half hours, on the average, to detect the shift. Another approach is to use a control chart that is more responsive to small shifts, such as the cumulative sum charts in Chapter 9.

7.3 Control Charts for Nonconformities (Defects)

A nonconforming item is a unit of product that does not satisfy one or more of the specifications for that product. Each specific point at which a specification is not satisfied results in a **defect** or **nonconformity**. Consequently, a nonconforming item will contain at least one nonconformity. However, depending on their nature and severity, it is quite possible for a unit to contain several nonconformities and *not* be classified as nonconforming. As an example, suppose we are manufacturing personal computers. Each unit could have one or more very minor flaws in the cabinet finish, and since these flaws do not seriously affect the unit's functional operation, it could be classified as conforming. However, if there are too many of these flaws, the personal computer should be classified as nonconforming, since the flaws would be very noticeable to the customer and might affect the sale of the unit. There are many practical situations in which we prefer to work directly with the number of defects or nonconformities rather than the fraction nonconforming. These include the number of defective welds in 100 m of oil pipeline, the number of broken rivets in an aircraft wing, the number of functional defects in an electronic logic device, the number of errors on a document, and so forth.

It is possible to develop control charts for either the **total number of nonconformities** in a unit or the **average number of nonconformities per unit**. These control charts usually assume that the occurrence of nonconformities in samples of constant size is well modeled by the Poisson distribution. Essentially, this requires that the number of opportunities or potential locations for nonconformities be infinitely large and that the probability of occurrence of a nonconformity at any location be small and constant. Furthermore, the **inspection unit** must be the same for each sample. That is, each inspection unit must always represent an identical **area of opportunity** for the occurrence of nonconformities. In addition, we can count nonconformities of several different types on one unit, as long as the above conditions are satisfied for each class of nonconformity.

In most practical situations, these conditions will not be satisfied exactly. The number of opportunities for the occurrence of nonconformities may be finite, or the probability of occurrence of nonconformities may not be constant. As long as these departures from the assumptions are not severe, the Poisson model will usually work reasonably well. There are cases, however, in which the Poisson model is completely inappropriate. These situations are discussed in more detail at the end of Section 7.3.1.

7.3.1 Procedures with Constant Sample Size

Consider the occurrence of nonconformities in an inspection unit of product. In most cases, the inspection unit will be a single unit of product, although this is not necessarily always so. The inspection unit is simply an entity for which it is convenient to keep records. It could be a group of 5 units of product, 10 units of product, and so on. Suppose that defects or nonconformities occur in this inspection unit according to the Poisson distribution; that is,

$$p(x) = \frac{e^{-c} c^x}{x!} \quad x = 0, 1, 2, \dots$$

where x is the number of nonconformities and $c > 0$ is the parameter of the Poisson distribution. From Section 3.2.3 we recall that both the mean and variance of the Poisson distribution are the parameter c . Therefore, a **control chart for nonconformities, or c chart** with three-sigma limits would be defined as follows,²

Control Chart for Nonconformities: Standard Given

$$\begin{aligned} \text{UCL} &= c + 3\sqrt{c} \\ \text{Center line} &= c \\ \text{LCL} &= c - 3\sqrt{c} \end{aligned} \tag{7.16}$$

assuming that a standard value for c is available. Should these calculations yield a negative value for the LCL, set $\text{LCL} = 0$.

If no standard is given, then c may be estimated as the observed average number of nonconformities in a preliminary sample of inspection units—say, $\bar{c}$. In this case, the control chart has parameters defined as follows.

²The α -risk for three-sigma limits is not equally allocated above the UCL and below the LCL, because the Poisson distribution is asymmetric. Some authors recommend the use of probability limits for this chart, particularly when c is small.

Control Chart for Nonconformities: No Standard Given

$$\begin{aligned}
 \text{UCL} &= \bar{c} + 3\sqrt{\bar{c}} \\
 \text{Center line} &= \bar{c} \\
 \text{LCL} &= \bar{c} - 3\sqrt{\bar{c}}
 \end{aligned}
 \tag{7.17}$$

When no standard is given, the control limits in equation (7.17) should be regarded as *trial* control limits, and the preliminary samples examined for lack of control in the usual phase I analysis. The control chart for nonconformities is also sometimes called the *c* chart.

EXAMPLE 7.3

Nonconformities in Printed Circuit Boards

Table 7.7 presents the number of nonconformities observed in 26 successive samples of 100 printed circuit boards. Note that, for reasons of convenience, the inspection unit is defined as 100 boards. Set up a *c* chart for these data.

TABLE 7.7
Data on the Number of Nonconformities in Samples of 100 Printed Circuit Boards

Sample Number	Number of Nonconformities	Sample Number	Number of Nonconformities
1	21	14	19
2	24	15	10
3	16	16	17
4	12	17	13
5	15	18	22
6	5	19	18
7	28	20	39
8	20	21	30
9	31	22	24
10	25	23	16
11	20	24	19
12	24	25	17
13	16	26	15

SOLUTION

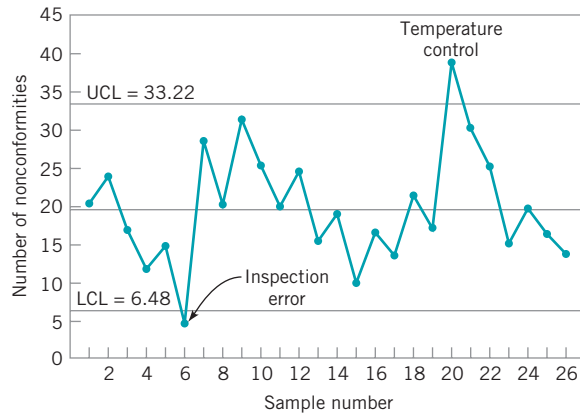
Since the 26 samples contain 516 total nonconformities, we estimate *c* by

$$\bar{c} = \frac{516}{26} = 19.85$$

Therefore, the trial control limits are given by

$$\begin{aligned}
 \text{UCL} &= \bar{c} + 3\sqrt{\bar{c}} = 19.85 + 3\sqrt{19.85} = 33.22 \\
 \text{Center line} &= \bar{c} = 19.85 \\
 \text{LCL} &= \bar{c} - 3\sqrt{\bar{c}} = 19.85 - 3\sqrt{19.85} = 6.48
 \end{aligned}$$

The control chart is shown in Fig. 7.12. The number of observed nonconformities from the preliminary samples is plotted on this chart. Two points plot outside the control limits, samples 6 and 20. Investigation of sample 6 revealed that a new inspector had examined the boards in this sample and that he did not recognize several of the types of nonconformities that could have been present. Furthermore, the unusually large number of nonconformities in sample 20 resulted from a temperature control problem in the wave soldering machine, which was subsequently repaired. Therefore, it seems reasonable to exclude these two



■ **FIGURE 7.12** Control chart for nonconformities for Example 7.3.

samples and revise the trial control limits. The estimate of c is now computed as

$$\bar{c} = \frac{472}{24} = 19.67$$

and the revised control limits are

$$UCL = \bar{c} + 3\sqrt{\bar{c}} = 19.67 + 3\sqrt{19.67} = 32.97$$

$$\text{Center line} = \bar{c} = 19.67$$

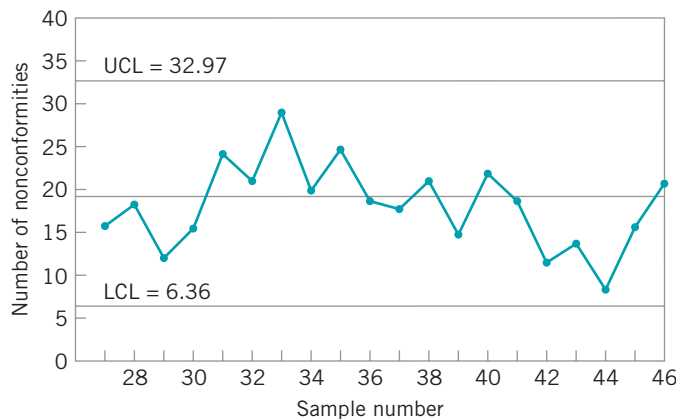
$$LCL = \bar{c} - 3\sqrt{\bar{c}} = 19.67 - 3\sqrt{19.67} = 6.36$$

These become the standard values against which production in the next period can be compared.

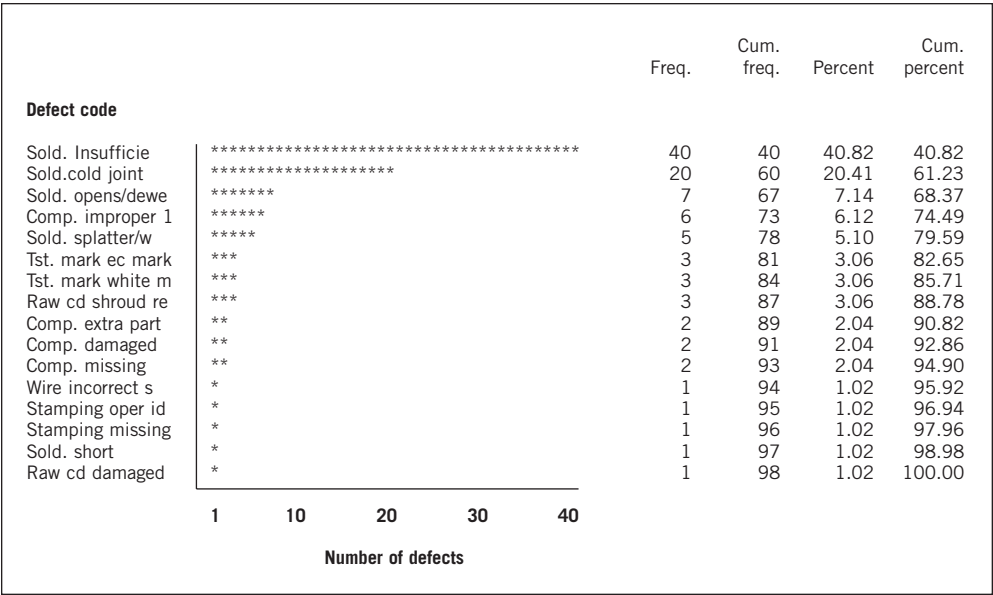
Twenty new samples, each consisting of one inspection unit (i.e., 100 boards), are subsequently collected. The number of nonconformities in each sample is noted and recorded in Table 7.8. These points are plotted on the control chart in Fig. 7.13. No lack of control is indicated; however, the number of nonconformities per board is still unacceptably high. Further action is necessary to improve the process.

■ **TABLE 7.8**
Additional Data for the Control Chart for Nonconformities, Example 7.3

Sample Number	Number of Nonconformities	Sample Number	Number of Nonconformities
27	16	37	18
28	18	38	21
29	12	39	16
30	15	40	22
31	24	41	19
32	21	42	12
33	28	43	14
34	20	44	9
35	25	45	16
36	19	46	21



■ **FIGURE 7.13** Continuation of the control chart for nonconformities. Example 7.3.



■ FIGURE 7.14 Pareto analysis of nonconformities for the printed circuit board process.

Further Analysis of Nonconformities. Defect or nonconformity data are always more informative than fraction nonconforming, because there will usually be several different *types* of nonconformities. By analyzing the nonconformities by type, we can often gain considerable insight into their cause. This can be of considerable assistance in developing the **out-of-control-action plans** (OCAPs) that must accompany control charts.

For example, in the printed circuit board process, there are sixteen different types of defects. Defect data for 500 boards are plotted on a Pareto chart in Fig. 7.14. Note that over 60% of the total number of defects is due to *two defect types*: solder insufficiency and solder cold joints. This points to further problems with the wave soldering process. If these problems can be isolated and eliminated, there will be a dramatic increase in process yield. Notice that the nonconformities follow the Pareto distribution; that is, most of the defects are attributable to a few (in this case, two) defect types.

This process manufactures several different types of printed circuit boards. Therefore, it may be helpful to examine the occurrence of defect type by type of printed circuit board (part number). Table 7.9 presents this information. Note that all 40 solder insufficiencies and all 20 solder cold joints occurred on the same part number, 0001285. This implies that this particular type of board is very susceptible to problems in wave soldering, and special attention must be directed toward improving this step of the process for this part number.

Another useful technique for further analysis of nonconformities is the **cause-and-effect diagram** discussed in Chapter 5. The cause-and-effect diagram is used to illustrate the various sources of nonconformities in products and their interrelationships. It is useful in focusing the attention of operators, manufacturing engineers, and managers on quality problems. Developing a good cause-and-effect diagram usually advances the level of technological understanding of the process.

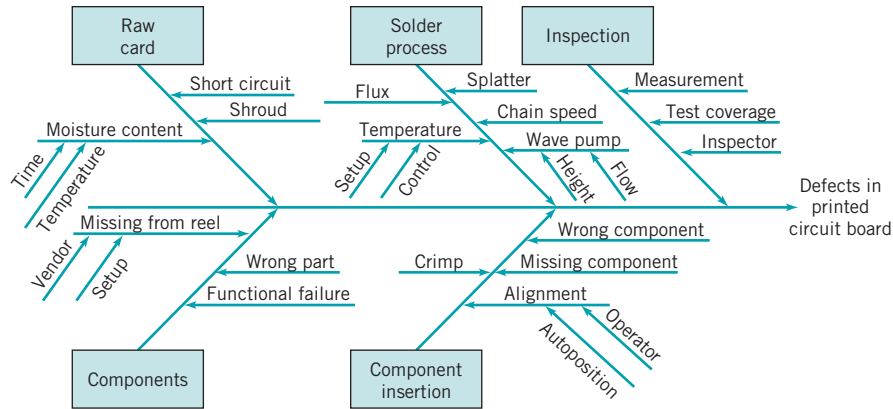
A cause-and-effect diagram for the printed circuit board assembly process is shown in Fig. 7.15. Since most of the defects in this example were solder-related, the cause-and-effect diagram could help choose the variables for a designed experiment to optimize the wave

■ TABLE 7.9

Table of Defects Classified by Part Number and Defect Code

Part Number Frequency Percent Row Percent Column Percent	Defect								
	Component	Damaged (NO)	Code	Component Improper I	Raw	Card Shroud RE	Card Damaged	Solder	Cold Joint
	Component Missing		Component Extra Part		Card			Short	
0001285	1	0	0	0	0	1	0	5	20
	1.02	0.00	0.00	0.00	0.00	1.02	0.00	5.10	20.41
	1.41	0.00	0.00	0.00	0.00	1.41	0.00	7.04	28.17
	50.00	0.00	0.00	0.00	0.00	100.00	0.00	71.43	100.00
0001481	1	2	2	6	3	0	1	2	0
	1.02	2.04	6.12	3.06	0.00	0.00	1.02	2.04	0.00
	3.70	7.41	22.22	11.11	0.00	0.00	3.70	7.41	0.00
	50.00	100.00	100.00	100.00	100.00	0.00	100.00	28.57	0.00
0006429	0	0	0	0	0	0	0	0	0
	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Total	2	2	2	6	3	1	1	7	20
	2.04	2.04	2.04	6.12	3.06	1.02	1.02	7.14	20.41

Part Number Frequency Percent Row Percent Column Percent	Defect								
	Solder	Solder	Code	Test	Test	Wire	Incorrect	Good	Total
	Insufficiencies	Spatter	Stamping Missing	Stamping Operator ID	Mark White M	Mark EC Mark		Unit(s)	
0001285	40	0	0	0	2	1	1	0	71
	40.82	0.00	0.00	0.00	2.04	1.02	1.02	0.00	72.45
	56.32	0.00	0.00	0.00	2.82	1.41	1.41	0.00	
	100.00	0.00	0.00	0.00	66.67	33.33	100.00	0.00	
0001481	0	5	1	1	1	2	0	0	27
	0.00	5.10	1.02	1.02	1.02	2.04	0.00	0.00	27.55
	0.00	18.52	3.70	3.70	3.70	7.41	0.00	0.00	
	0.00	100.00	100.00	100.00	33.33	66.67	0.00	0.00	
0006429	0	0	0	0	0	0	0	0	0
	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	
	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	
Total	40	5	1	1	3	3	1	0	98
	40.82	5.10	1.02	1.02	3.06	3.06	1.02	0.00	100.00



■ FIGURE 7.15 Cause-and-effect diagram.

soldering process. There are several ways to draw the diagram. This one focuses on the three main generic sources of nonconformities: materials, operators, and equipment. Another useful approach is to organize the diagram according to the flow of material through the process.

Choice of Sample Size: The *u* Chart. Example 7.3 illustrates a control chart for nonconformities with the sample size exactly equal to one inspection unit. The inspection unit is chosen for operational or data-collection simplicity. However, there is no reason why the sample size must be restricted to one inspection unit. In fact, we would often prefer to use *several* inspection units in the sample, thereby increasing the area of opportunity for the occurrence of nonconformities. The sample size should be chosen according to statistical considerations, such as specifying a sample size large enough to ensure a positive lower control limit or to obtain a particular probability of detecting a process shift. Alternatively, economic factors could enter into sample-size determination.

Suppose we decide to base the control chart on a sample size of n inspection units. Note that n does not have to be an integer. To illustrate this, suppose that in Example 7.3 we were to specify a subgroup size of $n = 2.5$ inspection units. Then the sample size becomes $(2.5)(100) = 250$ boards. There are two general approaches to constructing the revised chart once a new sample size has been selected. One approach is simply to redefine a new inspection unit that is equal to n times the old inspection unit. In this case, the center line on the new control chart is $n\bar{c}$ and the control limits are located at $n\bar{c} \pm 3\sqrt{n\bar{c}}$, where $\bar{c}$ is the observed mean number of nonconformities in the *original* inspection unit. Suppose that in Example 7.3, after revising the trial control limits, we decided to use a sample size of $n = 2.5$ inspection units. Then the center line would have been located at $n\bar{c} = (2.5)(19.67) = 49.18$ and the control limits would have been $49.18 \pm 3\sqrt{49.18}$ or LCL = 28.14 and UCL = 70.22.

The second approach involves setting up a control chart based on the average number of nonconformities per inspection unit. If we find x *total* nonconformities in a sample of n inspection units, then the *average* number of nonconformities per inspection unit is

$$u = \frac{x}{n} \quad (7.18)$$

Note that x is a Poisson random variable; consequently, the parameters of the control chart for the average number of nonconformities per unit are as follows,

Control Chart for Average Number of Nonconformities per Unit

$$\begin{aligned} \text{UCL} &= \bar{u} + 3\sqrt{\frac{\bar{u}}{n}} \\ \text{Center line} &= \bar{u} \\ \text{LCL} &= \bar{u} - 3\sqrt{\frac{\bar{u}}{n}} \end{aligned} \quad (7.19)$$

where $\bar{u}$ represents the observed average number of nonconformities per unit in a preliminary set of data. Control limits found from equation (7.19) would be regarded as trial control limits. This per-unit chart often is called the **control chart for nonconformities, or u chart**.

EXAMPLE 7.4 Control Charts in Supply Chain Operations

A supply chain engineering group monitors shipments of materials through the company distribution network. Errors on either the delivered material or the accompanying documentation are tracked on a weekly basis. Fifty randomly selected

shipments are examined and the errors recorded. Data for twenty weeks are shown in Table 7.10. Set up a u control chart to monitor this process.

SOLUTION

From the data in Table 7.10, we estimate the number of errors (nonconformities) per unit (shipment) to be:

$$\bar{u} = \frac{\sum_{i=1}^{20} u_i}{20} = \frac{1.48}{20} = 0.0740$$

Therefore, the parameters of the control chart are

$$\text{UCL} = \bar{u} + 3\sqrt{\frac{\bar{u}}{n}} = 0.0740 + 3\sqrt{\frac{0.0740}{50}} = 0.1894$$

$$\text{Center line} = \bar{u} = 1.93$$

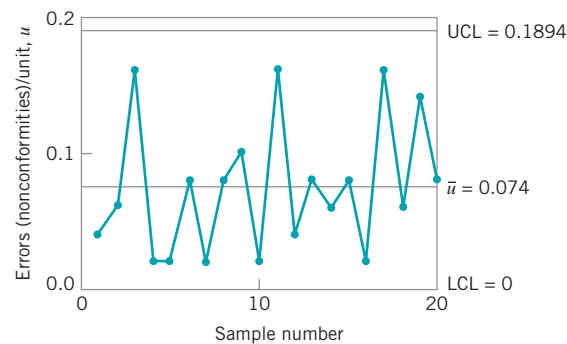
$$\text{LCL} = \bar{u} - 3\sqrt{\frac{\bar{u}}{n}} = 0.0740 - 3\sqrt{\frac{0.0740}{50}} = -0.0414$$

Since the $\text{LCL} < 0$, we would set $\text{LCL} = 0$ for the u chart. The control chart is shown in Fig. 7.16. The preliminary data do not exhibit lack of statistical control; therefore, the trial control limits given here would be adopted for phase II monitoring of future operations. Once again, note that, although the process is in control, the average number of errors per shipment is high. Action should be taken to improve the supply chain system.

(continued)

■ **TABLE 7.10**
Data on Number of Shipping Errors in a Supply Chain Network

Sample Number (week), i	Sample Size, n	Total Number of Errors (Nonconformities), x_i	Average Number of Errors (Nonconformities) per Unit, $u_i = x_i/n$
1	50	2	0.04
2	50	3	0.06
3	50	8	0.16
4	50	1	0.02
5	50	1	0.02
6	50	4	0.08
7	50	1	0.02
8	50	4	0.08
9	50	5	0.10
10	50	1	0.02
11	50	8	0.16
12	50	2	0.04
13	50	4	0.08
14	50	3	0.06
15	50	4	0.08
16	50	1	0.02
17	50	8	0.16
18	50	3	0.06
19	50	7	0.14
20	50	4	0.08
		74	1.48



■ **FIGURE 7.16** The control chart for nonconformities per unit from Minitab for Example 7.4.

Alternative Probability Models for Count Data. Most applications of the c chart assume that the Poisson distribution is the correct probability model underlying the process. However, it is not the only distribution that could be utilized as a model of “count” or nonconformities per unit-type data. Various types of phenomena can produce distributions of defects that are not well modeled by the Poisson distribution. In the Poisson

distribution, the mean and the variance are equal. When the sample data indicate that the sample variance is substantially different from the mean, the Poisson assumption is likely to be inappropriate.

The situation where the Poisson assumption is likely to be inappropriate is when nonconformities tend to occur in **clusters**; that is, if there is one nonconformity in some part of a product, then it is likely that there will be others. Note that there are at least two random processes at work here: one generating the number and location of clusters, and the second generating the number of nonconformities within each cluster. If the number of clusters has a Poisson distribution and the number of nonconformities within each cluster has a common distribution (say, f), then the total number of nonconformities has a **compound Poisson distribution**. Many types of compound or generalized distributions could be used as a model for count-type data. As an illustration, if the number of clusters has a Poisson distribution and the number of nonconformities within each cluster is also Poisson, then Neyman's type-A distribution models the total number of nonconformities. Alternatively, if the cluster distribution is gamma and the number of nonconformities within each cluster is Poisson, the negative binomial distribution results. Johnson and Kotz (1969) give a good summary of these and also discuss other discrete distributions that could be useful in modeling count-type data.

Mixtures of various types of nonconformities can lead to situations in which the total number of nonconformities is not adequately modeled by the Poisson distribution. Similar situations occur when the count data have either too many or too few zeros. A good discussion of this general problem is the paper by Jackson (1972). The use of the negative binomial distribution to model count data in inspection units of varying size has been studied by Sheaffer and Leavenworth (1976). The dissertation by Gardiner (1987) describes the use of various discrete distributions to model the occurrence of defects in integrated circuits.

As we noted in Section 3.2.4, the geometric distribution can also be useful as a model for count or "event" data. Kaminski et al. (1992) have proposed control charts for counts based on the geometric distribution. The probability model that they use for the geometric distribution is

$$p(x) = p(1 - p)^{x-a} \text{ for } x = a, a + 1, a + 2, \dots$$

where a is the known minimum possible number of events. Suppose that the data from the process is available as a subgroup of size n , say $x_1, x_2, \dots, x_n$. These observations are independently and identically distributed observations from a geometric distribution when the process is stable (in control). The two statistics that can be used to form a control chart are the **total** number of events

$$T = x_1 + x_2 + \dots + x_n$$

and the **average** number of events

$$\bar{x} = \frac{x_1 + x_2 + \dots + x_n}{n}$$

It turns out that the sum of independently and identically distributed geometric random variables is a negative binomial random variable. This would be useful information in constructing OC curves or calculating ARLs for the control charts for T or $\bar{x}$.

The mean and variance of the total number of events T are

$$\mu_T = n \left(\frac{1-p}{p} + a \right)$$

and

$$\sigma_T^2 = \frac{n(1-p)}{p^2}$$

and the mean and variance of the average number of events are

$$\mu_{\bar{x}} = \frac{1-p}{p} + a$$

and

$$\sigma_{\bar{x}}^2 = \frac{1-p}{np^2}$$

Consequently, the control charts can be constructed in the usual manner for Shewhart charts. Kaminski et al. (1992) refer to the control chart for the total number of events as a “*g* chart” and the control chart for the average number of events as an “*h* chart.” The center lines and control limits for each chart are shown in the following display.

g and *h* Control Charts, Standards Given

	Total number of events chart, <i>g</i> chart	Average number of events chart, <i>h</i> chart
Upper control limit (UCL)	$n\left(\frac{1-p}{p} + a\right) + L\sqrt{\frac{n(1-p)}{p^2}}$	$\frac{1-p}{p} + a + L\sqrt{\frac{1-p}{np^2}}$
Center line (CL)	$n\left(\frac{1-p}{p} + a\right)$	$\frac{1-p}{p} + a$
Lower control limit (LCL)	$n\left(\frac{1-p}{p} + a\right) - L\sqrt{\frac{n(1-p)}{p^2}}$	$\frac{1-p}{p} + a - L\sqrt{\frac{1-p}{np^2}}$

While we have assumed that a is known, in most situations the parameter p will likely be unknown. The estimator for p is

$$\hat{p} = \frac{1}{\bar{\bar{x}} - a + 1}$$

where $\bar{\bar{x}}$ is the average of all of the count data. Suppose that there are m subgroups available, each of size n , and let the total number of events in each subgroup be $t_1, t_2, \dots, t_m$. The average number of events per subgroup is

$$\bar{\bar{t}} = \frac{t_1 + t_2 + \dots + t_m}{m}$$

Therefore,

$$\bar{\bar{x}} = \frac{\bar{\bar{t}}}{n} = \frac{1 - \hat{p}}{\hat{p}} + a$$

and

$$\frac{1 - \hat{p}}{\hat{p}^2} = \left(\frac{\bar{\bar{t}}}{n} - a\right) \left(\frac{\bar{\bar{t}}}{n} - a + 1\right)$$

The center line and control limits for the g chart and the h chart based on an estimate of p are shown below.

g and h Control Charts, No Standards Given

	Total number of events chart, g chart	Average number of events chart, h chart
Upper control limit (UCL)	$\bar{t} + L\sqrt{n\left(\frac{\bar{t}}{n} - a\right)\left(\frac{\bar{t}}{n} - a + 1\right)}$	$\frac{\bar{t}}{n} + \frac{L}{\sqrt{n}}\sqrt{\left(\frac{\bar{t}}{n} - a\right)\left(\frac{\bar{t}}{n} - a + 1\right)}$
Center line (CL)	$\bar{t}$	$\frac{\bar{t}}{n}$
Lower control limit (LCL)	$\bar{t} - L\sqrt{n\left(\frac{\bar{t}}{n} - a\right)\left(\frac{\bar{t}}{n} - a + 1\right)}$	$\frac{\bar{t}}{n} - \frac{L}{\sqrt{n}}\sqrt{\left(\frac{\bar{t}}{n} - a\right)\left(\frac{\bar{t}}{n} - a + 1\right)}$

7.3.2 Procedures with Variable Sample Size

Control charts for nonconformities are occasionally formed using 100% inspection of the product. When this method of sampling is used, the number of inspection units in a sample will usually not be constant. For example, the inspection of rolls of cloth or paper often leads to a situation in which the size of the sample varies, because not all rolls are exactly the same length or width. If a control chart for nonconformities (c chart) is used in this situation, both the center line and the control limits will vary with the sample size. Such a control chart would be very difficult to interpret. The correct procedure is to use a control chart for nonconformities per unit (u chart). This chart will have a constant center line; however, the control limits will vary inversely with the square root of the sample size n .

EXAMPLE 7.5 Constructing a u chart

In a textile finishing plant, dyed cloth is inspected for the occurrence of defects per 50 square meters. The data on ten

rolls of cloth are shown in Table 7.11. Use these data to set up a control chart for nonconformities per unit.

■ **TABLE 7.11**
Occurrence of Nonconformities in Dyed Cloth

Roll Number	Number of Square Meters	Total Number of Nonconformities	Number of Inspection Units in Roll, n	Number of Nonconformities per Inspection Unit
1	500	14	10.0	1.40
2	400	12	8.0	1.50
3	650	20	13.0	1.54
4	500	11	10.0	1.10
5	475	7	9.5	0.74
6	500	10	10.0	1.00
7	600	21	12.0	1.75
8	525	16	10.5	1.52
9	600	19	12.0	1.58
10	625	23	12.5	1.84
		153	107.50	

(continued)

SOLUTION

The center line of the chart should be the average number of nonconformities per inspection unit—that is, the average number of nonconformities per 50 square meters, computed as

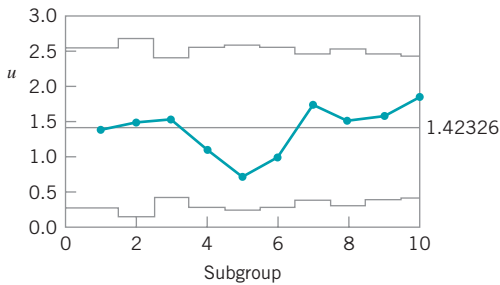
$$\bar{u} = \frac{153}{107.5} = 1.42$$

Note that $\bar{u}$ is the ratio of the total number of observed nonconformities to the total number of inspection units.

The control limits on this chart are computed from equation (7.19) with n replaced by n_i . The width of the control limits will vary inversely with n_i , the number of inspection units in the roll. The calculations for the control limits are displayed in Table 7.12. Figure 7.17 plots the control chart constructed by Minitab.

■ TABLE 7.12
Calculation of Control Limits, Example 7.5

Roll Number, i	n_i	$\text{UCL} = \bar{u} + 3\sqrt{\bar{u}/n_i}$	$\text{LCL} = \bar{u} - 3\sqrt{\bar{u}/n_i}$
1	10.0	2.55	0.29
2	8.0	2.68	0.16
3	13.0	2.41	0.43
4	10.0	2.55	0.29
5	9.5	2.58	0.26
6	10.0	2.55	0.29
7	12.0	2.45	0.39
8	10.5	2.52	0.32
9	12.0	2.45	0.39
10	12.5	2.43	0.41



■ FIGURE 7.17 Computer-generated (Minitab) control chart for Example 7.5.

As noted previously, the u chart should always be used when the sample size is variable. The most common implementation involves variable control limits, as illustrated in Example 7.5. There are, however, two other possible approaches:

- 1. Use control limits based on an average sample size

$$\bar{n} = \sum_{i=1}^m n_i / m$$

2. Use a standardized control chart (this is the preferred option). This second alternative would involve plotting a standardized statistic

$$Z_i = \frac{u_i - \bar{u}}{\sqrt{\frac{\bar{u}}{n_i}}} \quad (7.20)$$

on a control chart with LCL = -3 and UCL = +3 and the center line at zero. This chart is appropriate if tests for runs and other pattern-recognition methods are to be used in conjunction with the chart. Figure 7.18 shows the standardized version of the control chart in Example 7.5. This standardized control chart could also be useful in the short production run situation (see Chapter 10, Section 10.1).

7.3.3 Demerit Systems

With complex products such as automobiles, computers, or major appliances, we usually find that many different types of nonconformities or defects can occur. Not all of these types of defects are equally important. A unit of product having one very serious defect would probably be classified as nonconforming to requirements, but a unit having several minor defects might not necessarily be nonconforming. In such situations, we need a method to classify nonconformities or defects according to severity and to weight the various types of defects in a reasonable manner. **Demerit systems for attribute data** can be of value in these situations.

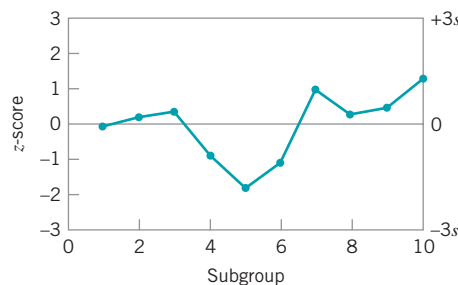
One possible demerit scheme is defined as follows.

Class A Defects—Very Serious. The unit is either completely unfit for service, or will fail in service in such a manner that cannot be easily corrected in the field, or will cause personal injury or property damage.

Class B Defects—Serious. The unit will possibly suffer a Class A operating failure, or will certainly cause somewhat less serious operating problems, or will certainly have reduced life or increased maintenance cost.

Class C Defects—Moderately Serious. The unit will possibly fail in service, or cause trouble that is less serious than operating failure, or possibly have reduced life or increased maintenance costs, or have a major defect in finish, appearance, or quality of work.

Class D Defects—Minor. The unit will not fail in service but has minor defects in finish, appearance, or quality of work.



■ **FIGURE 7.18** Standardized control chart for nonconformities per unit, Example 7.5.

Let c_{iA} , c_{iB} , c_{iC} , and c_{iD} represent the number of Class A, Class B, Class C, and Class D defects, respectively, in the i th inspection unit. We assume that each class of defect is independent, and the occurrence of defects in each class is well modeled by a Poisson distribution. Then we define the number of **demerits** in the inspection unit as

$$d_i = 100c_{iA} + 50c_{iB} + 10c_{iC} + c_{iD} \quad (7.21)$$

The demerit weights of Class A—100, Class B—50, Class C—10, and Class D—1 are used fairly widely in practice. However, any reasonable set of weights appropriate for a specific problem may also be used.

Suppose that a sample of n inspection units is used. Then the number of demerits per unit is

$$u_i = \frac{D}{n} \quad (7.22)$$

where $D = \sum_{i=1}^n d_i$ is the total number of demerits in all n inspection units. Since u_i is a linear combination of independent Poisson random variables, the statistics u_i could be plotted on a control chart with the following parameters:

	$\text{UCL} = \bar{u} + 3\hat{\sigma}_u$	
	$\text{Center line} = \bar{u}$	(7.23)
	$\text{LCL} = \bar{u} - 3\hat{\sigma}_u$	
where		
	$\bar{u} = 100\bar{u}_A + 50\bar{u}_B + 10\bar{u}_C + \bar{u}_D$	(7.24)
and		
	$\hat{\sigma}_u = \left[\frac{(100)^2 \bar{u}_A + (50)^2 \bar{u}_B + (10)^2 \bar{u}_C + \bar{u}_D}{n} \right]^{1/2}$	(7.25)

In the preceding equations, $\bar{u}_A$, $\bar{u}_B$, $\bar{u}_C$, and $\bar{u}_D$ represent the average number of Class A, Class B, Class C, and Class D defects per unit. The values of $\bar{u}_A$, $\bar{u}_B$, $\bar{u}_C$, and $\bar{u}_D$ are obtained from the analysis of preliminary data, taken when the process is supposedly operating in control. Standard values for u_A , u_B , u_C , and u_D may also be used, if they are available.

Jones, Woodall, and Conerly (1999) provide a very thorough discussion of demerit-based control charts. They show how probability-based limits can be computed as alternatives to the traditional three-sigma limits used above. They also show that, in general, the probability limits give superior performance. They are, however, more complicated to compute.

Many variations of this idea are possible. For example, we can classify nonconformities as either **functional defects** or **appearance defects** if a two-class system is preferred. It is also fairly common practice to maintain separate control charts on each defect class rather than combining them into one chart.

7.3.4 The Operating-Characteristic Function

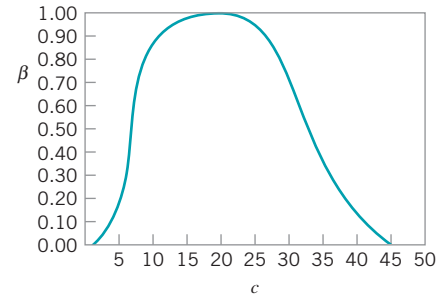
The operating-characteristic (OC) curves for both the c chart and the u chart can be obtained from the Poisson distribution. For the c chart, the OC curve plots the probability of type II error β against the true mean number of defects c . The expression for β is

$$\beta = P\{x < \text{UCL} | c\} - P\{x \leq \text{LCL} | c\} \quad (7.26)$$

■ TABLE 7.13

Calculation of the OC Curve for a c Chart with UCL = 33.22 and LCL = 6.48

c	$P\{x \leq 33 c\}$	$P\{x \leq 6 c\}$	$\beta = P\{x \leq 33 c\} - P\{x \leq 6 c\}$
1	1.000	0.999	0.001
3	1.000	0.966	0.034
5	1.000	0.762	0.238
7	1.000	0.450	0.550
10	1.000	0.130	0.870
15	0.999	0.008	0.991
20	0.997	0.000	0.997
25	0.950	0.000	0.950
30	0.744	0.000	0.744
33	0.546	0.000	0.546
35	0.410	0.000	0.410
40	0.151	0.000	0.151
45	0.038	0.000	0.038



■ FIGURE 7.19 OC curve of a c chart with LCL = 6.48 and UCL = 33.22.

where x is a Poisson random variable with parameter c . Note that if the LCL < 0 the second term on the right-hand side of equation (7.26) should be dropped.

We will generate the OC curve for the c chart in Example 7.3. For this example, since the LCL = 6.48 and the UCL = 33.22, equation (7.26) becomes

$$\beta = P\{x < 33.22|c\} - P\{x \leq 6.48|c\}$$

Since the number of nonconformities must be integer, this is equivalent to

$$\beta = P\{x \leq 33|c\} - P\{x \leq 6|c\}$$

These probabilities are evaluated in Table 7.13. The OC curve is shown in Fig. 7.19.

For the u chart, we may generate the OC curve from

$$\begin{aligned}
 \beta &= P\{x < \text{UCL}|u\} - P\{x \leq \text{LCL}|u\} \\
 &= P\{c < n\text{UCL}|u\} - P\{c \leq n\text{LCL}|u\} \\
 &= P\{n\text{LCL} < x \leq n\text{UCL}|u\} \\
 &= \sum_{x=\langle n\text{LCL} \rangle}^{[n\text{UCL}]} \frac{e^{-nu}(nu)^x}{x!}
 \end{aligned} \tag{7.27}$$

where $\langle n\text{LCL} \rangle$ denotes the smallest integer greater than or equal to $n\text{LCL}$ and $[n\text{UCL}]$ denotes the largest integer less than or equal to $n\text{UCL}$. The limits on the summation in equation (7.26) follow from the fact that the total number of nonconformities observed in a sample of n inspection units must be an integer. Note that n need not be an integer.

7.3.5 Dealing with Low Defect Levels

When defect levels or in general, **count rates**, in a process become very low—say, under 1000 occurrences per million—there will be very long periods of time between the occurrence of a nonconforming unit. In these situations, many samples will have zero defects, and a control chart with the statistic consistently plotting at zero will be relatively uninformative. Thus,

conventional c and u charts become ineffective as count rates are driven into the low parts per million (ppm) range.

One way to deal with this problem is adopt a **time between occurrence control chart**, which charts a new variable: the time between the successive occurrences of the count. The time-between-events control chart has been very effective as a process-control procedure for processes with low defect levels.

Suppose that defects or counts or “events” of interest occur according to a Poisson distribution. Then the probability distribution of the time between events is the exponential distribution. Therefore, constructing a time-between-events control chart is essentially equivalent to control charting an exponentially distributed variable. However, the exponential distribution is highly skewed, and as a result, the corresponding control chart would be very asymmetric. Such a control chart would certainly look unusual, and might present some difficulties in interpretation for operating personnel.

Nelson (1994) has suggested solving this problem by transforming the exponential random variable to a Weibull random variable such that the resulting Weibull distribution is well approximated by the normal distribution. If y represents the original exponential random variable, the appropriate transformation is

$$x = y^{1/3.6} = y^{0.2777} \quad (7.28)$$

One would now construct a control chart on x , assuming that x follows a normal distribution.

EXAMPLE 7.6

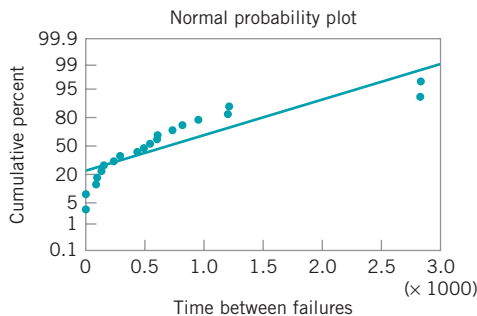
A chemical engineer wants to set up a control chart for monitoring the occurrence of failures of an important valve. She has decided to use the number of hours between failures as the vari-

able to monitor. Table 7.14 shows the number of hours between failures for the last twenty failures of this valve. Figure 7.20 is a normal probability plot of the time between failures.

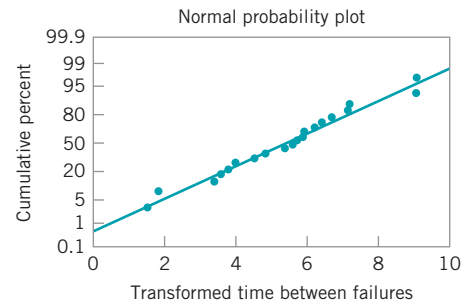
SOLUTION

Set up a time-between-events control chart for this process. Clearly, time between failures is not normally distributed. Table 7.14 also shows the values of the transformed time between events, computed from equation (7.27). Figure

7.21 is a normal probability plot of the transformed time between failures. Note that the plot indicates that the distribution of this transformed variable is well approximated by the normal.



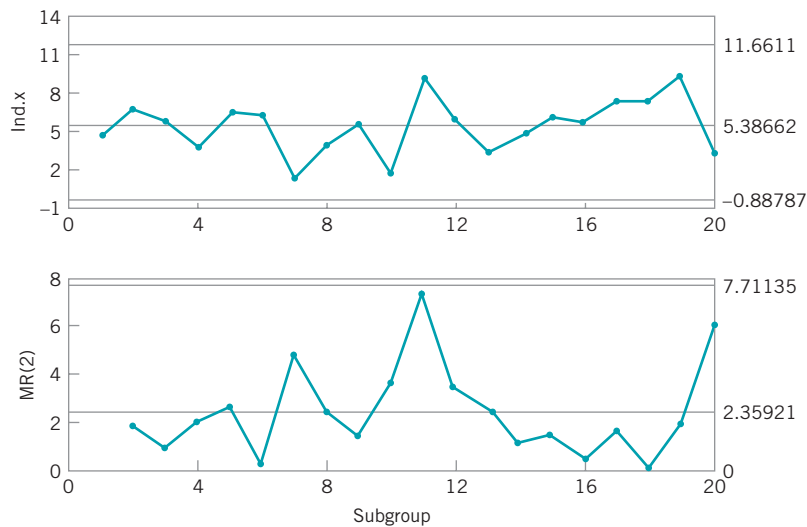
■ **FIGURE 7.20** Normal probability plot of time between failures, Example 7.6.



■ **FIGURE 7.21** Normal probability plot for the transformed failure data.

■ **TABLE 7.14**
Time between Failure Data, Example 7.6

Failure	Time between Failures, y (hr)	Transformed Value of Time between Failures, $x = y^{0.2777}$
1	286	4.80986
2	948	6.70903
3	536	5.72650
4	124	3.81367
5	816	6.43541
6	729	6.23705
7	4	1.46958
8	143	3.96768
9	431	5.39007
10	8	1.78151
11	2837	9.09619
12	596	5.89774
13	81	3.38833
14	227	4.51095
15	603	5.91690
16	492	5.59189
17	1199	7.16124
18	1214	7.18601
19	2831	9.09083
20	96	3.55203



■ **FIGURE 7.22** Control charts for individuals and moving-range control chart for the transformed time between failures, Example 7.6.

Figure 7.22 is a control chart for individuals and a moving range control chart for the transformed time between failures. Note that the control charts indicate a state of control, implying that the failure mechanism for this valve is constant. If a process change is made that improves the failure rate (such as

a different type of maintenance action), then we would expect to see the mean time between failures get longer. This would result in points plotting above the upper control limit on the individuals control chart in Fig. 7.22.

The previous example illustrated the use of the individuals control chart with time-between-events data. In many cases, the cusum and EWMA control charts in Chapter 4 would be better alternatives, because they are more effective in detecting small shifts in the mean.

Kittlitz (1999) has also investigated transforming the exponential distribution for control charting purposes. He notes that a log transformation will stabilize the variance of the exponential distribution, but produces a rather negatively skewed distribution. Kittlitz suggests using the transformation $x = y^{0.25}$, noting that it is very similar to Nelson's recommendation and it is also very easy to compute (simply push the square root key on the calculator twice!).

7.3.6 Nonmanufacturing Applications

The c chart and u chart are widely used in transactional and service business applications of statistical process control. In effect, we can treat **errors** in those environments the same as we treat defects or nonconformities in the manufacturing world. To give just a few examples, we can plot errors on engineering drawings, errors on plans and documents, and errors in computer software as c or u charts. An example using u charts to monitor errors in computer software during product development is given in Gardiner and Montgomery (1987).

7.4 Choice Between Attributes and Variables Control Charts

In many applications, the analyst will have to choose between using a variables control chart, such as the $\bar{x}$ and R charts, and an attributes control chart, such as the p chart. In some cases, the choice will be clear-cut. For example, if the quality characteristic is the color of the item, such as might be the case in carpet or cloth production, then attributes inspection would often be preferred over an attempt to quantify the quality characteristic "color." In other cases, the choice will not be obvious, and the analyst must take several factors into account in choosing between attributes and variables control charts.

Attributes control charts have the advantage that several quality characteristics can be considered jointly and the unit classified as nonconforming if it fails to meet the specification on any one characteristic. On the other hand, if the several quality characteristics are treated as variables, then each one must be measured, and either a separate $\bar{x}$ and R chart must be maintained on each or some multivariate control technique that considers all the characteristics must simultaneously be employed. There is an obvious simplicity associated with the attributes chart in this case. Furthermore, expensive and time-consuming measurements may sometimes be avoided by attributes inspection.

Variables control charts, in contrast, provide much more useful information about process performance than does an attributes control chart. Specific information about the process mean and variability is obtained directly. In addition, when points plot out of control on variables control charts, usually much more information is provided relative to the potential *cause* of that out-of-control signal. For a process capability study, variables control charts are almost always preferable to attributes control charts. The exceptions to this are studies relative to nonconformities produced by machines or operators in which there are a very limited number of sources of nonconformities, or studies directly concerned with process yields and fallout.

Perhaps the most important advantage of the $\bar{x}$ and R control charts is that they often provide an indication of impending trouble and allow operating personnel to take corrective action *before* any defectives are actually produced. Thus, $\bar{x}$ and R charts are *leading indicators* of trouble, whereas p charts (or c and u charts) will not react unless the process has already changed

8 *Process and Measurement System Capability Analysis*

CHAPTER OUTLINE

- | | | | |
|-------|---|-------|--|
| 8.1 | INTRODUCTION | 8.7.2 | The Analysis of Variance Method |
| 8.2 | PROCESS CAPABILITY ANALYSIS USING A HISTOGRAM OR A PROBABILITY PLOT | 8.7.3 | Confidence Intervals in Gauge R & R Studies |
| 8.2.1 | Using the Histogram | 8.7.4 | False Defectives and Passed Defectives |
| 8.2.2 | Probability Plotting | 8.7.5 | Attribute Gauge Capability |
| 8.3 | PROCESS CAPABILITY RATIOS | 8.8 | SETTING SPECIFICATION LIMITS ON DISCRETE COMPONENTS |
| 8.3.1 | Use and Interpretation of C_p | 8.8.1 | Linear Combinations |
| 8.3.2 | Process Capability Ratio for an Off-Center Process | 8.8.2 | Nonlinear Combinations |
| 8.3.3 | Normality and the Process Capability Ratio | 8.9 | ESTIMATING THE NATURAL TOLERANCE LIMITS OF A PROCESS |
| 8.3.4 | More about Process Centering | 8.9.1 | Tolerance Limits Based on the Normal Distribution |
| 8.3.5 | Confidence Intervals and Tests on Process Capability Ratios | 8.9.2 | Nonparametric Tolerance Limits |
| 8.4 | PROCESS CAPABILITY ANALYSIS USING A CONTROL CHART | | |
| 8.5 | PROCESS CAPABILITY ANALYSIS USING DESIGNED EXPERIMENTS | | |
| 8.6 | PROCESS CAPABILITY ANALYSIS WITH ATTRIBUTE DATA | | |
| 8.7 | GAUGE AND MEASUREMENT SYSTEM CAPABILITY STUDIES | | |
| 8.7.1 | Basic Concepts of Gauge Capability | | |
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|--|--|--|--|
| Supplemental Material for Chapter 8 | | | |
| S8.1 | Fixed versus Random Factors in the Analysis of Variance | | |
| S8.2 | More about Analysis of Variance Methods for Measurement Systems Capability Studies | | |

The supplemental material is on the textbook Website www.wiley.com/college/montgomery.

CHAPTER OVERVIEW AND LEARNING OBJECTIVES

In Chapter 6, we formally introduced the concept of process capability, or how the inherent variability in a process compares with the specifications or requirements for the product. Process capability analysis is an important tool in the DMAIC process, with application in both the analyze and improve steps. This chapter provides more extensive discussion of process capability, including several ways to study or analyze the capability of a process. We believe that the control chart is a simple and effective process capability analysis technique. We also extend the presentation of process capability ratios that we began in Chapter 6, showing how to interpret these ratios and discussing their potential dangers. The chapter also contains information on evaluating measurement system performance, illustrating graphical methods, as well as techniques based on the analysis of variance. Measurement systems analysis is used extensively in DMAIC, principally during the measure step. We also discuss setting specifications on individual discrete parts or components and estimating the natural tolerance limits of a process.

After careful study of this chapter you should be able to do the following:

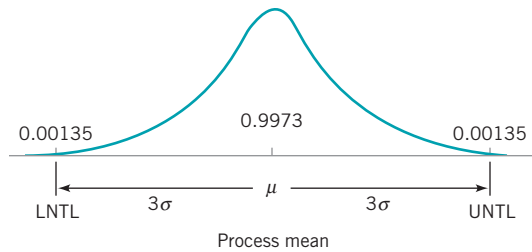
1. Investigate and analyze process capability using control charts, histograms, and probability plots
2. Understand the difference between process capability and process potential
3. Calculate and properly interpret process capability ratios
4. Understand the role of the normal distribution in interpreting most process capability ratios
5. Calculate confidence intervals on process capability ratios
6. Know how to conduct and analyze a measurement systems capability (or gauge R & R) experiment
7. Know how to estimate the components of variability in a measurement system
8. Know how to set specifications on components in a system involving interaction components to ensure that overall system requirements are met
9. Estimate the natural limits of a process from a sample of data from that process

8.1 Introduction

Statistical techniques can be helpful throughout the product cycle, including development activities prior to manufacturing, in quantifying process variability, in analyzing this variability relative to product requirements or specifications, and in assisting development and manufacturing in eliminating or greatly reducing this variability. This general activity is called **process capability analysis**.

Process capability refers to the **uniformity** of the process. Obviously, the variability of critical-to-quality characteristics in the process is a measure of the uniformity of output. There are two ways to think of this variability:

1. The natural or inherent variability in a critical-to-quality characteristic at a specified time; that is, “instantaneous” variability
2. The variability in a critical-to-quality characteristic over time



■ **FIGURE 8.1** Upper and lower natural tolerance limits in the normal distribution.

We present methods for investigating and assessing both aspects of process capability. Determining process capability is an important part of the DMAIC process. It is used primarily in the analyze step, but it also can be useful in other steps, such as improve.

It is customary to take the six-sigma spread in the distribution of the product quality characteristic as a measure of process capability. Figure 8.1 shows a process for which the quality characteristic has a normal distribution with mean μ and standard deviation σ . The upper and lower **natural tolerance limits of the process** fall at $\mu + 3\sigma$ and $\mu - 3\sigma$, respectively; that is,

$$\text{UNTL} = \mu + 3\sigma$$

$$\text{LNTL} = \mu - 3\sigma$$

For a normal distribution, the natural tolerance limits include 99.73% of the variable, or put another way, only 0.27% of the process output will fall outside the natural tolerance limits. Two points should be remembered:

1. 0.27% outside the natural tolerances sounds small, but this corresponds to 2700 nonconforming parts per million.
2. If the distribution of process output is non-normal, then the percentage of output falling outside $\mu \pm 3\sigma$ may differ considerably from 0.27%.

We define **process capability analysis** as a formal study to estimate **process capability**. The estimate of process capability may be in the form of a probability distribution having a specified shape, center (mean), and spread (standard deviation). For example, we may determine that the process output is normally distributed with mean $\mu = 1.0$ cm and standard deviation $\sigma = 0.001$ cm. In this sense, a process capability analysis may be performed **without regard to specifications on the quality characteristic**. Alternatively, we may express process capability as a percentage outside of specifications. However, specifications are not *necessary* to process capability analysis.

A process capability study usually measures functional parameters or critical-to-quality characteristics on the product, not the process itself. When the analyst can directly observe the process and can control or monitor the data-collection activity, the study is a true process capability study, because by controlling the data collection and knowing the time sequence of the data, inferences can be made about the stability of the process over time. However, when we have available only sample units of product, perhaps obtained from the supplier, and there is no direct observation of the process or time history of production, then the study is more properly called **product characterization**. In a product characterization study we can only estimate the distribution of the product quality characteristic or the process yield (fraction conforming to specifications); we can say nothing about the dynamic behavior of the process or its state of statistical control.

Process capability analysis is a vital part of an overall quality-improvement program. Among the major uses of data from a process capability analysis are the following:

1. Predicting how well the process will hold the tolerances
2. Assisting product developers/designers in selecting or modifying a process
3. Assisting in establishing an interval between sampling for process monitoring
4. Specifying performance requirements for new equipment
5. Selecting between competing suppliers and other aspects of supply chain management
6. Planning the sequence of production processes when there is an interactive effect of processes on tolerances
7. Reducing the variability in a process

Thus, process capability analysis is a technique that has application in many segments of the product cycle, including product and process design, supply chain management, production or manufacturing planning, and manufacturing.

Three primary techniques are used in process capability analysis: **histograms** or **probability plots**, **control charts**, and **designed experiments**. We will discuss and illustrate each of these methods in the next three sections. We will also discuss the process capability ratio (PCR) introduced in Chapter 6 and some useful variations of this ratio.

8.2 Process Capability Analysis Using a Histogram or a Probability Plot

8.2.1 Using the Histogram

The histogram can be helpful in estimating process capability. Alternatively, a stem-and-leaf plot may be substituted for the histogram. At least 100 or more observations should be available for the histogram (or the stem-and-leaf plot) to be moderately stable so that a reasonably reliable estimate of process capability may be obtained. If the quality engineer has access to the process and can control the data-collection effort, the following steps should be followed prior to data collection:

1. Choose the machine or machines to be used. If the results based on one (or a few) machines are to be extended to a larger population of machines, the machine selected should be representative of those in the population. Furthermore, if the machine has multiple workstations or heads, it may be important to collect the data so that head-to-head variability can be isolated. This may imply that designed experiments should be used.
2. Select the process operating conditions. Carefully define conditions, such as cutting speeds, feed rates, and temperatures, for future reference. It may be important to study the effects of varying these factors on process capability.
3. Select a representative operator. In some studies, it may be important to estimate *operator* variability. In these cases, the operators should be selected at random from the population of operators.
4. Carefully monitor the data-collection process, and record the time order in which each unit is produced.

The histogram, along with the sample average $\bar{x}$ and sample standard deviation s , provides information about process capability. You may wish to review the guidelines for constructing histograms in Chapter 3.

EXAMPLE 8.1 Estimating Process Capability with a Histogram

Figure 8.2 presents a histogram of the bursting strength of 100 glass containers. The data are shown in Table 8.1. What is the capability of the process?

SOLUTION

Analysis of the 100 observations gives

$$\bar{x} = 264.06 \quad s = 32.02$$

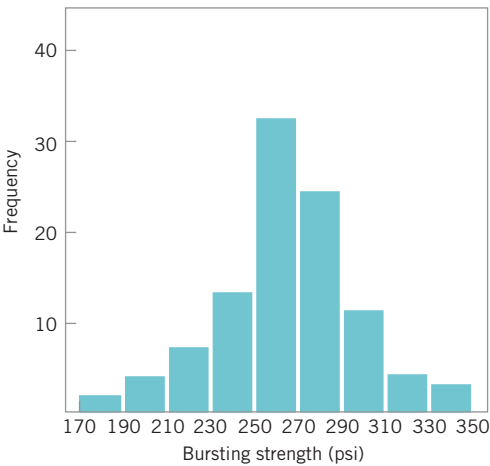
Consequently, the process capability would be estimated as

$$\bar{x} \pm 3s$$

or

$$264.06 \pm 3(32.02) \approx 264 \pm 96 \text{ psi}$$

Furthermore, the shape of the histogram implies that the distribution of bursting strength is approximately normal. Thus, we can estimate that approximately 99.73% of the bottles manufactured by this process will burst between 168 and 360 psi. Note that we can estimate process capability *independent of the specifications on bursting strength*.

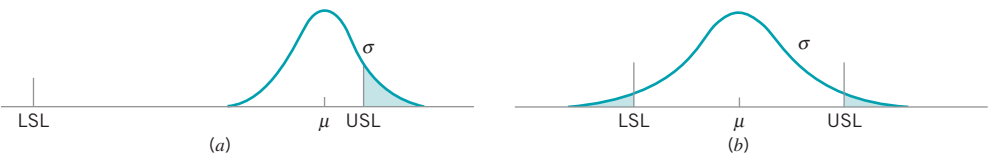


■ FIGURE 8.2 Histogram for the bursting-strength data.

■ TABLE 8.1
Bursting Strengths for 100 Glass Containers

265	197	346	280	265	200	221	265	261	278
205	286	317	242	254	235	176	262	248	250
263	274	242	260	281	246	248	271	260	265
307	243	258	321	294	328	263	245	274	270
220	231	276	228	223	296	231	301	337	298
268	267	300	250	260	276	334	280	250	257
260	281	208	299	308	264	280	274	278	210
234	265	187	258	235	269	265	253	254	280
299	214	264	267	283	235	272	287	274	269
215	318	271	293	277	290	283	258	275	251

An advantage of using the histogram to estimate process capability is that it gives an immediate, visual impression of process performance. It may also immediately show the reason for poor process performance. For example, Fig. 8.3a shows a process with adequate potential capability, but the process target is poorly located, whereas Fig. 8.3b shows a process with poor capability resulting from excess variability.



■ FIGURE 8.3 Some reasons for poor process capability. (a) Poor process centering. (b) Excess process variability.

8.2.2 Probability Plotting

Probability plotting is an alternative to the histogram that can be used to determine the shape, center, and spread of the distribution. It has the advantage that it is unnecessary to divide the range of the variable into class intervals, and it often produces reasonable results for moderately small samples (which the histogram will not). Generally, a probability plot is a graph of the ranked data versus the sample cumulative frequency on special paper with a vertical scale chosen so that the cumulative distribution of the assumed type is a straight line. In Chapter 3 we discussed and illustrated **normal probability plots**. These plots are very useful in process capability studies.

To illustrate the use of a normal probability plot in a process capability study, consider the following 20 observations on glass container bursting strength: 197, 200, 215, 221, 231, 242, 245, 258, 265, 265, 271, 275, 277, 278, 280, 283, 290, 301, 318, and 346. Figure 8.4 is the normal probability plot of strength. Note that the data lie nearly along a straight line, implying that the distribution of bursting strength is normal. Recall from Chapter 4 that the mean of the normal distribution is the fiftieth percentile, which we may estimate from Fig. 8.4 as approximately 265 psi, and the standard deviation of the distribution is the *slope* of the straight line. It is convenient to estimate the standard deviation as the difference between the eighty-fourth and the fiftieth percentiles. For the strength data shown above and using Fig. 8.4, we find that

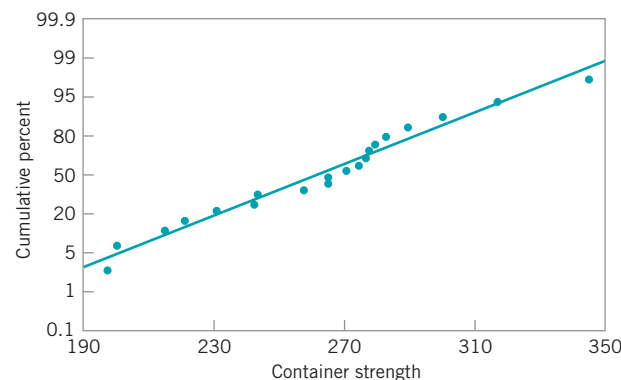
$$\hat{\sigma} = 84\text{th percentile} - 50\text{th percentile} = 298 - 265 \text{ psi} = 33 \text{ psi}$$

Note that $\hat{\mu} = 265$ psi and $\hat{\sigma} = 33$ psi are not far from the sample average $\bar{x} = 264.06$ and standard deviation $s = 32.02$.

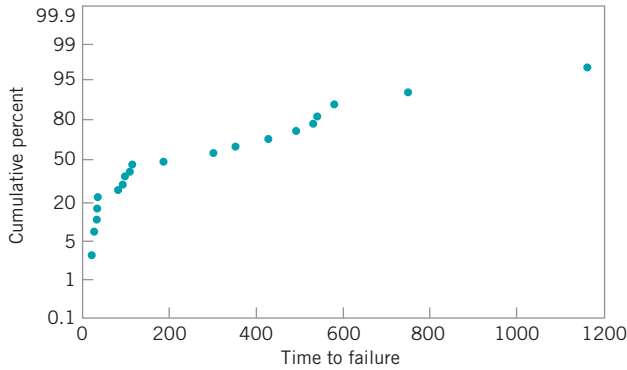
The normal probability plot can also be used to estimate process yields and fallouts. For example, the specification on container strength is $LSL = 200$ psi. From Fig. 8.4, we would estimate that about 5% of the containers manufactured by this process would burst below this limit.

Care should be exercised in using probability plots. If the data do not come from the assumed distribution, inferences about process capability drawn from the plot may be seriously in error. Figure 8.5 presents a normal probability plot of times to failure (in hours) of a valve in a chemical plant. From examining this plot, we can see that the distribution of failure time is not normal.

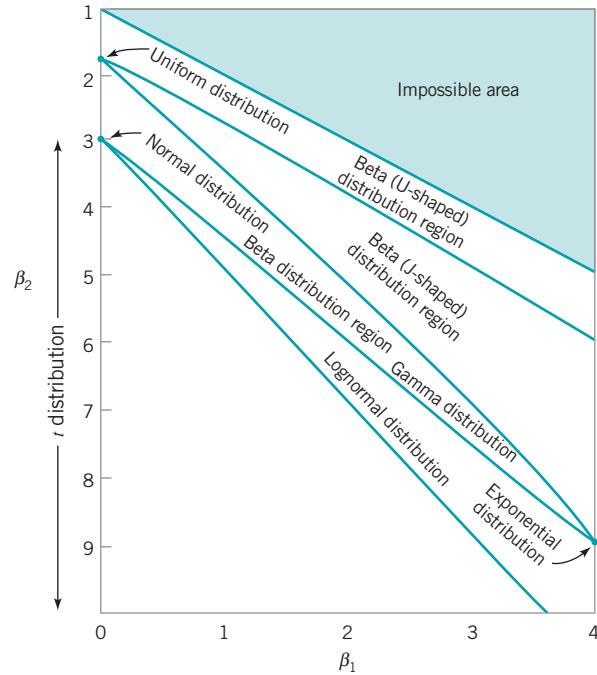
An obvious disadvantage of probability plotting is that it is not an objective procedure. It is possible for two analysts to arrive at different conclusions using the same data. For this reason, it is often desirable to supplement probability plots with more formal statistically



■ **FIGURE 8.4** Normal probability plot of the container-strength data.



■ **FIGURE 8.5** Normal probability plot of time to failure of a valve.



■ **FIGURE 8.6** Regions in the β_1, β_2 plane for various distributions. (From *Statistical Models in Engineering* by G. J. Hahn and S. S. Shapiro, John Wiley: New York, 1967.)

based goodness-of-fit tests. A good introduction to these tests is in Shapiro (1980). Augmenting the interpretation of a normal probability plot with the Shapiro-Wilk test for normality can make the procedure much more powerful and objective.

Choosing the distribution to fit the data is also an important step in probability plotting. Sometimes we can use our knowledge of the physical phenomena or past experience to suggest the choice of distribution. In other situations, the display in Fig. 8.6 may be useful in selecting a distribution that describes the data. This figure shows the regions in the β_1, β_2 plane for several standard probability distributions, where β_1 and β_2 are the measures of *skewness* and *kurtosis*, respectively. To use Fig. 8.6, calculate estimates of skewness and kurtosis from the sample—say,

$$\sqrt{\hat{\beta}_1} = \frac{M_3}{(M_2)^{3/2}} \quad (8.1)$$

and

$$\hat{\beta}_2 = \frac{M_4}{M_2^2} \quad (8.2)$$

where

$$M_j = \frac{\sum_{i=1}^n (x_i - \bar{x})^j}{n} \quad j = 1, 2, 3, 4 \quad (8.3)$$

and plot the point $(\hat{\beta}_1, \hat{\beta}_2)$ on the graph. If the plotted point falls close to a point, line, or region that corresponds to one of the distributions in the figure, then this distribution is a logical choice to use as a model for the data. If the point falls in regions of the β_1, β_2 plane where *none* of the distributions seems appropriate, other, more general probability distributions,

such as the Johnson or Pearson families of distributions, may be required. A note of caution should be sounded here; the skewness and kurtosis statistics are not reliable unless they are computed from very large samples. Procedures similar to that in Fig. 8.6 for fitting these distributions and graphs are in Hahn and Shapiro (1967).

8.3 Process Capability Ratios

8.3.1 Use and Interpretation of C_p

It is frequently convenient to have a simple, quantitative way to express process capability. One way to do so is through the **process capability ratio (PCR)** C_p first introduced in Chapter 6. Recall that

$$C_p = \frac{USL - LSL}{6\sigma} \quad (8.4)$$

where USL and LSL are the upper and lower specification limits, respectively. C_p and other process capability ratios are used extensively in industry. They are also widely *misused*. We will point out some of the more common abuses of process capability ratios. An excellent recent book on process capability ratios that is highly recommended is Kotz and Lovelace (1998). There is also extensive technical literature on process capability analysis and process capability ratios. The review paper by Kotz and Johnson (2002) and the bibliography by Spiring, Leong, Cheng, and Yeung (2003) are excellent sources.

In a practical application, the process standard deviation σ is almost always unknown and must be replaced by an estimate $\hat{\sigma}$. To estimate σ we typically use either the *sample standard deviation* s or $\bar{R}/d_2$ (when variables control charts are used in the capability study). This results in an estimate of C_p —say,

$$\hat{C}_p = \frac{USL - LSL}{6\hat{\sigma}} \quad (8.5)$$

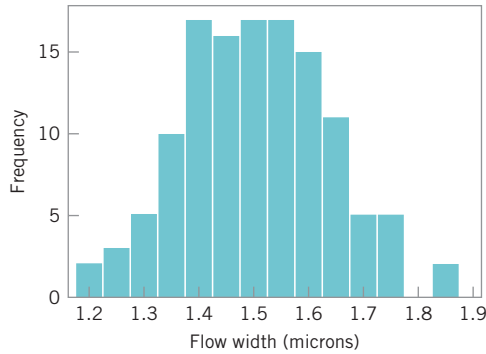
To illustrate the calculation of C_p , recall the semiconductor hard-bake process first analyzed in Example 6.1 using $\bar{x}$ and R charts. The specifications on flow width are USL = 1.00 microns and LSL = 2.00 microns, and from the R chart we estimated $\sigma = \bar{R}/d_2 = 0.1398$. Thus, our estimate of the PCR C_p is

$$\hat{C}_p = \frac{USL - LSL}{6\hat{\sigma}} = \frac{2.00 - 1.00}{6(0.1398)} = 1.192$$

In Chapter 6, we assumed that flow width is approximately normally distributed (a reasonable assumption, based on the histogram in Fig. 8.7) and the cumulative normal distribution table in the Appendix was used to estimate that the process produces approximately 350 ppm (parts per million) defective.

The PCR C_p in equation (8.4) has a useful practical interpretation—namely,

$$P = \left(\frac{1}{C_p} \right) 100 \quad (8.6)$$



■ **FIGURE 8.7** Histogram of flow width from Example 6.1.

is the percentage of the specification band used up by the process. The hard-bake process uses

$$P = \left(\frac{1}{1.192} \right) 100 = 83.89$$

percent of the specification band.

Equations (8.4) and (8.5) assume that the process has both upper and lower specification limits. For one-sided specifications, **one-sided process-capability ratios** are used. One-sided PCRs are defined as follows.

$$C_{pu} = \frac{USL - \mu}{3\sigma} \quad (\text{upper specification only}) \quad (8.7)$$

$$C_{pl} = \frac{\mu - LSL}{3\sigma} \quad (\text{lower specification only}) \quad (8.8)$$

Estimates $\hat{C}_{pu}$ and $\hat{C}_{pl}$ would be obtained by replacing μ and σ in equations (8.7) and (8.8) by estimates $\hat{\mu}$ and $\hat{\sigma}$, respectively.

EXAMPLE 8.2 One-Sided Process-Capability Ratios

Construct a one-sided process-capability ratio for the container bursting-strength data in Example 8.1. Suppose that the lower specification limit on bursting strength is 200 psi.

SOLUTION

We will use $\bar{x} = 264$ and $s = 32$ as estimates of μ and σ , respectively, and the resulting estimate of the one-sided lower process-capability ratio is

$$\hat{C}_{pl} = \frac{\hat{\mu} - LSL}{3\hat{\sigma}} = \frac{264 - 200}{3(32)} = 0.67$$

The fraction of defective containers produced by this process is estimated by finding the area to the left of $Z = (LSL$

$- \mu)/\sigma = (200 - 264)/32 = -2$ under the standard normal distribution. The estimated fallout is about 2.28% defective, or about 22,800 nonconforming containers per million. Note that if the normal distribution were an inappropriate model for strength, then this last calculation would have to be performed using the appropriate probability distribution.

■ TABLE 8.2

Values of the Process Capability Ratio (C_p) and Associated Process Fallout for a Normally Distributed Process (in Defective ppm) That Is in Statistical Control

PCR	Process Fallout (in defective ppm)	
	One-Sided Specifications	Two-Sided Specifications
0.25	226,628	453,255
0.50	66,807	133,614
0.60	35,931	71,861
0.70	17,865	35,729
0.80	8,198	16,395
0.90	3,467	6,934
1.00	1,350	2,700
1.10	484	967
1.20	159	318
1.30	48	96
1.40	14	27
1.50	4	7
1.60	1	2
1.70	0.17	0.34
1.80	0.03	0.06
2.00	0.0009	0.0018

The process capability ratio is a measure of the ability of the process to manufacture product that meets the specifications. Table 8.2 presents several values of the PCR C_p along with the associated values of process fallout, expressed in defective parts or nonconforming units of product per million (ppm). To illustrate the use of Table 8.2, notice that a PCR for a normally distributed stable process of $C_p = 1.00$ implies a fallout rate of 2700 ppm for two-sided specifications, whereas a PCR of $C_p = 1.50$ for this process implies a fallout rate of 4 ppm for one-sided specifications.

The ppm quantities in Table 8.2 were calculated using the following **important assumptions**:

1. The quality characteristic has a normal distribution.
2. The process is in statistical control.
3. In the case of two-sided specifications, the process mean is centered between the lower and upper specification limits.

These assumptions are absolutely **critical** to the **accuracy** and **validity** of the reported numbers, and if they are not valid, then the reported quantities may be seriously in error. For example, Somerville and Montgomery (1996) report an extensive investigation of the errors in using the normality assumption to make inferences about the ppm level of a process when in fact the underlying distribution is non-normal. They investigated various non-normal distributions and observed that errors of several orders of magnitude can result in predicting ppm by erroneously making the normality assumption. Even when using a t distribution with as many as 30 degrees of freedom, substantial errors result. Thus even though a t distribution

■ **TABLE 8.3**
Recommended Minimum Values of the Process Capability Ratio

	Two-Sided Specifications	One-Sided Specifications
Existing processes	1.33	1.25
New processes	1.50	1.45
Safety, strength, or critical parameter, existing process	1.50	1.45
Safety, strength, or critical parameter, new process	1.67	1.60

with 30 degrees of freedom is symmetrical and almost visually indistinguishable from the normal, the longer and heavier tails of the t distribution make a significant difference when estimating the ppm. Consequently, symmetry in the distribution of process output alone is insufficient to ensure that any PCR will provide a reliable prediction of process ppm. We will discuss the non-normality issue in more detail in Section 8.3.3.

Stability or **statistical control** of the process is also **essential** to the correct interpretation of any PCR. Unfortunately, it is fairly common practice to compute a PCR from a sample of historical process data without any consideration of whether or not the process is in statistical control. If the process is not in control, then of course its parameters are unstable, and the value of these parameters in the future is uncertain. Thus the predictive aspects of the PCR regarding process ppm performance are lost.

Finally, remember that what we actually observe in practice is an **estimate** of the PCR. This estimate is subject to error in estimation, since it depends on sample statistics. English and Taylor (1993) report that large errors in estimating PCRs from sample data can occur, so the estimate one actually has at hand may not be very reliable. It is always a good idea to report the estimate of any PCR in terms of a **confidence interval**. We will show how to do this for some of the commonly used PCRs in Section 8.3.5.

Table 8.3 presents some recommended **guidelines for minimum values** of the PCR. The bottle-strength characteristic is a parameter closely related to the safety of the product; bottles with inadequate pressure strength may fail and injure consumers. This implies that the PCR should be at least 1.45. Perhaps one way the PCR could be improved would be by increasing the mean strength of the containers—say, by pouring more glass in the mold.

We point out that the values in Table 8.3 are only **minimums**. In recent years, many companies have adopted criteria for evaluating their processes that include process capability objectives that are more stringent than those of Table 8.3. For example, a six-sigma company would require that when the process mean is in control, it will not be closer than six standard deviations from the nearest specification limit. This, in effect, requires that the minimum acceptable value of the process capability ratio will be at least 2.0.

8.3.2 Process Capability Ratio for an Off-Center Process

The process capability ratio C_p does not take into account *where* the process mean is located relative to the specifications. C_p simply measures the spread of the specifications relative to the six-sigma spread in the process. For example, the top two normal distributions in Fig. 8.8 both have $C_p = 2.0$, but the process in panel (b) of the figure clearly has lower capability than the process in panel (a) because it is not operating at the midpoint of the interval between the specifications.

This situation may be more accurately reflected by defining a new process capability ratio—**PRC** C_{pk} —that takes process centering into account. This quantity is

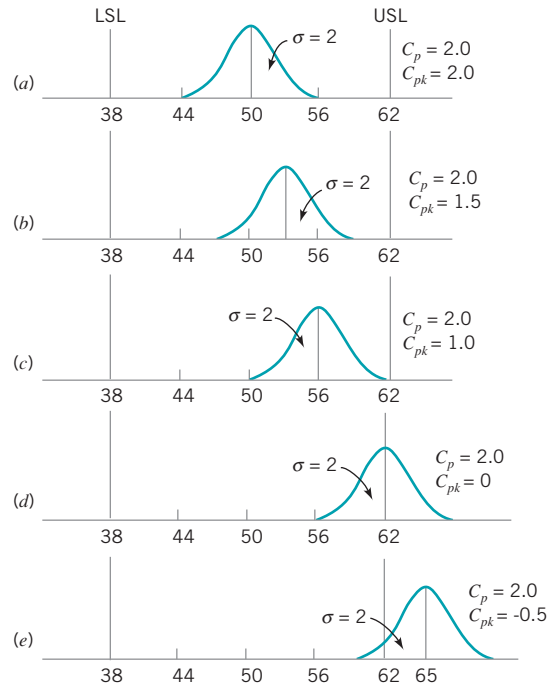
$$C_{pk} = \min(C_{pu}, C_{pl}) \quad (8.9)$$

Note that C_{pk} is simply the one-sided PCR for the specification limit nearest to the process average. For the process shown in Fig. 8.8*b*, we would have

$$\begin{aligned} C_{pk} &= \min(C_{pu}, C_{pl}) \\ &= \min\left(C_{pu} = \frac{USL - \mu}{3\sigma}, C_{pl} = \frac{\mu - LSL}{3\sigma}\right) \\ &= \min\left(C_{pu} = \frac{62 - 53}{3(2)} = 1.5, C_{pl} = \frac{53 - 38}{3(2)} = 2.5\right) \\ &= 1.5 \end{aligned}$$

Generally, if $C_p = C_{pk}$, the process is centered at the midpoint of the specifications, and when $C_{pk} < C_p$ the process is off-center.

The magnitude of C_{pk} relative to C_p is a direct measure of how off-center the process is operating. Several commonly encountered cases are illustrated in Fig. 8.8. Note in panel (c) of Fig. 8.8 that $C_{pk} = 1.0$ while $C_p = 2.0$. One can use Table 8.2 to get a quick estimate of potential improvement that would be possible by centering the process. If we take $C_p = 1.0$ in Table 8.2 and read the fallout from the one-sided specifications column, we can estimate the *actual* fallout as 1350 ppm. However, if we can center the process, then $C_p = 2.0$ can be achieved, and Table 8.2 (using $C_p = 2.0$ and two-sided specifications) suggests that the *potential*



■ FIGURE 8.8 Relationship of C_p and C_{pk} .

fallout is 0.0018 ppm, an improvement of several orders of magnitude in process performance. Thus, we usually say that C_p measures **potential capability** in the process, whereas C_{pk} measures **actual capability**.

Panel (d) of Fig. 8.8 illustrates the case in which the process mean is exactly equal to one of the specification limits, leading to $C_{pk} = 0$. As panel (e) illustrates, when $C_{pk} < 0$ the implication is that the process mean lies outside the specifications. Clearly, if $C_{pk} < -1$, the entire process lies outside the specification limits. Some authors define C_{pk} to be nonnegative, so that values less than zero are defined as zero.

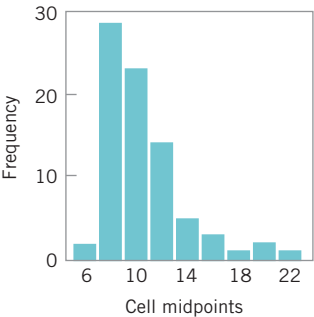
Many quality-engineering authorities have advised against the **routine use** of process capability ratios such as C_p and C_{pk} (or the others discussed later in this section) on the grounds that they are an oversimplification of a complex phenomenon. Certainly, any statistic that combines information about both location (the mean and process centering) and variability and that requires the assumption of normality for its meaningful interpretation is likely to be misused (or abused). Furthermore, as we will see, point estimates of process capability ratios are virtually useless if they are computed from small samples. Clearly, these ratios need to be used and interpreted very carefully.

8.3.3 Normality and the Process Capability Ratio

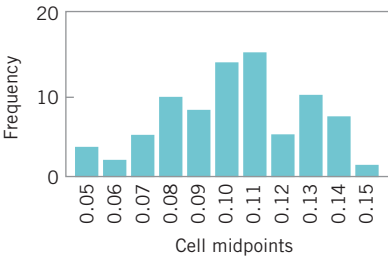
An important assumption underlying our discussion of process capability and the ratios C_p and C_{pk} is that their usual interpretation is based on a **normal distribution** of process output. If the underlying distribution is non-normal, then as we previously cautioned, the statements about expected process fallout attributed to a particular value of C_p or C_{pk} may be in error.

To illustrate this point, consider the data in Fig. 8.9, which is a histogram of 80 measurements of surface roughness on a machined part (measured in microinches). The upper specification limit is at $USL = 32$ microinches. The sample average and standard deviation are $\bar{x} = 10.44$ and $S = 3.053$, implying that $\hat{C}_{pu} = 2.35$, and Table 8.2 would suggest that the fallout is less than one part per billion. However, since the histogram is highly skewed, we are fairly certain that the distribution is non-normal. Thus, this estimate of capability is unlikely to be correct.

One approach to dealing with this situation is to **transform the data** so that in the new, transformed metric the data have a normal distribution appearance. There are various graphical and analytical approaches to selecting a transformation. In this example, a reciprocal transformation was used. Figure 8.10 presents a histogram of the reciprocal values $x^* = 1/x$. In the transformed scale, $\bar{x}^* = 0.1025$ and $s^* = 0.0244$, and the original upper specification limit becomes $1/32 = 0.03125$. This results in a value of $\hat{C}_{pl} = 0.97$, which implies that about 1350 ppm are outside of specifications. This estimate of process performance is clearly much more realistic than the one resulting from the usual “normal theory” assumption.



■ **FIGURE 8.9** Surface roughness in microinches for a machined part.



■ **FIGURE 8.10** Reciprocals of surface roughness. (Adapted from data in the “Statistics Corner” column in *Quality Progress*, March 1989, with permission of the American Society for Quality Control.)

Other approaches have been considered in dealing with non-normal data. There have been various attempts to extend the definitions of the standard capability indices to the case of non-normal distributions. Luceño (1996) introduced the index C_{pc} , defined as

$$C_{pc} = \frac{USL - LSL}{6\sqrt{\frac{\pi}{2}}E|X - T|} \quad (8.10)$$

where the process target value $T = \frac{1}{2}(USL + LSL)$. Luceño uses the second subscript in C_{pc} to stand for *confidence*, and he stresses that the confidence intervals based on C_{pc} are reliable; of course, this statement should be interpreted cautiously. The author has also used the constant $6\sqrt{\pi/2} = 7.52$ in the denominator, to make it equal to 6σ when the underlying distribution is normal. We will give the confidence interval for C_{pc} in Section 8.3.5.

There have also been attempts to modify the usual capability indices so that they are appropriate for two **general families** of distributions, the Pearson and Johnson families. This would make PCRs broadly applicable for both normal and non-normal distributions. Good discussions of these approaches are in Rodríguez (1992) and Kotz and Lovelace (1998).

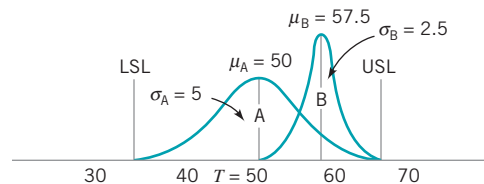
The general idea is to use appropriate **quantiles** of the process distribution—say, $x_{0.00135}$ and $x_{0.99865}$ —to define a quantile-based PCR—say,

$$C_p(q) = \frac{USL - LSL}{x_{0.99865} - x_{0.00135}} \quad (8.11)$$

Now since in the normal distribution $x_{0.00135} = \mu - 3\sigma$ and $x_{0.99865} = \mu + 3\sigma$, we see that in the case of a normal distribution $C_p(q)$ reduces to C_p . Clements (1989) proposed a method for determining the quantiles based on the Pearson family of distributions. In general, however, we could fit *any* distribution to the process data, determine its quantiles $x_{0.99865}$ and $x_{0.00135}$, and apply equation (8.11). Refer to Kotz and Lovelace (1998) for more information.

8.3.4 More about Process Centering

The process capability ratio C_{pk} was initially developed because C_p does not adequately deal with the case of a process with mean μ that is not centered between the specification limits. However, C_{pk} alone is still an inadequate measure of process centering. For example, consider the two processes shown in Fig. 8.11. Both processes A and B have $C_{pk} = 1.0$, yet their centering is clearly different. To characterize process centering satisfactorily, C_{pk} must be compared to C_p . For process A, $C_{pk} = C_p = 1.0$, implying that the process is centered, whereas for process B, $C_p = 2.0 > C_{pk} = 1.0$, implying that the process is off-center. For any fixed value of μ in the interval from LSL to USL, C_{pk} depends inversely on σ and becomes large as σ



■ **FIGURE 8.11** Two processes with $C_{pk} = 1.0$.

approaches zero. This characteristic can make C_{pk} unsuitable as a measure of centering. That is, a large value of C_{pk} does not really tell us anything about the location of the mean in the interval from LSL to USL.

One way to address this difficulty is to use a process capability ratio that is a better indicator of centering. PCR C_{pm} is one such ratio,

$$C_{pm} = \frac{USL - LSL}{6\tau} \quad (8.12)$$

and τ is the square root of expected squared deviation from target $T = \frac{1}{2}(USL + LSL)$,

$$\begin{aligned} \tau^2 &= E[(x - T)^2] \\ &= E[(x - \mu)^2] + (\mu - T)^2 \\ &= \sigma^2 + (\mu - T)^2 \end{aligned}$$

Thus, equation (8.12) can be written as

$$C_{pm} = \frac{USL - LSL}{6\sqrt{\sigma^2 + (\mu - T)^2}} = \frac{C_p}{\sqrt{1 + \xi^2}} \quad (8.13)$$

where

$$\xi = \frac{\mu - T}{\sigma} \quad (8.14)$$

A logical way to estimate C_{pm} is by

$$\hat{C}_{pm} = \frac{\hat{C}_p}{\sqrt{1 + V^2}} \quad (8.15)$$

where

$$V = \frac{\bar{x} - T}{s} \quad (8.16)$$

Chan et al. (1988) discussed this ratio, various estimators of C_{pm} , and their sampling properties. Boyles (1991) has provided a definitive analysis of C_{pm} and its usefulness in measuring process centering. He notes that both C_{pk} and C_{pm} coincide with C_p when $\mu = T$ and decrease as μ moves away from T . However, $C_{pk} < 0$ for $\mu > USL$ or $\mu < LSL$, whereas C_{pm} approaches zero asymptotically as $|(\mu - T)| \rightarrow \infty$. Boyles also shows that the C_{pm} of a process with $|(\mu - T)| = \Delta > 0$ is strictly bounded above by the C_p value of a process with $\sigma = \Delta$. That is,

$$C_{pm} < \frac{USL - LSL}{6|\mu - T|} \quad (8.17)$$

Thus, a necessary condition for $C_{pm} \geq 1$ is

$$|\mu - T| < \frac{1}{6}(USL - LSL)$$

This statistic says that if the target value T is the midpoint of the specifications, a C_{pm} of one or greater implies that the mean μ lies within the middle third of the specification band. A similar statement can be made for any value of C_{pm} . For instance, $C_{pm} \geq \frac{4}{3}$ implies that $|(\mu - T)| < \frac{1}{8}(USL - LSL)$. Thus, a given value of C_{pm} places a constraint on the difference between μ and the target value T .

EXAMPLE 8.3 Process Centering

To illustrate the use of C_{pm} , consider the two processes A and B in Fig. 8.11. For process A we find that

$$C_{pm} = \frac{C_p}{\sqrt{1 + \xi^2}} = \frac{1.0}{\sqrt{1 + 0}} = 1.0$$

since process A is centered at the target value $T = 50$. Note that $C_{pm} = C_{pk}$ for process A. Now consider process B:

$$C_{pm} = \frac{C_p}{\sqrt{1 + \xi^2}} = \frac{2}{\sqrt{1 + (3)^2}} = 0.63$$

If we use equation (8.17), this is equivalent to saying that the process mean lies approximately within the middle half of the specification range. Visual examination of Fig. 8.11 reveals this to be the case.

Pearn et al. (1992) proposed the process capability ratio

$$C_{pkm} = \frac{C_{pk}}{\sqrt{1 + \left(\frac{\mu - T}{\sigma}\right)^2}} = \frac{C_{pk}}{\sqrt{1 + \xi^2}} \quad (8.18)$$

This is sometimes called a “third generation” process capability ratio, since it is constructed from the “second generation” ratios C_{pk} and C_{pm} in the same way that they were generated from the “first generation” ratio C_p . The motivation of this new ratio is increased sensitivity to departures of the process mean μ from the desired target T . For more details, see Kotz and Lovelace (1998).

8.3.5 Confidence Intervals and Tests on Process Capability Ratios

Confidence Intervals on Process Capability Ratios. Much of the industrial use of process capability ratios focuses on computing and interpreting the **point estimate** of the desired quantity. It is easy to forget that $\hat{C}_p$ or $\hat{C}_{pk}$ (for examples) are simply point estimates, and, as such, are subject to statistical fluctuation. An alternative that should become standard practice is to report **confidence intervals for process capability ratios**.

It is easy to find a confidence interval for the “first generation” ratio C_p . If we replace σ by s in the equation for C_p , we produce the usual point estimator $\hat{C}_p$. If the quality characteristic follows a normal distribution, then a $100(1 - \alpha)\%$ confidence interval on C_p is obtained from

$$\frac{USL - LSL}{6s} \sqrt{\frac{\chi_{1-\alpha/2, n-1}^2}{n-1}} \leq C_p \leq \frac{USL - LSL}{6s} \sqrt{\frac{\chi_{\alpha/2, n-1}^2}{n-1}} \quad (8.19)$$

or

$$\hat{C}_p \sqrt{\frac{\chi_{1-\alpha/2, n-1}^2}{n-1}} \leq C_p \leq \hat{C}_p \sqrt{\frac{\chi_{\alpha/2, n-1}^2}{n-1}} \quad (8.20)$$

where $\chi_{1-\alpha/2, n-1}^2$ and $\chi_{\alpha/2, n-1}^2$ are the lower $\alpha/2$ and upper $\alpha/2$ percentage points of the chi-square distribution with $n - 1$ degrees of freedom. These percentage points are tabulated in Appendix Table III.

EXAMPLE 8.4 A Confidence Interval in C_p

Suppose that a stable process has upper and lower specifications at $USL = 62$ and $LSL = 38$. A sample of size $n = 20$ from this process reveals that the process mean is centered approxi-

mately at the midpoint of the specification interval and that the sample standard deviation $s = 1.75$. Find a 95% confidence interval on C_p .

SOLUTION

A point estimate of C_p is

$$\hat{C}_p = \frac{USL - LSL}{6s} = \frac{62 - 38}{6(1.75)} = 2.29$$

The 95% confidence interval on C_p is found from equation (8.20) as follows:

$$\begin{aligned} \hat{C}_p \sqrt{\frac{\chi^2_{1-0.025, n-1}}{n-1}} &\leq C_p \leq \hat{C}_p \sqrt{\frac{\chi^2_{0.025, n-1}}{n-1}} \\ 2.29 \sqrt{\frac{8.91}{19}} &\leq C_p \leq 2.29 \sqrt{\frac{32.85}{19}} \\ 1.57 &\leq C_p \leq 3.01 \end{aligned}$$

where $\chi^2_{0.975, 19} = 8.91$ and $\chi^2_{0.025, 19} = 32.85$ were taken from Appendix Table III.

The confidence interval on C_p in Example 8.4 is relatively wide because the sample standard deviation s exhibits considerable fluctuation in small to moderately large samples. This means, in effect, that confidence intervals on C_p based on small samples will be wide.

Note also that the confidence interval uses s rather than $\bar{R}/d_2$ to estimate σ . This further emphasizes that the process must be **in statistical control** for PCRs to have any real meaning. If the process is not in control, s and $\bar{R}/d_2$ could be very different, leading to very different values of the PCR.

For more complicated ratios such as C_{pk} and C_{pm} , various authors have developed approximate confidence intervals; for example, see Zhang, Stenbeck, and Wardrop (1990), Bissell (1990), Kushler and Hurley (1992), and Pearn et al. (1992). If the quality characteristic is normally distributed, then an approximate $100(1 - \alpha)\%$ confidence interval on C_{pk} is given as follows.

$$\begin{aligned} \hat{C}_{pk} \left[1 - Z_{\alpha/2} \sqrt{\frac{1}{9n\hat{C}_{pk}^2} + \frac{1}{2(n-1)}} \right] &\leq C_{pk} \\ &\leq \hat{C}_{pk} \left[1 + Z_{\alpha/2} \sqrt{\frac{1}{9n\hat{C}_{pk}^2} + \frac{1}{2(n-1)}} \right] \end{aligned} \quad (8.21)$$

Kotz and Lovelace (1998) give an extensive summary of confidence intervals for various PCRs.

EXAMPLE 8.5 A Confidence Interval in C_{pk}

A sample of size $n = 20$ from a stable process is used to estimate C_{pk} , with the result that $\hat{C}_{pk} = 1.33$. Find an approximate 95% confidence interval on C_{pk} .

SOLUTION

Using equation (8.21), an approximate 95% confidence interval on C_{pk} is

$$\begin{aligned} \hat{C}_{pk} \left[1 - Z_{\alpha/2} \sqrt{\frac{1}{9n\hat{C}_{pk}^2} + \frac{1}{2(n-1)}} \right] \\ \leq C_{pk} \leq \hat{C}_{pk} \left[1 + Z_{\alpha/2} \sqrt{\frac{1}{9n\hat{C}_{pk}^2} + \frac{1}{2(n-1)}} \right] \\ 1.33 \left[1 - 1.96 \sqrt{\frac{1}{9(20)(1.33)^2} + \frac{1}{2(19)}} \right] \\ \leq C_{pk} \leq 1.33 \left[1 + 1.96 \sqrt{\frac{1}{9(20)(1.33)^2} + \frac{1}{2(19)}} \right] \end{aligned}$$

or

$$0.88 \leq C_{pk} \leq 1.78$$

This is an extremely wide confidence interval. Based on the sample data, the ratio C_{pk} could be less than 1 (a very bad situation), or it could be as large as 1.78 (a very good situation). Thus,

we have learned very little about actual process capability, because C_{pk} is very imprecisely estimated. The reason for this, of course, is that a very small sample ($n = 20$) has been used.

For non-normal data, the PCR C_{pc} developed by Luceño (1996) can be employed. Recall that C_{pc} was defined in equation (8.10). Luceño developed the confidence interval for C_{pc} as follows: First, evaluate $|x - T|$, whose expected value is estimated by

$$\bar{c} = \frac{1}{n} \sum_{i=1}^n |x_i - T|$$

leading to the estimator

$$\hat{C}_{pc} = \frac{USL - LSL}{6\sqrt{\frac{\pi}{2}}\bar{c}}$$

A $100(1 - \alpha)\%$ confidence interval for $E|x - T|$ is given as

$$\bar{c} \pm t_{\alpha/2, n-1} \frac{s_c}{\sqrt{n}}$$

where

$$s_c^2 = \frac{1}{n-1} \sum_{i=1}^n (|x_i - T| - \bar{c})^2 = \frac{1}{n-1} \left(\sum_{i=1}^n |x_i - T|^2 - n\bar{c}^2 \right)$$

Therefore, a $100(1 - \alpha)\%$ confidence interval for C_{pc} is given by

$$\frac{\hat{C}_{pc}}{1 + t_{\alpha/2, n-1} \left[s_c / (\bar{c} \sqrt{n}) \right]} \leq C_{pc} \leq \frac{\hat{C}_{pc}}{1 - t_{\alpha/2, n-1} \left[s_c / (\bar{c} \sqrt{n}) \right]} \quad (8.22)$$

Testing Hypotheses about PCRs. A practice that is becoming increasingly common in industry is to require a supplier to **demonstrate** process capability as part of the contractual agreement. Thus, it is frequently necessary to demonstrate that the process capability ratio C_p meets or exceeds some particular target value—say, C_{p0} . This problem may be formulated as a hypothesis testing problem

$$H_0: C_p = C_{p0} \text{ (or the process is not capable)}$$

$$H_1: C_p \geq C_{p0} \text{ (or the process is capable)}$$

We would like to reject H_0 (recall that in statistical hypothesis testing rejection of H_0 is always a strong conclusion), thereby demonstrating that the process is capable. We can formulate the statistical test in terms of $\hat{C}_p$, so that we will reject H_0 if $\hat{C}_p$ exceeds a critical value C .

Kane (1986) has investigated this test, and provides a table of sample sizes and critical values for C to assist in testing process capability. We may define $C_p(\text{High})$ as a process capability that we would like to accept with probability $1 - \alpha$ and $C_p(\text{Low})$ as a process capability that we would like to reject with probability $1 - \beta$. Table 8.4 gives values of $C_p(\text{High})/C_p(\text{Low})$ and $C/C_p(\text{Low})$ for varying sample sizes and $\alpha = \beta = 0.05$ or $\alpha = \beta = 0.10$. Example 8.6 illustrates the use of this table.

EXAMPLE 8.6 Supplier Qualification

A customer has told his supplier that, in order to qualify for business with his company, the supplier must demonstrate that his process capability exceeds $C_p = 1.33$. Thus, the supplier is interested in establishing a procedure to test the hypotheses

$$H_0: C_p = 1.33$$

$$H_1: C_p > 1.33$$

The supplier wants to be sure that if the process capability is below 1.33 there will be a high probability of detecting this (say, 0.90), whereas if the process capability exceeds 1.66 there will be a high probability of judging the process capable (again, say, 0.90). This would imply that $C_p(\text{Low}) = 1.33$, $C_p(\text{High}) = 1.66$, and $\alpha = \beta = 0.10$. To find the sample size and critical value for C from Table 8.4, compute

$$\frac{C_p(\text{High})}{C_p(\text{Low})} = \frac{1.66}{1.33} = 1.25$$

■ **TABLE 8.4**
Sample Size and Critical Value Determination for Testing
 $H_0: C_p = C_{p0}$

Sample Size, n	(a) $\alpha = \beta = 0.10$		(b) $\alpha = \beta = 0.05$	
	$C_p(\text{High})/$ $C_p(\text{Low})$	$C/C_p(\text{Low})$	$C_p(\text{High})/$ $C_p(\text{Low})$	$C/C_p(\text{Low})$
10	1.88	1.27	2.26	1.37
20	1.53	1.20	1.73	1.26
30	1.41	1.16	1.55	1.21
40	1.34	1.14	1.46	1.18
50	1.30	1.13	1.40	1.16
60	1.27	1.11	1.36	1.15
70	1.25	1.10	1.33	1.14
80	1.23	1.10	1.30	1.13
90	1.21	1.10	1.28	1.12
100	1.20	1.09	1.26	1.11

Source: Adapted from Kane (1986), with permission of the American Society for Quality Control.

and enter the table value in panel (a) where $\alpha = \beta = 0.10$. This yields

$$n = 70$$

and

$$C/C_p(\text{Low}) = 1.10$$

from which we calculate

$$C = C_p(\text{Low})1.10 = 1.33(1.10) = 1.46$$

Thus, to demonstrate capability, the supplier must take a sample of $n = 70$ parts, and the sample process capability ratio $\hat{C}_p$ must exceed $C \leq 1.46$.

This example shows that, in order to demonstrate that process capability is at least equal to 1.33, the *observed sample* $\hat{C}_p$ will have to exceed 1.33 by a considerable amount. This illustrates that some common industrial practices may be questionable statistically. For example, it is fairly common practice to accept the process as capable at the level $C_p \geq 1.33$ if the *sample* $\hat{C}_p \geq 1.33$ based on a sample size of $30 \leq n \leq 50$ parts. Clearly, this procedure does not account for sampling variation in the estimate of σ , and larger values of n and/or higher acceptable values of $\hat{C}_p$ may be necessary in practice.

Process Performance Indices. In 1991, the Automotive Industry Action Group (AIAG) was formed and consists of representatives of the “big three” (Ford, General Motors, and Chrysler) and the American Society for Quality Control (now the American Society for Quality). One of their objectives was to standardize the reporting requirements from suppliers and in general of their industry. The AIAG recommends using the process capability indices C_p and C_{pk} when the process is in control, with the process standard deviation estimated by $\hat{\sigma} = R/d_2$. When the process is *not* in control, the AIAG recommends using **process performance indices** P_p and P_{pk} , where, for example,

$$\hat{P}_p = \frac{USL - LSL}{6s}$$

and s is the usual sample standard deviation $s = \sqrt{\sum_{i=1}^n (x_i - \bar{x})^2 / (n - 1)}$. Even the American National Standards Institute in ANSI Standard Z1 on Process Capability Analysis (1996) states that P_p and P_{pk} should be used when the process is not in control.

Now it is clear that when the process is normally distributed and in control, $\hat{P}_p$ is essentially $\hat{C}_p$ and $\hat{P}_{pk}$ is essentially $\hat{C}_{pk}$ because for a stable process the difference between s and $\hat{\sigma} = R/d_2$ is minimal. However, please note that if the process is **not** in control, the indices P_p and P_{pk} have no meaningful interpretation relative to process capability, because they cannot **predict** process performance. Furthermore, their statistical properties are not determinable, and so no valid inference can be made regarding their true (or population) values. Also, P_p and P_{pk} provide no motivation or incentive to the companies that use them to bring their processes into control.

Kotz and Lovelace (1998) strongly recommend **against** the use of P_p and P_{pk} , indicating that these indices are actually a step **backward** in quantifying process capability. They refer to the mandated use of P_p and P_{pk} through quality standards or industry guidelines as undiluted “statistical terrorism” (i.e., the use or misuse of statistical methods along with threats and/or intimidation to achieve a business objective).

This author agrees completely with Kotz and Lovelace. The process performance indices P_p and P_{pk} are actually more than a step backward. **They are a waste of engineering and management effort—they tell you nothing.** Unless the process is stable (in control), no index is going to carry useful predictive information about process capability or convey any

information about future performance. Instead of imposing the use of meaningless indices, organizations should devote effort to developing and implementing an effective process characterization, control, and improvement plan. This is a much more reasonable and effective approach to process improvement.

8.4 Process Capability Analysis Using a Control Chart

Histograms, probability plots, and process capability ratios summarize the performance of the process. They do not necessarily display the **potential capability** of the process because they do not address the issue of **statistical control**, or show systematic patterns in process output that, if eliminated, would reduce the variability in the quality characteristic. Control charts are very effective in this regard. The control chart should be regarded as the primary technique of process capability analysis.

Both attributes and variables control charts can be used in process capability analysis. The $\bar{x}$ and R charts should be used whenever possible, because of the greater power and better information they provide relative to attributes charts. However, both p charts and c (or u) charts are useful in analyzing process capability. Techniques for constructing and using these charts are given in Chapters 6 and 7. Remember that to use the p chart there must be specifications on the product characteristics. The $\bar{x}$ and R charts allow us to study processes without regard to specifications.

The $\bar{x}$ and R control charts allow both the instantaneous variability (short-term process capability) and variability across time (long-term process capability) to be analyzed. It is particularly helpful if the data for a process capability study are collected in two to three different time periods (such as different shifts, different days, etc.).

Table 8.5 presents the container bursting-strength data in 20 samples of five observations each. The calculations for the $\bar{x}$ and R charts are summarized here:

$\bar{x}$ Chart
Center line = $\bar{\bar{x}} = 264.06$
UCL = $\bar{\bar{x}} + A_2\bar{R} = 264.06 + (0.577)(77.3) = 308.66$
LCL = $\bar{\bar{x}} - A_2\bar{R} = 264.06 - (0.577)(77.3) = 219.46$
R Chart
Center line = $\bar{R} = 77.3$
UCL = $D_4\bar{R} = (2.115)(77.3) = 163.49$
LCL = $D_3\bar{R} = (0)(77.3) = 0$

Figure 8.12 presents the $\bar{x}$ and R charts for the 20 samples in Table 8.5. Both charts exhibit statistical control. The process parameters may be estimated from the control chart as

$$\hat{\mu} = \bar{\bar{x}} = 264.06$$

$$\hat{\sigma} = \frac{\bar{R}}{d_2} = \frac{77.3}{2.326} = 33.23$$

TABLE 8.5
Glass Container Strength Data (psi)

Sample	Data					$\bar{x}$	R
1	265	205	263	307	220	252.0	102
2	268	260	234	299	215	255.2	84
3	197	286	274	243	231	246.2	89
4	267	281	265	214	318	269.0	104
5	346	317	242	258	276	287.8	104
6	300	208	187	264	271	246.0	113
7	280	242	260	321	228	266.2	93
8	250	299	258	267	293	273.4	49
9	265	254	281	294	223	263.4	71
10	260	308	235	283	277	272.6	73
11	200	235	246	328	296	261.0	128
12	276	264	269	235	290	266.8	55
13	221	176	248	263	231	227.8	87
14	334	280	265	272	283	286.8	69
15	265	262	271	245	301	268.8	56
16	280	274	253	287	258	270.4	34
17	261	248	260	274	337	276.0	89
18	250	278	254	274	275	266.2	28
19	278	250	265	270	298	272.2	48
20	257	210	280	269	251	253.4	70
						$\bar{\bar{x}} = 264.06$	$\bar{R} = 77.3$

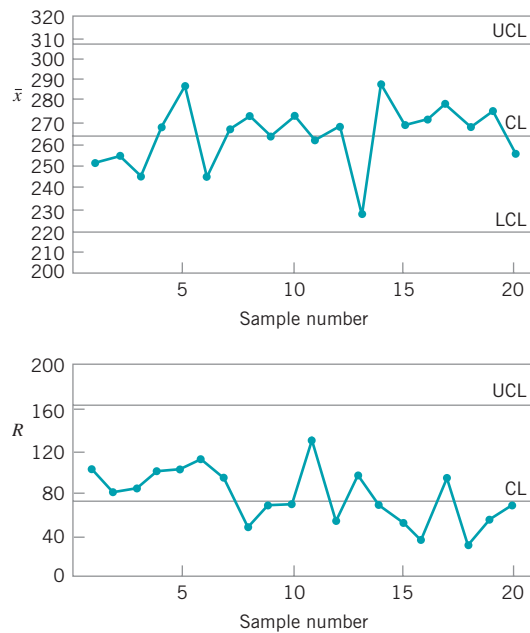


FIGURE 8.12 $\bar{x}$ and R charts for the bottle-strength data.

Thus, the one-sided lower process capability ratio is estimated by

$$\hat{C}_{pl} = \frac{\mu - \text{LSL}}{3\hat{\sigma}} = \frac{264.06 - 200}{3(33.23)} = 0.64$$

Clearly, since strength is a safety-related parameter, the process capability is inadequate.

This example illustrates a process that is in control but operating at an unacceptable level. There is no evidence to indicate that the production of nonconforming units is **operator-controllable**. Engineering and/or management intervention will be required either to improve the process or to change the requirements if the quality problems with the bottles are to be solved. The objective of these interventions is to increase the process capability ratio to at least a minimum acceptable level. The control chart can be used as a monitoring device or logbook to show the effect of changes in the process on process performance.

Sometimes the process capability analysis indicates an out-of-control process. It is **unsafe** to estimate process capability in such cases. The process must be stable in order to produce a reliable estimate of process capability. When the process is out of control in the early stages of process capability analysis, the first objective is finding and eliminating the assignable causes in order to bring the process into an in-control state.

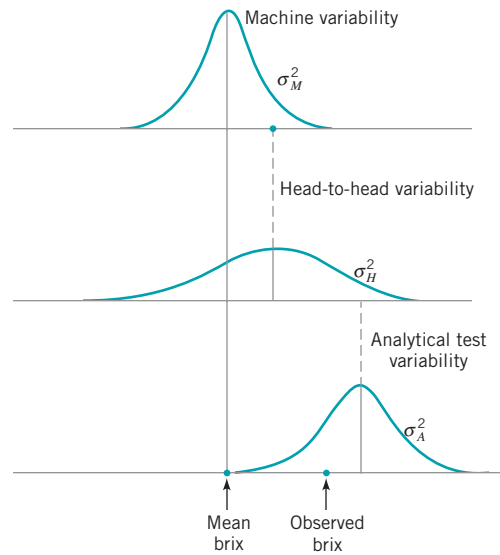
8.5 Process Capability Analysis Using Designed Experiments

A designed experiment is a systematic approach to varying the input **controllable** variables in the process and analyzing the effects of these process variables on the output. Designed experiments are also useful in discovering *which* set of process variables is influential on the output, and at what levels these variables should be held to optimize process performance. Thus, design of experiments is useful in more general problems than merely estimating process capability. For an introduction to design of experiments, see Montgomery (2005). Part V of this textbook provides more information on experimental design methods and on their use in process improvement.

One of the major uses of designed experiments is in isolating and estimating the **sources of variability** in a process. For example, consider a machine that fills bottles with a soft-drink beverage. Each machine has a large number of filling heads that must be independently adjusted. The quality characteristic measured is the syrup content (in degrees brix) of the finished product. There can be variation in the observed brix (σ_B^2) because of machine variability (σ_M^2), head variability (σ_H^2), and analytical test variability (σ_A^2). The variability in the observed brix value is

$$\sigma_B^2 = \sigma_M^2 + \sigma_H^2 + \sigma_A^2$$

An experiment can be designed, involving sampling from several machines and several heads on each machine, and making several analyses on each bottle, which would allow estimation of the variances (σ_M^2), (σ_H^2), and (σ_A^2). Suppose that the results appear as in Fig. 8.13. Since a substantial portion of the total variability in observed brix is due to variability between heads, this indicates that the process can perhaps best be improved by reducing the head-to-head variability. This could be done by more careful setup or by more careful control of the operation of the machine.



■ **FIGURE 8.13** Sources of variability in the bottling line example.

8.6 Process Capability Analysis with Attribute Data

Often process performance is measured in terms of **attribute** data; that is, nonconforming units or defectives, or nonconformities or defects. When a fraction nonconforming is the measure of performance, it is typical to use the parts per million (ppm) defective as a measure of process capability. In some organizations, this ppm defective is converted to an equivalent sigma level. For example, a process producing 2700 ppm defective would be equivalent to a three-sigma process (without the “usual” 1.5σ shift in the mean that many six-sigma organizations employ in the calculations taken into account).

When dealing with nonconformities or defects, a defects per unit (DPU) statistic is often used as a measure of capability, where

$$\text{DPU} = \frac{\text{Total number of defects}}{\text{Total number of units}}$$

Here the unit is something that is delivered to a customer and can be evaluated or judged as to its suitability. Some examples include:

1. An invoice
2. A shipment
3. A customer order
4. An enquiry or call

The defects or nonconformities are anything that does not meet the customer requirements, such as:

1. An error on an invoice
2. An incorrect or incomplete shipment
3. An incorrect or incomplete customer order
4. A call that is not satisfactorily completed

Obviously, these quantities are estimated from sample data. Large samples need to be used to obtain reliable estimates.

The DPU measure does not directly take the complexity of the unit into account. A widely used way to do this is the defect per million opportunities (DPMO) measure

$$\text{DPMO} = \frac{\text{Total number of defects}}{\text{Number of units} \times \text{Number of opportunities}}$$

Opportunities are the number of potential chances within a unit for a defect to occur. For example, on a purchase order, the number of opportunities would be the number of fields in which information is recorded times two, because each field can either be filled out incorrectly or blank (information is missing). It is important to be consistent about how opportunities are defined, as a process may be artificially improved simply by increasing the number of opportunities over time.

8.7 Gauge and Measurement System Capability Studies

8.7.1 Basic Concepts of Gauge Capability

Determining the capability of the measurement system is an important aspect of many quality and process improvement activities. Generally, in any activity involving measurements, some of the observed variability will be inherent in the units or items that are being measured, and some of the variability will result from the measurement system that is used. The measurement system will consist (minimally) of an instrument or gauge, and it often has other components, such as the operator(s) that uses it and the conditions or different points in time under which the instrument is used. There may also be other factors that impact measurement system performance, such as setup or calibration activities. The purpose of most measurement systems capability studies is to:

1. Determine how much of the total observed variability is due to the gauge or instrument
2. Isolate the components of variability in the measurement system
3. Assess whether the instrument or gauge is capable (that is, is it suitable for the intended application)

Measurements are a significant component of any quality system. Measurement is an integral component of the DMAIC problem-solving process, but it's even more important than that. An ineffective measurement system can dramatically impact business performance because it leads to uninformed (and usually bad) decision making.

In this section we will introduce the two R's of measurement systems capability: **repeatability** (do we get the same observed value if we measure the same unit several times under identical conditions), and **reproducibility** (how much difference in observed values do we experience when units are measured under different conditions, such as different operators, time periods, and so forth).

These quantities answer only indirectly the fundamental question: Is the system able to distinguish between good and bad units? That is, what is the probability that a good unit is judged to be defective and, conversely, that a bad unit is passed along to the customer as good? These misclassification probabilities are fairly easy to calculate from the results of a standard measurement systems capability study, and give reliable, useful, and easy-to-understand information about measurement systems performance.

In addition to repeatability and reproducibility, there are other important aspects of measurement systems capability. The **linearity** of a measurement system reflects the differences in observed accuracy and/or precision experienced over the range of measurements made by the system. A simple linear regression model is often used to describe this feature.

on the acceptance control chart. Obviously, to use this approach, n must not be seriously restricted by cost or other factors such as rational subgrouping considerations.

10.3 Control Charts for Multiple-Stream Processes

10.3.1 Multiple-Stream Processes

A **multiple-stream process (MSP)** is a process with data at a point in time consisting of measurements from several individual sources or streams. When the process is in control, the sources or streams are assumed to be identical. Another characteristic of the MSP is that we can monitor and adjust each of the streams individually or in small groups.

The MSP occurs often in practice. For example, a machine may have several heads, with each head producing (we hope) identical units of product. In such situations several possible control procedures may be followed. One possibility is to use separate control charts on each stream. This approach usually results in a prohibitively large number of control charts. If the output streams are highly correlated—say, nearly perfectly correlated—then control charts on only one stream may be adequate. The most common situation is that the streams are only moderately correlated, so monitoring only one of the streams is not appropriate. There are at least two types of situations involving the occurrence of assignable causes in the MSP.

1. The output of one stream (or a few streams) has shifted off target.
2. The output of all streams has shifted off target.

In the first case, we are trying to detect an assignable cause that affects only one stream (or at most a few streams), whereas in the second, we are looking for an assignable cause that impacts *all* streams (such as would result from a change in raw materials).

The standard control chart for the MSP is the group control chart, introduced by Boyd (1950). We will also discuss other approaches to monitoring the MSP. Throughout this section we assume that the process has s streams and that the output quality characteristic from each stream is normally distributed.

10.3.2 Group Control Charts

The **group control chart (GCC)** was introduced by Boyd (1950) and remains the basic procedure for monitoring an MSP. To illustrate the methods of construction and use, suppose that the process has $s = 6$ streams and that each stream has the same target value and inherent variability. Variables measurement is made on the items produced, and the distribution of the measurement is well approximated by the normal. To establish a group control chart, the sampling is performed as if separate control charts were to be set up on each stream. Suppose, for purposes of illustration, that a sample size of $n = 4$ is used. This means that 4 units will be taken from each of the $s = 6$ streams over a short period of time. This will be repeated until about 20 such groups of samples have been taken. At this point we would have $20 \times 6 = 120$ averages of $n = 4$ observations each and 120 corresponding ranges. These averages and ranges would be averaged to produce a grand average $\bar{\bar{x}}$ and an average range $\bar{\bar{R}}$. The limits on the group control charts would be at

$$UCL = \bar{\bar{x}} + A_2 \bar{\bar{R}}$$

$$LCL = \bar{\bar{x}} - A_2 \bar{\bar{R}}$$

for the $\bar{x}$ chart and at

$$UCL = D_4 \bar{\bar{R}}$$

$$LCL = D_3 \bar{\bar{R}}$$

for the R chart, with $A_2 = 0.729$, $D_3 = 0$, and $D_4 = 2.282$. Note that the sample size $n = 4$ determines the control chart constants.

When the group control chart is used to control the process, we plot only the largest and smallest of the $s = 6$ means observed at any time period on the $\bar{x}$ chart. If these means are inside the control limits, then all other means will also lie inside the limits. Similarly, only the largest range will be plotted on the range chart. Each plotted point is identified on the chart by the number of the stream that produced it. The process is out of control if a point exceeds a three-sigma limit. Runs tests cannot be applied to these charts, because the conventional runs tests were not developed to test averages or ranges that are the extremes of a group of averages or ranges.

It is useful to examine the stream numbers on the chart. In general, if a stream consistently gives the largest (or smallest) value several times in a row, that may constitute evidence that this stream is different from the others. If the process has s streams and if r is the number of consecutive times that a particular stream is the largest (or smallest) value, then the one-sided in-control average run length for this event is given by Nelson (1986) as

$$\text{ARL}(1)_0 = \frac{s^r - 1}{s - 1} \quad (10.6)$$

if all streams are identical. To illustrate the use of this equation, if $s = 6$ and $r = 4$, then

$$\text{ARL}(1)_0 = \frac{6^4 - 1}{6 - 1} = 259$$

That is, if the process is in control, we will expect to see the same stream producing an extreme value on the group control chart four times in a row only once every 259 samples.

One way to select the value of r to detect the presence of one stream that is different from the others is to use equation (10.6) to find an ARL that is roughly consistent with the ARL of a conventional control chart. The ARL for an in-control process for a single point beyond the upper control limit (say) is 740. Thus, using $r = 4$ for a six-stream process results in an ARL that is too short and that will give too many false alarms. A better choice is $r = 5$, since

$$\text{ARL}(1)_0 = \frac{6^5 - 1}{6 - 1} = 1555$$

Thus, if we have six streams and if the same stream produces an extreme value on the control chart in five consecutive samples, then we have strong evidence that this stream is different from the others.

Using equation (10.6), we can generate some general guidelines for choosing r given the number of streams s . Suitable pairs (s, r) would include (3, 7), (4, 6), (5–6, 5), and (7–10, 4). All of these combinations would give reasonable values of the one-sided ARL when the process is in control.

The two-sided in-control average run length $\text{ARL}(2)_0$ is defined as the expected number of trials until r consecutive largest or r consecutive smallest means come from the same stream while the MSP is in control. Mortell and Runger (1995) and Nelson and Stephenson (1996) used the *Markov chain approach* of Brook and Evans (1972) to compute $\text{ARL}(2)_0$ numerically. A closed-form expression is not possible for $\text{ARL}(2)_0$, but Nelson and Stephenson (1996) give a lower bound on $\text{ARL}(2)_0$ as

$$\text{ARL}(2)_0 = \frac{s^r - 1}{2(r - 1)} \quad (10.7)$$

This approximation agrees closely with the numerical computations from the Markov chain approach.

There are some drawbacks connected with the GCC. These may be summarized as follows:

1. At each point in time where samples are taken, all s streams must be sampled. When s is small, this may be a satisfactory approach. However, some high-speed filling and packaging machines may have $100 \leq s \leq 200$ streams. The GCC is not practical for these situations.
2. There is no information about nonextreme streams at each trial, thus, we cannot utilize past values to form an EWMA or a cusum statistic to improve on the GCC performance.
3. The supplementary run tests fail if more than one stream shifts simultaneously (all up or all down) to the same off-target level, because the shifted streams will most likely alternate having the extreme value, and no single stream shift can be identified.
4. The selection of the run or pattern length r for a specific problem with s streams is a discrete process, and the difference in ARL_0 for $r = r^*$ or $r = r^* + 1$ choice can be substantial, as observed by Mortell and Runger (1995). This problem is evident in the original run rule devised by Nelson (1986), but can be alleviated somewhat by using the modified run rules in Nelson and Stephenson (1996), which are based on an $r - 1$ out of r in a row approach. In fact, it can be easily shown that the difference in the ARL_0 between $r = r^*$ and $r = r^* + 1$ choices equals s^{r^*} , which becomes very large for problems with large numbers of streams.
5. Although using an $r - 1$ out of r in a row run rule allows some flexibility in the choice of the critical value $r = r^*$ and ARL_0 , it adds some complexity since it is no longer a simple counting procedure.

10.3.3 Other Approaches

Mortell and Runger (1995) proposed the following model for the MSP to accommodate the practical case of dependent (cross-correlated) streams:

$$x_{ijk} = \mu + A_t + \varepsilon_{ijk} \quad (10.8)$$

where x_{ijk} is the k th measurement on the j th stream at time t , $A_t \sim N(0, \sigma_a^2)$ represents the difference of the mean over all the streams at time t from the MSP target μ , and ε_{ijk} is a $N(0, \sigma^2)$ random variable that represents the difference of the k th measurements on the j th stream from the process mean over all streams at time t . In this representation of MSP data, the total variation is allocated into two sources: σ_a^2 accounting for the variation over time common to all streams, and σ^2 accounting for the variation between the streams at specific time t . From this model, the cross correlation between any pair of streams is given by $\rho = \sigma_a^2 / (\sigma_a^2 + \sigma^2)$.

Mortell and Runger proposed monitoring the average at time t of the means across all streams with an individuals control chart to detect an overall assignable cause. Also, they proposed monitoring the range of the stream's means at time t denoted by $R_t = \max(\bar{x}_{ij}) - \min(\bar{x}_{ij})$ or the maximum **residual** at time t given by $\max(\bar{x}_{ij}) - \bar{\bar{x}}_t$, with any individuals control chart to detect a relative assignable cause affecting one or a few streams. These proposed **control charts on residuals** are better than the GCC, especially when the variation in the process means over time is greater than the between-stream variability (i.e., when $\sigma_a > \sigma$). Mortell and Runger compared the performance of these proposed charts with the runs test of the GCC by Nelson (1986) using simulations of various MSPs and concluded that the cusum chart on the range outperforms the Shewhart and EWMA on either the range or the maximum residual. On the other hand, little difference in performance exists between the runs test of Nelson (1986)

a set of historical data is analyzed. Comprehensive reviews of many scan-based procedures are in Balakrishnan and Koutras (2002) and Glaz et al. (2001). Also see Kulldorff (1997), (2001), (2003), (2005), Sonesson and Bock (2003), and Naus and Wallenstein (2006).

An important feature of health care monitoring is the human element of the process. Differences between the people in the system are likely to be great, and these differences have great potential impact on the outcomes. For example, when evaluating hospital performance and physician care, we have to take into account that the patients will vary (from patient to patient, from hospital to hospital, from physician to physician) with regard to their general state of health and other demographics and factors. Risk-adjusted control charts can be devised to deal with this situation. Grigg and Farewell (2004) give a review of risk-adjusted monitoring. Grigg and Spiegelhalter (2007) recently proposed a risk-adjusted EWMA method. Risk adjusted control charts are also discussed by Winkel and Zhang (2007).

10.11 Overview of Other Procedures

There are many other process monitoring and control techniques in addition to those presented previously. This section gives a brief overview of some of these methods, along with some basic references. The selection of topics is far from exhaustive but does reflect a collection of ideas that have found practical application.

10.11.1 Tool Wear

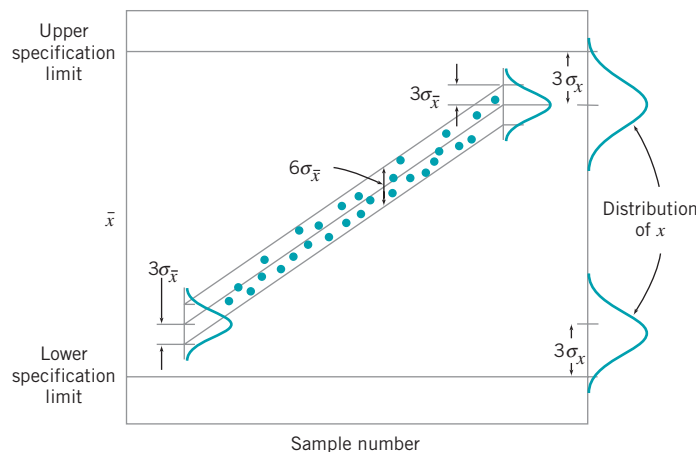
Many production processes are subject to tool wear. When tool wear occurs, we usually find that the process variability at any one point in time is considerably less than the allowable variability over the entire life of the tool. Furthermore, as the tool wears out, there will generally be an upward drift or trend in the mean caused by the worn tool producing larger dimensions. In such cases, the distance between specification limits is generally much greater than, say, 6σ . Consequently, the modified control chart concept can be applied to the tool-wear problem. The procedure is illustrated in Fig. 10.29.

The initial setting for the tool is at some multiple of σ_x above the lower specification limit—say, $3\sigma_x$ —and the maximum permissible process average is at the same multiple of σ_x below the upper specification limit. If the rate of wear is known or can be estimated from the data, we can construct a set of slanting control limits about the tool-wear trend line. If the sample values of $\bar{x}$ fall within these limits, the tool wear is in control. When the trend line exceeds the maximum permissible process average, the process should be reset or the tool replaced.

Control charts for tool wear are discussed in more detail by Duncan (1986) and Manuele (1945). The regression control chart [see Mandel (1969)] can also be adapted to the tool-wear problem. Quesenberry (1988) points out that these approaches essentially assume that resetting the process is expensive and that they attempt to minimize the number of adjustments made to keep the parts within specifications rather than reducing overall variability. Quesenberry develops a two-part tool-wear compensator that centers the process periodically and protects against assignable causes, as well as adjusting for the estimated mean tool wear since the previous adjustment.

10.11.2 Control Charts Based on Other Sample Statistics

Some authors have suggested the use of sample statistics other than the average and range (or standard deviation) for construction of control charts. For example, Ferrell (1953) proposed that subgroup midranges and ranges be used, with control limits determined by the median midrange and the median range. The author noted that ease of computation would be



■ **FIGURE 10.29** Control chart for tool wear.

a feature of such control charts and that they would do a better job of detecting “outlier” points than conventional control charts. The median has been used frequently instead of $\bar{x}$ as a center line on charts of individuals when the underlying distribution is skewed. Similarly, medians of R and s have been proposed as the center lines of those charts so that the asymmetrical distribution of these statistics will not influence the number of runs above and below the center line.

The recent interest in robust statistical methods has generated some application of these ideas to control charts. Generally speaking, the presence of assignable causes produces outlier values that stretch or extend the control limits, thereby reducing the sensitivity of the control chart. One approach to this problem has been to develop control charts using statistics that are themselves outlier-resistant. Examples include the median and midrange control charts [see Clifford (1959)] and plotting subgroup boxplots [(see Ingelwiz and Hoaglin (1987) and White and Schroeder (1987)]. These procedures are typically not as effective in assignable-cause or outlier detection as are conventional $\bar{x}$ and R (or S) charts.

A better approach is to plot a sample statistic that is sensitive to assignable causes ($\bar{x}$ and R or s), but to base the control limits on some outlier-resistant method. The paper by Ferrell (1953) mentioned above is an example of this approach, as is plotting $\bar{x}$ and R on charts with control limits determined by the trimmed mean of the sample means and the trimmed mean of the ranges, as suggested by Langenberg and Ingelwiz (1986).

Rocke (1989) has reported that plotting an outlier-sensitive statistic on a control chart with control limits determined using an outlier-resistant method works well in practice. The suggested procedures in Ferrell (1953), Langenberg and Ingelwiz (1986), and his own method are very effective in detecting assignable causes. Interestingly enough, Rocke also notes that the widely used two-stage method of setting control limits, wherein the initial limits are treated as trial control limits, the samples that plot outside these trial limits are then removed, and a final set of limits are then calculated, performs nearly as well as the more complex robust methods. In other words, the use of this two-stage method creates a robust control chart.

In addition to issues of robustness, other authors have suggested control charts for other sample statistics for process-specific reasons. For example, when pairs of measurements are made on each unit or when comparison with a standard unit is desired, one may plot the difference $x_{1j} - x_{2j}$ on a **difference control chart** [see Grubbs (1946) and the supplemental text material]. In some cases, the largest and smallest sample values may be of interest. These charts have been developed by Howell (1949).

10.11.3 Fill Control Problems

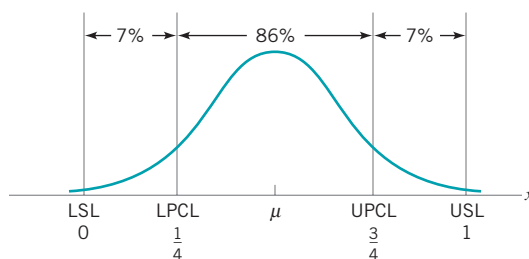
Many products are filled on a high-speed, multiple-head circular filling machine that operates continuously. It is not unusual to find machines in the beverage industry that have from 40 to 72 heads and operate at speeds of from 800 to 1000 bottles per minute. In such cases, it is difficult to sample products from specific heads because there is no automatic method of identifying a filled container with its filling head. Furthermore, in addition to assignable causes that affect all filling heads simultaneously, some assignable causes affect only certain heads. Special sampling and control charting methods are needed for these types of fill control problems. Ott and Snee (1973) present some techniques useful for this problem, particularly for filling machines with a moderate number of heads. Many of the methods described in Section 10.3 for multiple-stream processes are also useful in fill control problems.

10.11.4 Precontrol

Precontrol is a technique that is used to detect shifts or upsets in the process that may result in the production of nonconforming units. The technique differs from statistical process control in that conventional control charts are designed to detect shifts in process parameters that are statistically significant, and precontrol requires no plotting of graphs and no computations. Precontrol uses the normal distribution in determining changes in the process mean or standard deviation that could result in increased production of nonconforming units. Only three units are required to give control information.

To demonstrate the procedure, suppose that the quality characteristic is normally distributed and that the natural tolerance limits ($\mu \pm 3\sigma$) exactly coincide with the specification limits. Furthermore, the process average μ is halfway between the specifications, so that the process is producing 0.27% fallout. Construct two precontrol limits (called upper and lower PC lines), each of which is one-fourth the distance in from the modified limit, as in Fig. 10.30. Since the distribution of the quality characteristic is normal, approximately 86% of the process output will lie inside the PC lines, and approximately 7% will lie in each of the regions between the PC line and specification limit. This means that only about one item in 14 will fall outside a PC line if the process mean and standard deviation are on the target values.

If the probability is one-fourteenth that one unit falls outside a PC line, the probability that two consecutive units fall outside a PC line is $\left(\frac{1}{14}\right)\left(\frac{1}{14}\right) = \frac{1}{196} = \frac{1}{200}$. That is, if the process is operating correctly, the probability of finding two consecutive units outside of a given PC line is only about $\frac{1}{200}$. When two such consecutive units are found, it is likely that the process has shifted to an out-of-control state. Similarly, it is unlikely to find the first unit beyond one PC line and the next beyond the other PC line. In this case, we suspect that process variability has increased.



■ FIGURE 10.30 Location of precontrol lines.

A set of rules follow that describe the operation of precontrol. These rules assume that 1% to 3% nonconforming production is acceptable and that the process capability ratio is at least 1.15.

1. Start the process. If the first item is outside specifications, reset and start again.
2. If an item is inside specifications but outside a PC line, check the next item.
3. If the second item is outside the same PC line, reset the process.
4. If the second item is inside the PC line, continue. The process is reset only when two consecutive items are outside a given PC line.
5. If one item is outside a PC line and the next item is outside the other PC line, the process variability is out of control.
6. When five consecutive units are inside the PC lines, shift to frequency gauging.
7. When frequency gauging, do not adjust process until an item exceeds a PC line. Then examine the next consecutive item, and proceed as in step 4.
8. When the process is reset, five consecutive items must fall inside the PC lines before frequency gauging can be resumed.
9. If the operator samples from the process more than 25 times without having to reset the process, reduce the gauging frequency so that more units are manufactured between samples. If we must reset before 25 samples are taken, increase the gauging frequency. An average of 25 samples to a reset indicates that the sampling frequency is satisfactory.

Precontrol is closely related to a technique called **narrow-limit gauging** (or **compressed-limit gauging**), in which inspection procedures are determined using tightened limits located so as to meet established risks of accepting nonconforming product. Narrow-limit gauging is discussed in more general terms by Ott (1975).

Although precontrol has the advantage of simplicity, it should not be used indiscriminately. The procedure has serious drawbacks. First, because no control chart is usually constructed, all the aspects of pattern recognition associated with the control chart cannot be used. Thus, the diagnostic information about the process contained in the pattern of points on the control chart, along with the logbook aspect of the chart, is lost. Second, the small sample sizes greatly reduce the ability of the procedure to detect even moderate-to-large shifts. Third, precontrol does not provide information that is helpful in bringing the process into control or that would be helpful in reducing variability (which is the goal of statistical process control). Finally, the assumption of an in-control process and adequate process capability is extremely important. Precontrol should only be considered in manufacturing processes where the process capability ratio is much greater than one (perhaps at least two or three), and where a near-zero defects environment has been achieved. Ledolter and Swersey (1997) in a comprehensive analysis of precontrol also observe that its use will likely lead to unnecessary tampering with the process; this can actually increase variability. This author believes that precontrol is a poor substitute for standard control charts and would never recommend it in practice.

10.11.5 Tolerance Interval Control Charts

The usual intent of standard control limits for phase II control charts based on normal distribution is that they contain six standard deviations of the distribution of the statistic that is plotted on the chart. For the normal distribution, this is the familiar quantity $0.9973 = 1 - 2(0.00135)$, where 0.00135 is the probability outside each control limit or the area in the tail of the distribution of the plotted statistic. Standard control limits do not satisfy this because while a normal distribution is assumed, many plotted statistics do not follow the normal distribution. The sample range R and sample standard deviation s are good examples. The